

CHEMISTRY

Consolidated Class Notes — A4 Edition

The complete class-wise notes of Chemistry by Sakshi Ma'am — re-arranged topic-wise, duplicate lectures merged, every chart & visual explanation kept.

12 UNITS INSIDE

01 Matter & Its States

02 Classification of Matter

03 Atomic Structure

04 Periodic Table & Classi...

05 Chemical Bonding & Mole...

06 Chemical Reactions

07 Acids, Bases & Salts

08 Metals, Non-metals & Me...

09 Important s, p, d & f B...

10 Organic Chemistry — Bas...

11 Polymers

12 Chemistry in Everyday L...

Re-arranged & de-duplicated edition

Source: Eduteria Chem Notes Class Wise.pdf (157 pages, 12 lecture files)

Chemistry by Sakshi Ma'am · Contact: 8252405793, 9431216685

Print-ready A4 · 2026

About this edition

These are the **complete class-wise chemistry notes of Sakshi Ma'am (Eduteria, 72nd BPSC batch)**, re-arranged into 12 clean topic-wise units for A4 printing. **Every fact, table, equation, mnemonic and visual explanation of the original notes is kept** — only the repetition is gone.

The source PDF teaches the same topics several times across its 12 lecture files (the batch revised them). Where a topic appeared more than once, the most complete version is kept and the repeats are merged into it. Green **"De-dup note"** strips record exactly what was merged, so nothing is lost silently. Lecture cover pages, watermarks and page headers were removed.

#	Source lecture file (as in the original PDF)	Pages	Now in
1	01 Basic Chemistry Part-I	2-10	Unit 1
2	02 Basic Chemistry Part-II	12-23	Units 2, 1 & 8
3	03 Atomic Structure	25-40	Unit 3 (+ history → Unit 4)
4	04 Periodic Table & Classification	42-46	Unit 4
5	05 Chemical Bonding & Molecular Structure	48-56	Unit 5
6	06 Chemical Bond and Hydrogen Bond	58-64	Unit 5
7	07 Chemical Reaction	66-71	Unit 6
8	08 Acids, Bases and Salts	73-81	Unit 7 (+ carbon intro → Unit 10)
9	09 "Acids, Bases..." (actual content: Periodic Table revision)	83-85	Unit 4
10	10 Important S, P, D, F Elements	87-110	Unit 9
11	11 Metal & Nonmetal	112-124	Unit 8
12	12 Organic Chemistry Part-1 & 2 (+ Polymers, Everyday life, Fuels)	126-157	Units 10-12

Major duplicate topics resolved

- ✓ States of matter — taught in Part-I (pp. 4-10) and again in Part-II (p. 12): merged in Unit 1; Part-II's examples kept.
- ✓ Metals / Non-metals / Metalloids — Part-II (pp. 13-16) vs Metal & Nonmetal lecture (pp. 112-117): merged in Unit 8; alloy tables of p. 17 attached there.
- ✓ History of periodic table (Döbereiner, Newlands, Mendeleev) — Atomic Structure (pp. 25-27) and Periodic Table (pp. 42-43) lectures: kept once, in Unit 4.
- ✓ Atomic number, electron distribution ($2n^2$ rule, octet) and valency — repeated in both lectures: kept once, in Unit 3; shell-capacity table merged.
- ✓ Periodic trends — three versions (p. 31 table, p. 46 table, pp. 83-85 revision): merged into one master table + factor lists in Unit 4.
- ✓ Bond types & comparison tables — Bonding (p. 48) vs Chemical Bond & H-bond (pp. 58, 63): merged into one set in Unit 5.
- ✓ Bases (Arrhenius/Brønsted definitions and examples) printed twice on source p. 75 — kept once.
- ✓ Organic-compound definition & nature of carbon — p. 81 vs pp. 126: kept once, in Unit 10.
- ✓ Refining methods (distillation/liquation/electrolytic) repeated on pp. 123-124 — kept once in Unit 8.
- ✓ d-block alloy-formation block printed twice on p. 104; $K_2Cr_2O_7$ properties wrongly repeated under $KMnO_4$ on p. 107 — each kept once in Unit 9.
- ✓ Detergent/biodegradability blocks repeated on p. 153; octane & TEL blocks repeated on pp. 156-157 — merged in Units 12.
- ✓ Ionization-energy factor list appears under both IE and electron-gain-enthalpy on p. 85 — kept once.

How to read: coloured bands open each unit, green strips tell you what was merged, boxed panels highlight exam points, and every original chart keeps its "Fig. U-x" caption with the source page.

Contents

Unit 01 — Matter & Its States	6
1.1 What is Matter?	6
1.2 Physical Nature of Matter — Particulate, not Continuous	6
1.3 The Three States of Matter	7
1.4 Can Matter Change Its State?	8
1.5 Evaporation & Cooling	9
1.6 The Five States of Matter	10
1.7 Important Measurement Units	11
1.8 Fundamental Gas Laws □ (from Basic Chemistry Part-II)	11
Unit 02 — Classification of Matter	13
2.1 Classification at a Glance	13
2.2 Pure Substances	13
2.3 Mixtures	14
2.4 Colloids	15
2.5 Separation of Mixtures	16
Unit 03 — Atomic Structure	17
3.1 Foundations — Laws of Chemical Combination	17
3.2 Dalton's Atomic Theory	17
3.3 Thomson's Model (Plum-Pudding)	17
3.4 Rutherford's Model (Nuclear Model)	18
3.5 Bohr's Model of the Atom	18
3.6 Sub-atomic Particles	19
3.7 Atomic Number, Mass Number & Electron Distribution	19
3.8 Isotopes, Isobars & Atomic Species	21
3.9 Atomic Spectrum of Hydrogen	22
3.10 Planck's Quantum Theory & Photoelectric Effect	22
3.11 Towards the Quantum Model	23
Unit 04 — Periodic Table & Classification of Elements	25
4.1 Introduction	25
4.2 Döbereiner's Triads (1829)	25
4.3 Newlands' Law of Octaves (1866)	25
4.4 Mendeleev's Periodic Table (1869)	25
4.5 Modern Periodic Table	26
4.6 Periodic Trends — Master Table	27

Unit 05 — Chemical Bonding & Molecular Structure	29
5.1 What is Chemical Bonding & Why do Atoms Combine?	29
5.2 Types of Chemical Bonds — One Combined Picture	30
5.3 Ionic (Electrovalent) Bond	32
5.4 Covalent Bond & Its Parameters	33
5.5 Molecular Orbital Theory (MOT)	37
5.6 Hydrogen Bond (Short Notes)	39
Unit 06 — Chemical Reactions	41
6.1 What is a Chemical Reaction?	41
6.2 Core Types of Chemical Reactions	42
6.3 Effects of Oxidation in Everyday Life	44
6.4 Exothermic vs Endothermic Reactions	44
Unit 07 — Acids, Bases & Salts	47
7.1 Acids — Definitions	47
7.2 Acid Rain	48
7.3 Bases	48
7.4 pH — Strength of Acids & Bases	49
7.5 Salts	49
7.6 Carbon — The Gateway to Organic Chemistry	51
Unit 08 — Metals, Non-metals & Metallurgy	52
8.1 Metals	52
8.2 Non-metals	54
8.3 Hydrogen & Water (from Metal & Nonmetal lecture)	56
8.4 Sulphur & Its Important Compounds	57
8.5 Metallurgy	57
Unit 09 — Important s, p, d & f Block Elements	61
9.1 s-Block Elements	61
9.2 p-Block Elements — General Properties	63
9.3 Group 13 — Boron Family (B, Al, Ga, In, Tl)	64
9.4 Group 14 — Carbon Family (C, Si, Ge, Sn, Pb)	65
9.5 Group 15 — Nitrogen Family (N, P, As, Sb, Bi)	65
9.6 Group 16 — Oxygen Family (O, S, Se, Te, Po)	66
9.7 Group 17 — Halogens (F, Cl, Br, I)	67
9.8 Group 18 — Noble Gases	68
9.9 d-Block Elements (Transition Metals)	69
9.10 f-Block Elements (Inner Transition Metals)	72

Unit 10 — Organic Chemistry — Basics & Hydrocarbons	75
10.1 Allotropy & Allotropes of Carbon	75
10.2 Organic Compounds & the Uniqueness of Carbon	75
10.3 Hydrocarbons	76
10.4 Cyclic (Closed-Chain) Compounds	77
10.5 Functional Groups	78
10.6 Isomerism	78
10.7 Chemical Properties of Carbon Compounds	80
10.8 Methane, Ethene, Ethyne, Ethanol & Ethanoic Acid	81
Unit 11 — Polymers	83
11.1 Polymers & Polymerisation	83
11.2 Classification of Polymers	83
11.3 Types of Polymerisation	84
11.4 Important Addition Polymers	85
11.5 Resins (Condensation Polymers)	86
11.6 Natural & Synthetic Rubber	87
11.7 Polyesters & Biodegradable Polymers	87
11.8 Some More Important Polymers	88
Unit 12 — Chemistry in Everyday Life & Fuels	89
12.1 Medicine & Drugs	89
12.2 Dyes	89
12.3 Rocket Propellants	90
12.4 Chemicals in Food	91
12.5 Soaps & Synthetic Detergents	91
12.6 Fuels	92
12.7 Quick Revision — Everyday Chemistry	94

UNIT 01

Matter & Its States

Consolidated from: Basic Chemistry Part-I (pp. 2-10) + Fundamental Gas Laws (from Part-II, pp. 22-23)

1.1 What is Matter?

Matter is everything in this universe that **occupies space and has mass** — it offers resistance and can be perceived directly by our senses. Air, water, stones, sand, clouds, pencils, books — everything is made up of matter (e.g. book, pen, pencil, water, air, all living beings).

- Which of these is matter — happiness, air, sandwich, thoughts, juice, eraser? → **Air, sandwich, juice and eraser**, because they have mass, occupy space and can be broken into further particles. Happiness and thoughts are not matter.

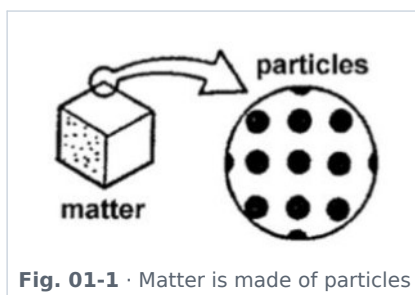


Fig. 01-1 · Matter is made of particles

Constituents of Matter

- According to the **early Indian philosophers**, every living and non-living thing is made of five basic elements called the **Panchtatva — Air, Water, Earth, Sky and Fire**. Matter is therefore a composition of these five constituents.

1.2 Physical Nature of Matter — Particulate, not Continuous

- Matter is **particulate in nature** — it consists of particles (visible in the microscopic image of a cube).
- If we put a drop of red colour in water, the water turns red — the particles of the colour mix with the particles of water.

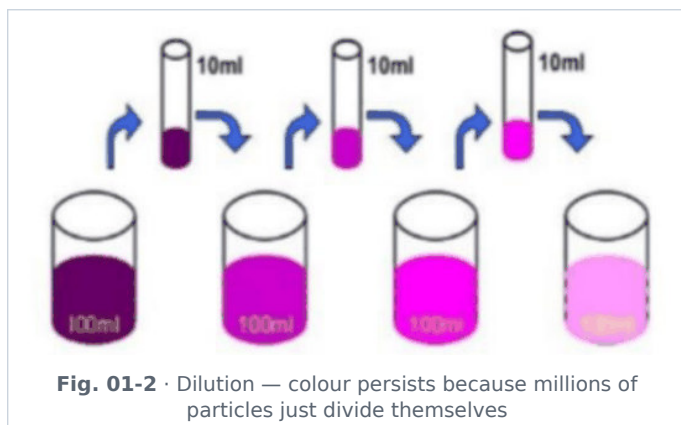
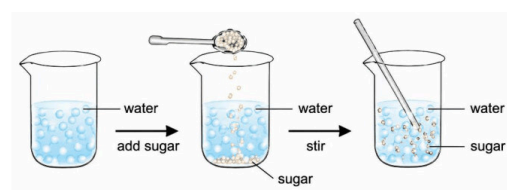


Fig. 01-2 · Dilution — colour persists because millions of particles just divide themselves



➤ We cannot see these particles through naked eyes.

Fig. 01-3 · Dissolving sugar/salt — particles slip into the spaces between water particles

- The particles of matter are **very small** and can be broken into further particles. On dilution of a colourful solution we can still see the colour — millions of particles divide themselves on dilution. We cannot see these particles with the naked eye.

Characteristics of Particles of Matter

- Particles of matter have **spaces between them** — salt dissolves in water because its particles get into the spaces between water particles and a mixture is formed.
- Particles of matter are **moving all the time** — hence they possess **kinetic energy** (energy due to motion). Particles intermix on their own: salt in water, various gases in air, ink in water.
 - Diffusion** — the process of mixing two different types of particles together. Diffusion becomes **faster on heating**, because kinetic energy of particles increases on heating.

- Particles of matter **attract each other** — the force of attraction varies in different forms of matter:
 - **Solids** have the highest force of attraction — we cannot move our hands through a solid object; the particles are so tightly bound.
 - **Gases** have the least — we can move our hands easily in air because the particles are loosely bound.
 - Liquids are intermediate — we can move our hands through water but not as freely as in air.

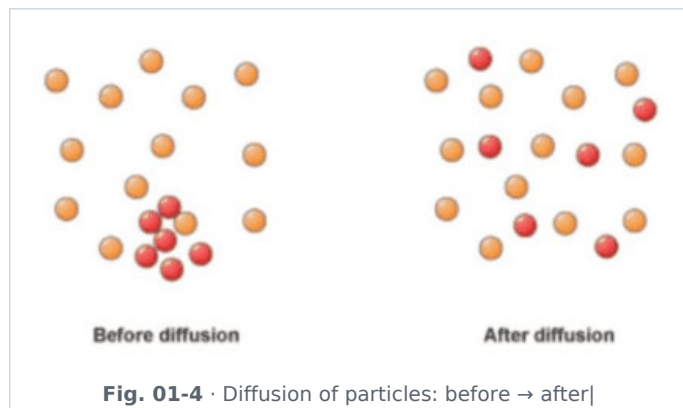


Fig. 01-4 · Diffusion of particles: before → after|

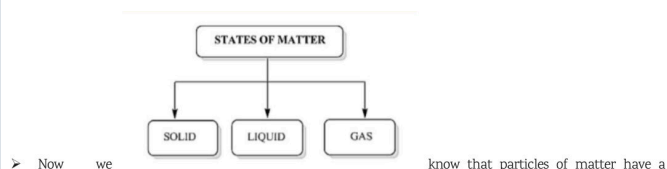


Fig. 01-5 · States of matter — based on force of attraction

Increasing order of force of attraction: Gas < Liquid < Solid

1.3 The Three States of Matter

Based on the force of attraction between particles, matter exists in three states — **solid, liquid and gaseous**.

The Solid State

- Solids have (i) a **specific shape**, (ii) **distinct boundaries** and (iii) a **fixed volume**. e.g. sugar, iron, gold, wood.
- Less kinetic energy; particles arranged in order; force of attraction is **maximum**; very little space between particles → **cannot be compressed**.
- Rubber band, sponge and salt are all solids — a rubber band and sponge change shape only when force is applied, and each grain of salt has its own definite shape (it only appears to take the container's shape).

The Liquid State

- Liquids have a **fixed volume but no fixed shape** — they take the shape of the container (also called **fluids**, as they can flow / be poured). e.g. water, milk, oil, mercury, alcohol.
- Force of attraction is less than in solids → space between particles → they can flow easily; cannot be compressed. Particles have high kinetic energy and keep moving.
- **Can other matter diffuse into liquids?** Yes — solids, liquids and gases can, because there is space between liquid particles:
 - Solid into liquid → mixing sugar in tea; Liquid into liquid → mixing ink in water; Gas into liquid → O₂ and CO₂ present in water.

The Gaseous State

- Gases have **neither fixed volume nor fixed shape** — they completely fill the container. e.g. hydrogen, oxygen.
- Particles have the least (almost no) force of attraction, large spaces between them, **maximum kinetic energy**, move freely at high speed in any direction.
- Gases **can be compressed** easily and put into a small container, unlike solids and liquids; different gases diffuse into each other easily.

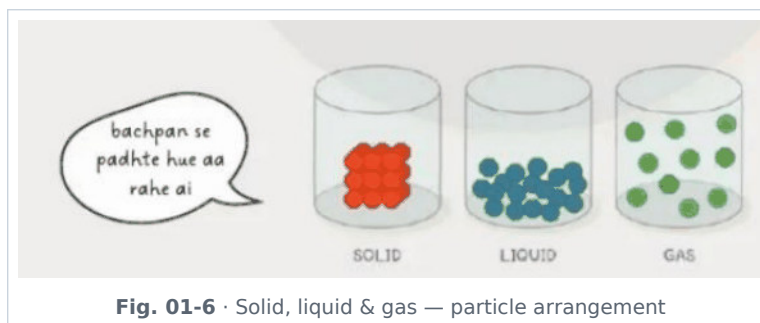


Fig. 01-6 · Solid, liquid & gas — particle arrangement

Differences between the Three States

Property	Solid	Liquid	Gas
Shape	Definite	Indefinite (takes container's shape)	Indefinite
Volume	Definite	Definite	Indefinite
Force of attraction	Maximum	Fewer than solid	Negligible
Compressibility	Cannot be compressed	Cannot be compressed	Can be compressed
Kinetic energy	Minimum	More than solid	Maximum
Motion of particles	Only vibrate at fixed positions	Slide over one another	Move freely
Density	Highest	Lower than solid	Lowest
Flow	Cannot flow	Flow	Flow

1.4 Can Matter Change Its State?

Water exists in three states — **Ice (solid)**, **Water (liquid)**, **Water vapour (gas)** — proof that matter can change state.

Effect of Temperature (heating)

- **Solids → Liquids:** on heating, kinetic energy increases and force of attraction decreases; particles vibrate, change positions and become free.
 - **Melting point** — temperature at which a solid melts to become a liquid at atmospheric pressure (melting point of ice = 273.16 K).
 - **Fusion** — the process of melting a solid into liquid. During melting the temperature stops rising; the heat is used to diminish the force of attraction — it is "hidden".
 - **Latent heat** — the heat energy used to break the force of attraction between particles. **Latent heat of fusion** — heat energy required to change 1 kg of solid into liquid at its melting point at atmospheric pressure.
 - **Atmospheric pressure** — pressure exerted by the weight of the atmosphere.
- **Liquids → Gases:** kinetic energy rises until particles overcome all attraction and the liquid boils.
 - **Boiling point** — temperature at which a liquid starts boiling at atmospheric pressure (boiling point of water = 373 K).
 - **Latent heat of vaporisation** — heat energy required to change 1 kg of liquid into gas at its boiling point at atmospheric pressure.



Fig. 01-7 · Adding heat to a solid creates a liquid (fusion)

2. Liquids:

- Just like in solids, the kinetic energy of particles of liquid increases, the force of attraction among them decreases and they start moving freely.
- As we keep on supplying the heat, a point comes when the particles overcome the forces of attraction completely.
- This is when a liquid starts changing into gas.
- **Boiling Point** - The temperature at which a liquid starts boiling at the atmospheric pressure is known as its Boiling Point. For Example, The boiling point of water is 373 Kelvin.
- **Latent Heat of Vaporisation** - the amount of heat energy required to

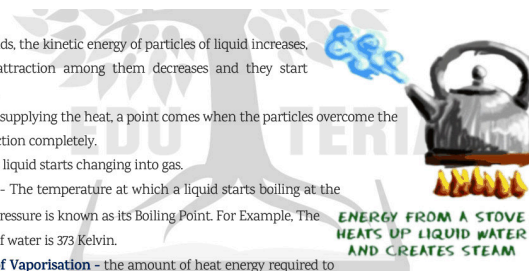
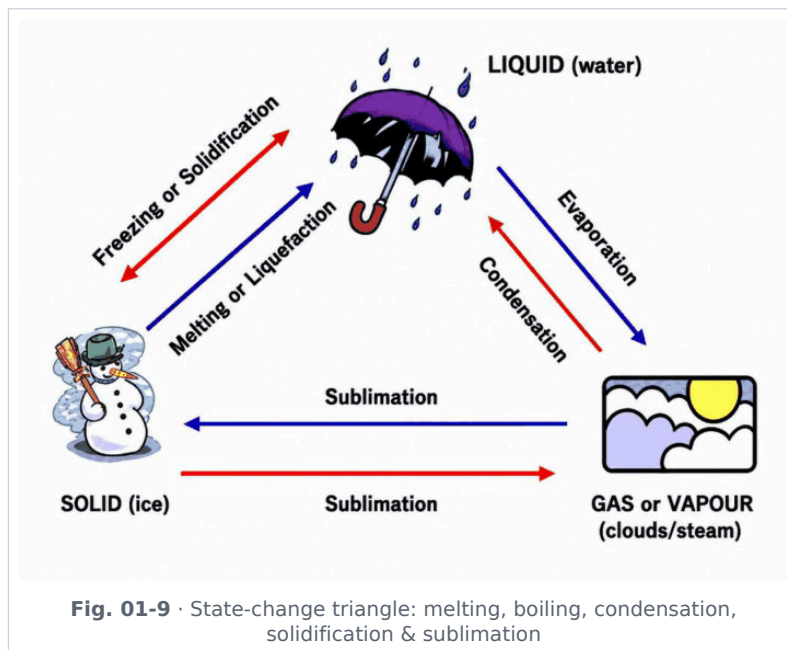


Fig. 01-8 · Boiling — latent heat of vaporisation

Effect of Temperature (cooling)

- **Gases → Liquids:** kinetic energy falls; the gas turns into liquid.
 - **Condensation / Liquefaction** — converting a gas into liquid by cooling (clouds form by condensation of water vapour from Earth).
- **Liquids → Solids:** kinetic energy falls; liquid becomes solid (formation of ice).
- **Sublimation** — change of state of gas directly into solid and vice-versa. e.g. **camphor** directly evaporates into air without becoming liquid.

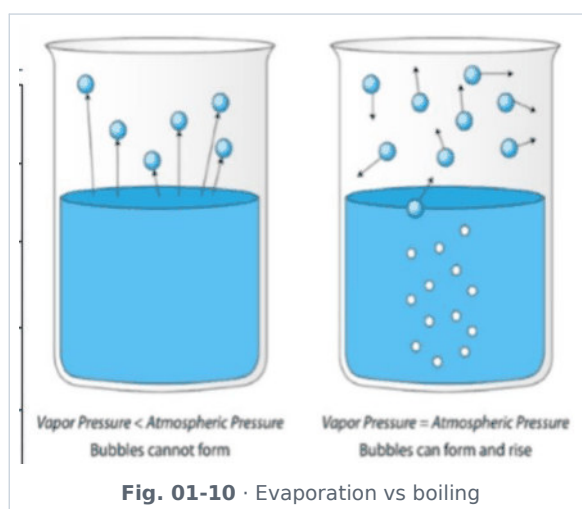


Effect of Pressure

- Applying pressure brings particles closer, increasing the force of attraction. On compressing and cooling, a gas changes into liquid.
- **Dry ice** — solid CO_2 ; directly turns into gas when pressure decreases to 1 atmosphere.

1.5 Evaporation & Cooling

- **Evaporation** — change of a liquid into vapours at any temperature **below its boiling point**. Liquid particles with high kinetic energy overcome the forces of attraction without any external help.



Why does evaporation cause cooling?

- Evaporation uses the energy of liquid particles; particles absorb energy from the surroundings to compensate → surroundings cool down. e.g.
- Palms feel cool when acetone (nail-paint remover) evaporates from them.

- People sprinkle water on roofs/ground on sunny days to cool the area.
- Hot tea sips faster from a saucer than from a cup.

Everyday applications

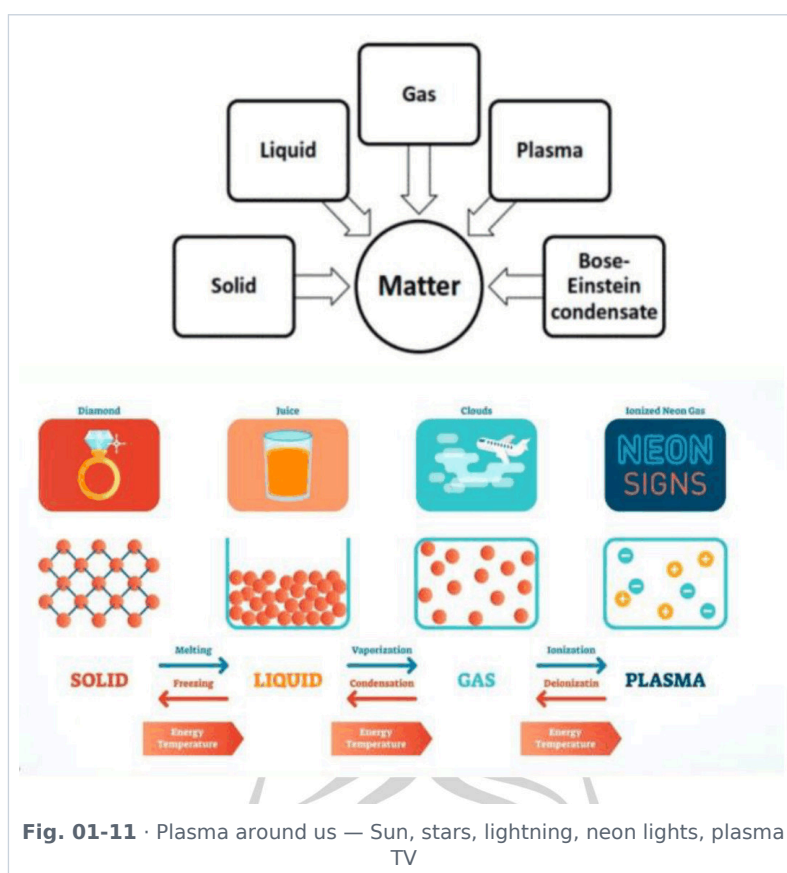
- **Cotton clothes in summer:** sweat evaporates taking energy from the body; cotton absorbs sweat and exposes it to the atmosphere → evaporation → cooling.
- **Water droplets on a cold glass:** water vapour in air condenses on contact with the cold walls of the glass (gas → liquid droplets).

1.6 The Five States of Matter

Besides solid, liquid and gas, scientists have discovered two more states: **Plasma** and **Bose-Einstein Condensate (BEC)**.

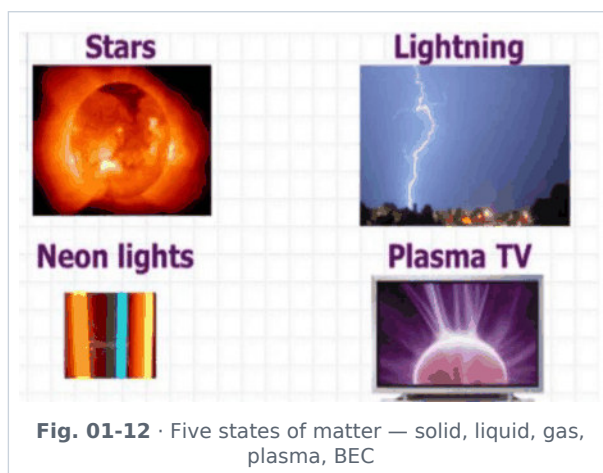
Plasma

- Particles are **super-excited and super-energetic**, in the form of **ionised gases**.
- e.g. fluorescent tubes and neon bulbs — the gas inside gets ionised when electricity passes and the plasma glows. **The Sun and stars glow because of plasma.**



Bose-Einstein Condensate (BEC)

- Fifth state, discovered by **Albert Einstein** from the studies of Indian scientist **Satyendra Nath Bose**.
- Formed by condensing gases of extremely low densities to much lower temperatures.



⚡ **De-dup note:** "The five states" also summarised the classification of matter — that flowchart now lives in Unit 2 (Classification of Matter), so it is not repeated here.

1.7 Important Measurement Units

Quantity	SI unit	Common / other units
Mass	Kilogram (kg)—	
Volume	Cubic metre (m ³)	Litre (L); 1 mL = 1 cm ³
Temperature	Kelvin (K)	°C; 0 °C = 273.16 K ≈ 273 K; K = °C + 273
Pressure	Pascal (Pa)	atmosphere (atm); 1 atm = 1.01 × 10 ⁵ Pa
Normal atmospheric pressure	1 atm at sea level—	

⚡ **De-dup note:** In the source this table carried merged dashes in single cells; values are unchanged.

1.8 Fundamental Gas Laws □ (from Basic Chemistry Part-II)

⚡ **De-dup note:** The gas laws were printed at the end of the classification lecture (Part-II, pp. 22–23). They belong with states of matter, so they are placed here — content unchanged.

Boyle's, Charles's, Gay-Lussac's & Avogadro's Laws

Law	Relationship	Statement	Constants	Mathematical form
Boyle's law	Pressure-Volume	Volume inversely proportional to pressure (P ↑ → V ↓)	T, n	$V \propto 1/P$ (PV = k)
Charles's law	Volume-Temperature	Volume directly proportional to Kelvin temperature	P, n	$V \propto T$
Gay-Lussac's law	Pressure-Temperature	Pressure directly proportional to Kelvin temperature	V, n	$P \propto T$
Avogadro's law	Volume-Amount	Volume directly proportional to moles of gas	T, P	$V \propto n$

Gas Laws Cheat sheet!				
Relationship between the physical properties	Boyle's Law Pressure - Volume	Charles's Law Volume - Temperature	Gay Lussac's Law Pressure - Temperature	Avagadro's Law Volume - Amount
Constant	Temperature and mass	Pressure and mass	Volume	Temperature and Pressure
Mathematical representation	$V \propto \frac{1}{P}$	$V_t = V_0 \left[1 + \frac{t}{273.15} \right]$ $V_0 = \text{Volume of given mass of gas at } 0^\circ\text{C}$ $V_t = \text{Volume of given mass of gas at } t^\circ\text{C}$	$P \propto T$	$V \propto T$ $M = k_d \frac{m}{V} = k_d d$ $M = \text{Molar mass of the gas}$ $d = \text{Density}$
Quantitative relationship	$\frac{P_1}{P_2} = \frac{V_2}{V_1}$	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$	

□□□

Fig. 01-13 · Combined gas laws chart — Boyle, Charles, Gay-Lussac, Avogadro & ideal-gas equation

Combined & Ideal Gas Law

$$P_1 V_1 / T_1 = P_2 V_2 / T_2 \quad PV = nRT$$

- **Combined gas law** joins Boyle's, Charles's and Gay-Lussac's laws for a fixed amount of gas.
- **Ideal gas equation** $PV = nRT$ connects all four variables: P = pressure, V = volume, n = number of moles, R = universal gas constant, T = temperature (in Kelvin).

Ideal vs Real Gases

Ideal gas	Real gas
Theoretical gas that perfectly obeys the gas laws	Deviates from ideal behaviour at high pressure & low temperature
Particles assumed to have no volume	Particles have finite volume
No attractive forces between particles	Intermolecular forces exist

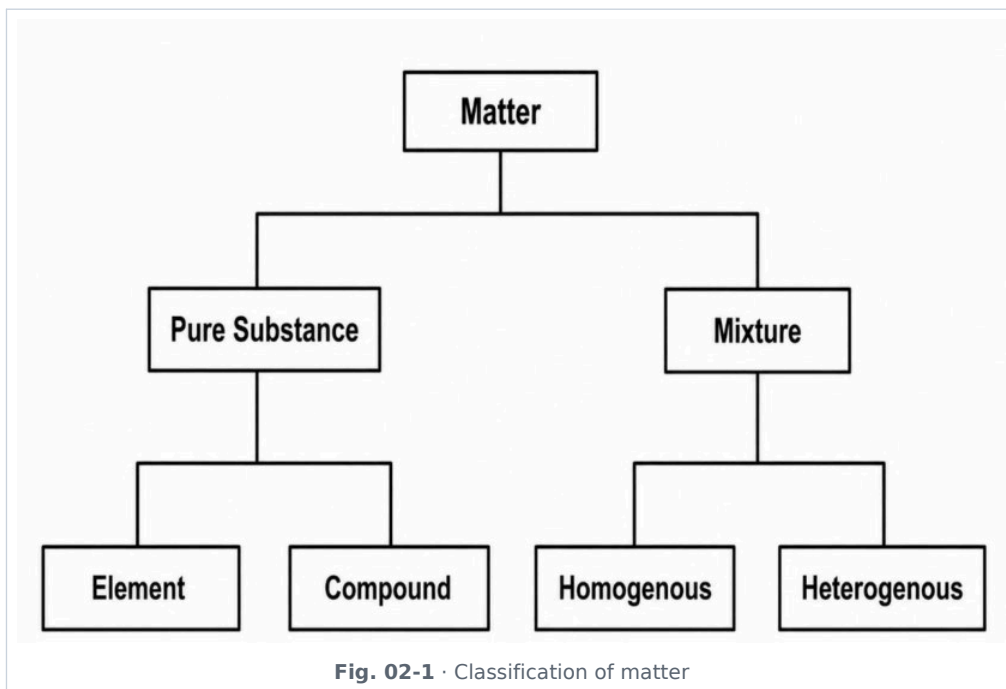
UNIT 02

Classification of Matter

Consolidated from: Basic Chemistry Part-II (pp. 12–23) — mixtures, pure substances, colloids & separation methods

2.1 Classification at a Glance

At the bulk (macroscopic) level matter is classified into **(a) Mixtures** and **(b) Pure Substances**, which are further classified as shown below.



🔗 **De-dup note:** The three phases (solid / liquid / gas) with their examples and the gas laws of this lecture are consolidated in Unit 1; the detailed element classes (metals, non-metals, metalloids) are consolidated in Unit 8 — no repetition here.

2.2 Pure Substances

- A material containing **only one type of particle** is a pure substance. Pure substances are further divided into **elements** and **compounds**.

Element

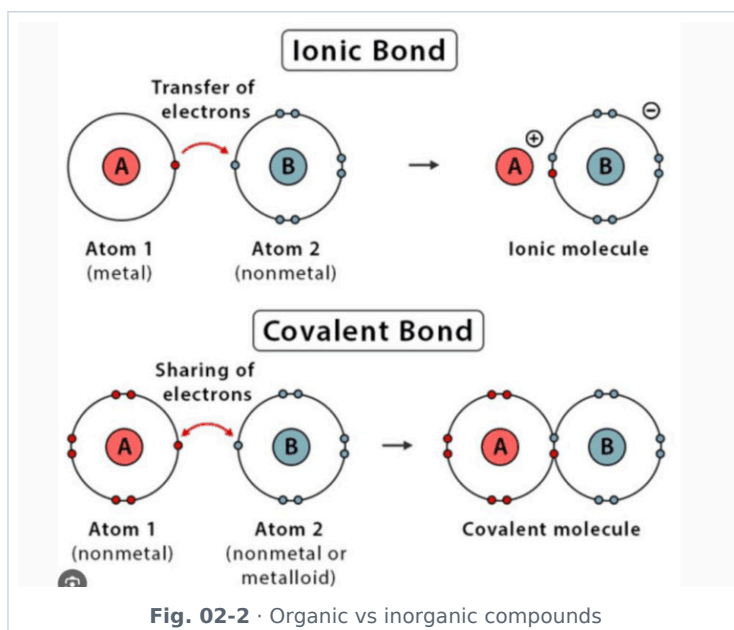
- A pure substance that **cannot be broken down** into simpler substances by chemical means; made of atoms with the same number of protons.
- Elements are split into three classes on the basis of physical & chemical properties: **(1) Metals (2) Non-metals (3) Metalloids** — position, properties and the full comparison table are given in **Unit 8**.

Remember

Most of the elements in the periodic table are metals. There are six metalloids — boron, silicon, germanium, arsenic, antimony, tellurium. Of the 17 non-metals, 12 are gases; all others are solids except **bromine**, a red volatile liquid.

Compound

- A substance composed of **two or more different elements chemically bonded in a fixed ratio**; formed by chemical reactions.



Organic vs Inorganic Compounds

Point	Organic compounds	Inorganic compounds
Composition	Always contain C and H, often O, N, S or P	Typically lack C-H bonds
Bonding	Covalent bonds	Often ionic bonds
Properties	Gases, liquids or low-melting solids; generally insoluble in water	High-melting solids, often water-soluble
Source	Mostly from living organisms	From non-living sources like minerals
Examples	Glucose, methane, proteins, sugar, DNA	Salt, metals, water, CO ₂

- Organic: carbohydrates, proteins, lipids, plastics. Inorganic: NaCl, H₂O, HCl.
- Exceptions** — some carbon substances behave like mineral compounds and are classified as inorganic: CO₂, CO, carbonates (CaCO₃), cyanides.

Ionic vs Covalent Compounds

Point	Ionic compounds	Covalent compounds
Bonding mechanism	Electron transfer (electrostatic attraction)	Electron sharing (directional shared pairs)
Elements	Usually metal + non-metal	Two or more non-metals
State & m.p./b.p.	Hard, brittle crystalline solids; high m.p./b.p.	Gases, liquids or soft solids; low m.p./b.p.
Conductivity	Conduct when molten or dissolved (not as solids)	Generally poor conductors
Solubility	Often soluble in water	Tend to be insoluble (polar covalent can be exceptions)
Structure	Crystal lattices	Discrete molecules or network solids
Examples	NaCl, CaO, (NH ₄) ₃ P	H ₂ O, CO ₂ , CH ₄ , SF ₆

Physical & Chemical Properties of Compounds

- Physical properties** — measured without changing chemical identity: melting/boiling points (defined, distinct), density, solubility, electrical conductivity (important for ionic compounds in liquid/aqueous state), appearance (colour, odour, state).
- Chemical properties** — ability to transform into new substances: reactivity (with oxygen/acids), flammability, toxicity, heat of combustion, corrosion resistance.

2.3 Mixtures

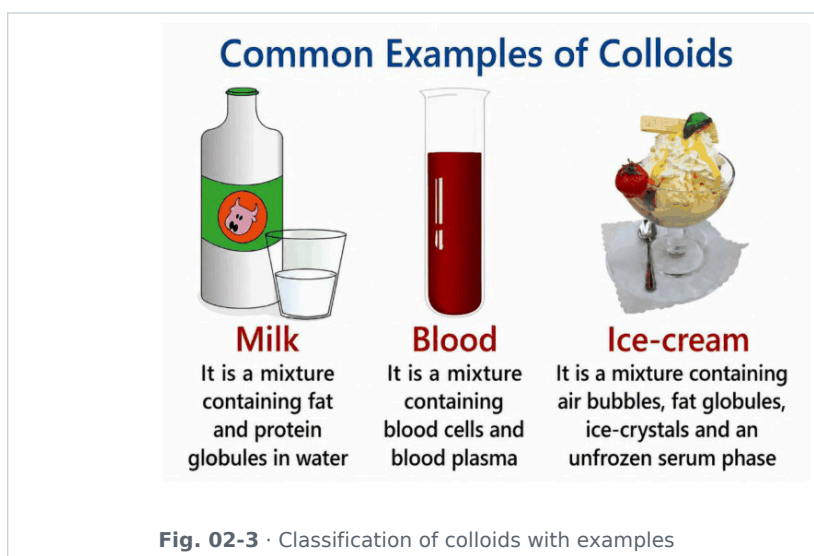
- A mixture has **two or more substances in any ratio**; primarily of two types — **homogeneous** and **heterogeneous**.

Point	Homogeneous mixture	Heterogeneous mixture
Composition	Uniform throughout (single phase); components indistinguishable	Non-uniform; two or more distinct phases
Visibility	Components not seen by naked eye	Components easily seen
Separation	Difficult — needs distillation etc.	Easy — filtration etc.
Also called	Solutions	Suspensions or colloids
Examples	Sugar solution, air, salt water, saline, vodka, brass	Oil & water, salt + sugar, suspensions, salad, pizza, soil, trail mix

- Phase** — any distinct portion of matter uniform throughout in composition and properties.

2.4 Colloids

- Colloids** are heterogeneous mixtures where microscopic insoluble particles (**1 nm - 1000 nm**, i.e. 10^{-7} – 10^{-4} cm) of one substance (**dispersed phase**) are scattered throughout another (**dispersion medium**). They appear homogeneous but are not.
- Key characteristics:**
 - Stability** — particles do not settle on standing.
 - Filterability** — pass through ordinary filter paper but separated by semipermeable membranes / centrifuges.
 - Tyndall effect** — scatter light, making a light beam visible (fog, milk).
 - Brownian movement** — continuous zig-zag motion due to bombardment by dispersion-medium molecules.



Types of Colloids

Colloid	Dispersed phase in medium	Examples
Sol	Solid in liquid	Paint, cell fluids, blood
Solid sol	Solid in solid	Coloured gemstones, alloys
Emulsion	Liquid in liquid	Milk, hair cream, mayonnaise
Foam	Gas in liquid	Whipped cream, soap lather
Solid foam	Gas in solid	Pumice stone, foam rubber
Aerosol	Liquid in gas	Fog, mist, hairspray
Solid aerosol	Solid in gas	Smoke, dust
Gel	Liquid in solid	Jelly, butter, cheese

- **Components:** dispersed phase = "solute-like" particles (smaller amount); dispersion medium = "solvent-like" medium (larger amount).
- **Applications:** medicines are colloidal for better absorption (argyrol); water treatment (alum coagulation); paint, ink, rubber and food industries; blood is a colloid — specialized colloidal solutions used in medical treatments.

2.5 Separation of Mixtures

Methods use physical properties — size, solubility, boiling point, magnetism — to isolate components **without changing their chemical identity**.

Method	What it separates	Example
Filtration	Insoluble solid from liquid	Sand from water
Evaporation	Soluble solid from liquid	Salt from water
Distillation	Liquids with different boiling points	Alcohol & water
Fractional distillation	Liquids with closer boiling points	Crude-oil components
Magnetic separation	Magnetic from non-magnetic material	Iron filings
Chromatography	Different rates of movement across a stationary phase	Separating dyes
Centrifugation	Particles by density at high spin	Blood components
Sublimation	Sublimable solid from non-sublimable	Iodine / ammonium chloride
Crystallisation	Purification by hot-solvent recrystallisation	Purifying solids

🔗 **De-dup note:** The metallurgy-related dressing/concentration methods for ores (gravity, magnetic, leaching) are kept with extraction of metals in Unit 8.

UNIT 03

Atomic Structure

Consolidated from: Atomic Structure lecture (pp. 32–40) + electron rules merged from the Periodic lecture (pp. 27–28, 44)

3.1 Foundations — Laws of Chemical Combination

🔖 **De-dup note:** The "What is atomic structure?" introduction of the source opens with the mass laws; they are kept here in full.

Law of Multiple Proportions — John Dalton

- If two elements form more than one compound, the masses of one element that combine with a fixed mass of the other are in a ratio of **small whole numbers**. e.g. in CO and CO₂, fixed carbon (12 g) combines with 16 g and 32 g of oxygen — a simple **1 : 2** ratio.
- Does not apply to non-stoichiometric compounds, polymers or oligomers; inapplicable for isotopes. It solidified the concept of atoms and helped determine relative atomic masses.

Law of Definite Proportions — Joseph Proust (1799)

- A pure chemical compound always contains the **same elements in the same fixed proportion by mass**, regardless of source, method of preparation or quantity.
- Not applicable to compounds containing isotopes (e.g. ³⁵Cl₂ vs ³⁷Cl₂) and non-stoichiometric compounds.

Law of Conservation of Mass — Antoine Lavoisier

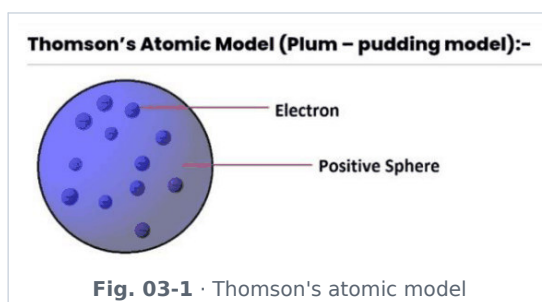
- Matter can neither be created nor destroyed in a chemical reaction or physical change; **mass of reactants = mass of products** (atoms are only rearranged). Basis of balancing equations and stoichiometry.
- Holds only for low-energy chemical reactions — in nuclear reactions mass converts to energy.

3.2 Dalton's Atomic Theory

- John Dalton: all matter is made of **atoms** — indivisible and indestructible; atoms of an element are exactly the same, atoms of different elements differ in size and mass.
- **Postulates:**
 - Every matter is made up of atoms; atoms are indivisible.
 - Specific elements have only one type of atom; each atom has its own constant mass that varies element to element.
 - Atoms undergo rearrangement during a chemical reaction; atoms can neither be created nor destroyed but transformed.
- Successfully explained the laws of chemical combination — conservation of mass, constant properties, multiple proportions and reciprocal proportions.
- **Demerits:** could not explain isotopes; said nothing about the structure of the atom; later sub-atomic particles proved atoms are divisible.

3.3 Thomson's Model (Plum-Pudding)

- Postulate — the atom is a **sphere of positive charge with electrons embedded** in it.
- Limitation — could not satisfactorily explain the results of Rutherford's scattering experiment.



3.4 Rutherford's Model (Nuclear Model)

- Experiment — thin **gold foil** bombarded with fast **alpha (α) particles**; he expected the gold atoms to deflect them.

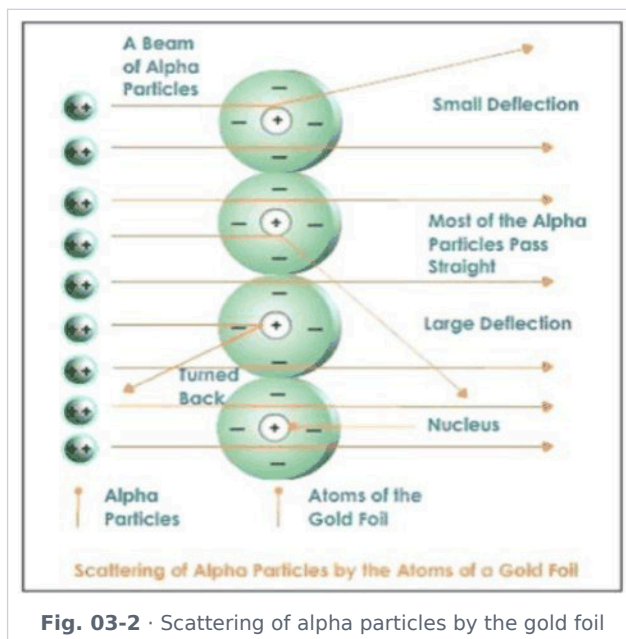


Fig. 03-2 · Scattering of alpha particles by the gold foil

	Inference	
passed straight through the gold foil	Atom contains a lot of empty space	
deflected by small angles	Positive charges do not occupy much of the atom's space	
	12000 particles bounced back	Positive charge is concentrated in a very small region (nucleus)

- Nuclear model:** atoms contain a lot of unoccupied space; a heavy **positively charged nucleus** sits at the centre; electrons revolve around it in circular paths called **orbits**; nucleus is much smaller than the atom and holds almost all the mass.
- Drawback (Maxwell's objection):** a charged particle moving in a circle radiates energy — the electron would lose speed and spiral into the nucleus, so the atom would be unstable. The model could not explain atomic stability.
- Nucleons** — protons and neutrons collectively.

3.5 Bohr's Model of the Atom

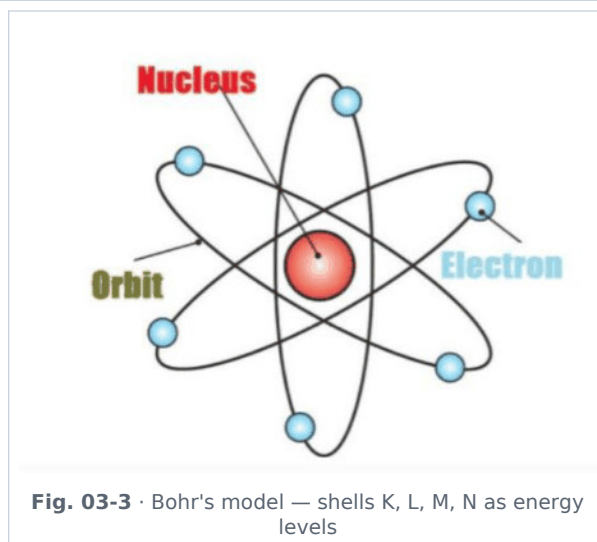


Fig. 03-3 · Bohr's model — shells K, L, M, N as energy levels

- Electrons spin around the nucleus only in **certain separate allowed orbits (energy levels)** — they do **not** emit energy while in these special orbits. Orbits are labelled K, L, M, N... ($n = 1, 2, 3, 4...$).
- Electron absorbs energy as EMR when it jumps from a lower energy level (ground state) to a higher one (excited state) and vice-versa: $\Delta E = E_2 - E_1 = h\nu$.
- Quantitative results for hydrogen-like species (H, He^+, Li^{2+} , nuclear charge Z):

$$\text{Radius of } n\text{-th orbit: } r_n = 52.9 \times n^2/Z \text{ pm}$$

$$\text{Energy of } n\text{-th orbit: } E_n = -2.18 \times 10^{-18} (Z^2/n^2) \text{ J} = -13.6 (Z^2/n^2) \text{ eV} = -313.6 (Z^2/n^2) \text{ kcal/mol}$$

$$\text{Velocity of electron: } v = (2.18 \times 10^8) Z/n \text{ cm s}^{-1} (n = 1, 2, 3, 4...)$$

- Limitations of Bohr's theory:** failed for atoms with more than one electron; could not explain splitting of spectral lines in strong magnetic (Zeeman) or electric (Stark) fields; ignored dual nature of particles and the uncertainty principle.

3.6 Sub-atomic Particles

- Electron** — J.J. Thomson, cathode-ray-tube experiment. **Proton** — E. Rutherford (gold-foil work). **Neutron** — James Chadwick (a neutral particle inside the atom).

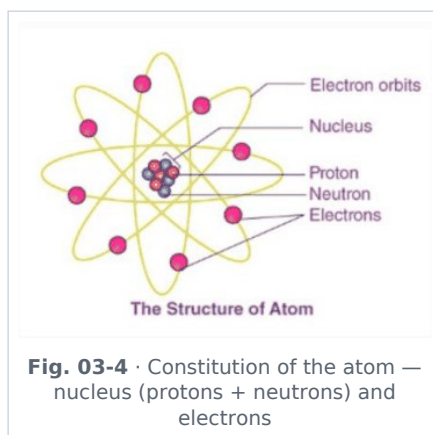


Fig. 03-4 · Constitution of the atom — nucleus (protons + neutrons) and electrons

Term	Electron (e^-)	Proton (p)	Neutron (n)
Symbol	e^-	p	n
Approximate relative mass	1/1836	1	1
Mass in kg	9.109×10^{-31}	1.673×10^{-27}	1.675×10^{-27}
Mass in amu	5.485×10^{-4}	1.007	1.008
Charge (coulomb)	1.602×10^{-19}	1.602×10^{-19}	0
Actual charge (e.s.u.)	4.8×10^{-10}	4.8×10^{-10}	0

- Atoms consist of protons and electrons in balanced proportion; protons sit inside, electrons outside — so electrons can be removed from an atom.

3.7 Atomic Number, Mass Number & Electron Distribution

- Atomic number (Z)** = number of protons in an atom. Atoms of different elements always have different proton counts; elements are defined by Z (hydrogen: $Z = 1$).
- Mass number (A)** = number of protons + number of neutrons.

Rules for Electron Distribution

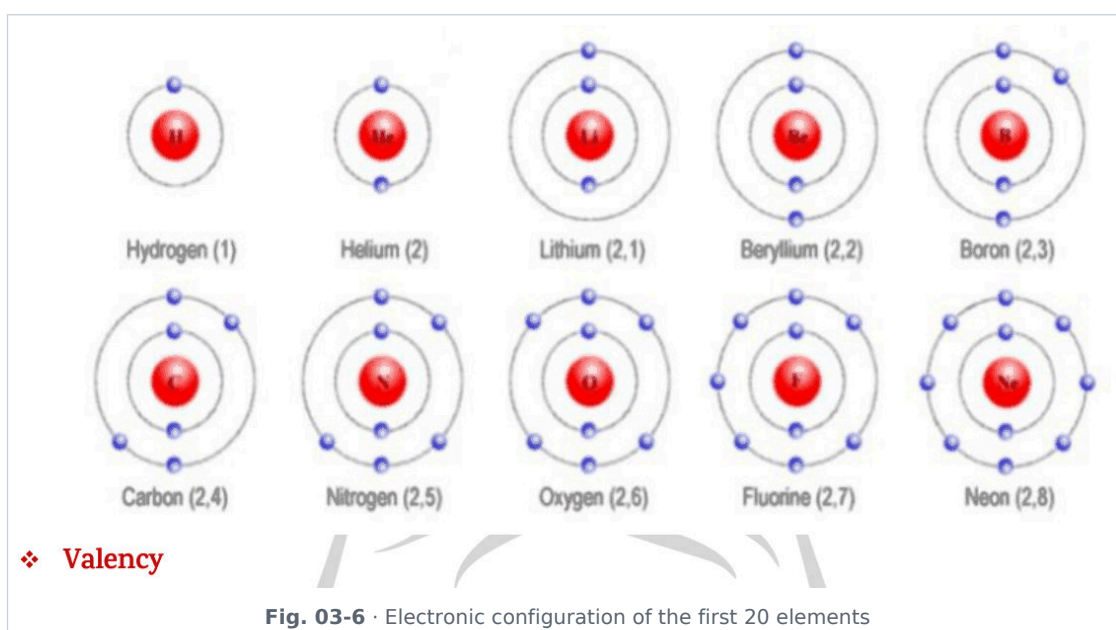
- (i) An orbit can hold a maximum of $2n^2$ electrons; (ii) orbits fill from inside out ($n = 1$ first); (iii) the outermost shell cannot hold more than **8 electrons** even if it has capacity — the **Octet rule** (8 electrons in the outermost shell makes the atom very stable).

No. of electrons	Electron Shell	Maximum Capacity
1	K Shell	$2(1)^2 = 2$ electrons
2	L Shell	$2(2)^2 = 8$ electrons
3	M Shell	$2(3)^2 = 18$ electrons
4	N Shell	$2(4)^2 = 32$ electrons

(ii) Orbits are filled from inside to outside. First, $n = 1$ shell is filled, then $n = 2$ shell, and so on.

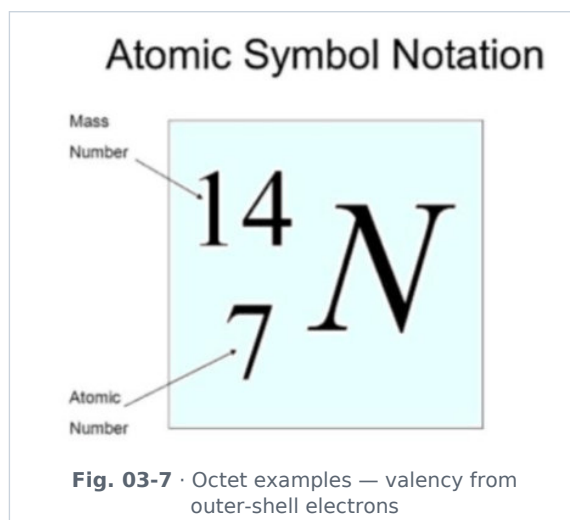
Fig. 03-5 · Rules for electron distribution

Electron shell	Maximum capacity
K shell ($n = 1$)	$2(1)^2 = 2$ electrons
L shell ($n = 2$)	$2(2)^2 = 8$ electrons
M shell ($n = 3$)	$2(3)^2 = 18$ electrons
N shell ($n = 4$)	$2(4)^2 = 32$ electrons



Valency

- **Valence electrons** — electrons in the outermost orbit (valence shell). Valency rises 1→4 then falls to 0 across a period and **remains the same down a group**.
- **Stable configuration:** $2 e^-$ if K is the outermost shell, otherwise $8 e^-$. Atoms react to achieve it; noble gases (He, Ne, Ar) are inert as they already have it.
- Valency = electrons an atom must give away or take to attain a stable configuration. Outermost-shell electrons decide it: silver has 1 outer electron → valency 1. Helium (2 outer electrons) can share both → valency 2. If the outer shell is nearly full, valency = $8 - \text{outer electrons}$ (oxygen: $8 - 6 = 2$).

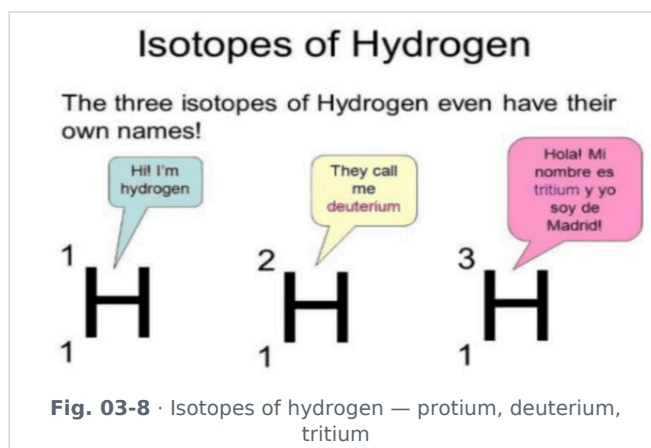


⚡ **De-dup note:** The $2n^2$ rule, octet rule and valency paragraphs were printed in both the Atomic-Structure and Periodic-Table lectures; they are merged here once.

3.8 Isotopes, Isobars & Atomic Species

Isotopes

- Atoms of the **same element** with the **same atomic number but different mass numbers**; pure substances; similar chemical nature but distinct physical characteristics.



- Uses of isotopes:** fuel of nuclear reactors — isotope of **uranium**; treatment of cancer — isotope of **cobalt**; treatment of goitre — isotope of **iodine**.

Isobars

- Atoms of **different elements** with the **same mass number** but different atomic numbers (e.g. ^{18}Ar , ^{19}K , ^{40}Ca — all $A = 40$).

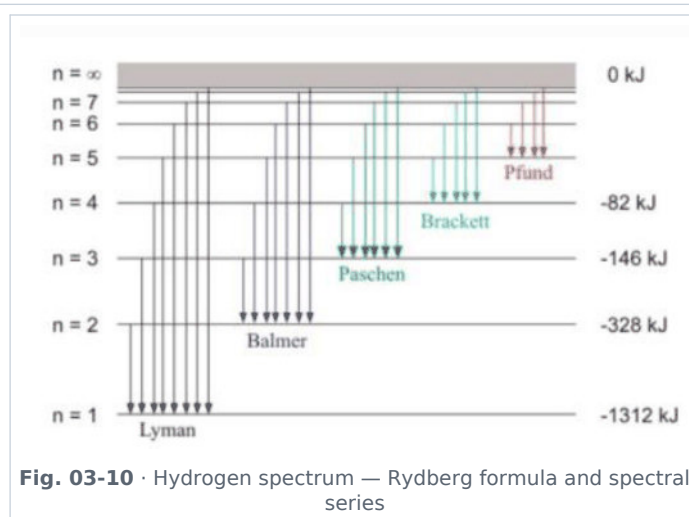
	U	V
Protons	5	5
Neutrons	5	6
Mass Number	$5 + 5 = 10$	$5 + 6 = 11$
Atomic Number	5	5

Fig. 03-9 · Isobars — same A, different Z

Atomic species to remember

Terms	Definition / explanation
Atomic number (Z)	Number of protons (or electrons) of a neutral atom
Mass number (A)	Total number of protons and neutrons
Nucleons	Protons + neutrons present in the nucleus
Isotopes	Same Z, different A — e.g. ${}^1\text{H}^1$, ${}^1\text{H}^2$, ${}^1\text{H}^3$
Isobars	Same A, different Z — e.g. ${}^{15}\text{P}^{32}$ and ${}^{16}\text{S}^{32}$
Isotones	Same number of neutrons, different Z or A — e.g. ${}^6\text{C}^{14}$, ${}^8\text{O}^{16}$, ${}^7\text{N}^{15}$
Isoelectronic	Atoms/molecules/ions with the same electron count — e.g. N_2 , CO , CN^-
Nuclear isomers	Same Z and A but different radioactive properties — U-X (half-life 1.4 min) and U-Z (half-life 6.7 h)
Isosters	Molecules with same number of atoms and same number of electrons — e.g. N_2 and CO

3.9 Atomic Spectrum of Hydrogen



$$\bar{\nu} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ where } R_H = \text{Rydberg constant } (108978 \text{ cm}^{-1})$$

n_1	n_2	Spectral series	Spectral region
2, 3, 4...	Lyman	UV	
3, 4, 5...	Balmer	Visible	
4, 5, 6...	Paschen	IR	
5, 6, 7...	Brackett	IR	
6, 7, 8...	Pfund	IR	

3.10 Planck's Quantum Theory & Photoelectric Effect

- Substances radiate or absorb energy **discontinuously** in the form of energy packets; the smallest packet is a **quantum** (in light: **photon**).
- $E \propto \nu \rightarrow E = h\nu$, h = Planck's constant = $6.626 \times 10^{-27} \text{ erg}\cdot\text{s}$ = $6.626 \times 10^{-34} \text{ J}\cdot\text{s}$. A body radiates/absorbs whole-number multiples: $h\nu$, $2h\nu$, $3h\nu$... $nh\nu$.
- Photoelectric effect:** light of suitable frequency ejects electrons from a metal surface; minimum frequency needed = **threshold frequency (ν_0)**; kinetic energy of ejected electrons $\propto$ frequency; number of electrons per second $\propto$ intensity of radiation.

$$h\nu - h\nu_0 = \frac{1}{2}mv^2$$

3.11 Towards the Quantum Model

de-Broglie equation

$$\lambda = h/mv = h/p \text{ (dual nature of matter)}$$

Heisenberg's Uncertainty Principle

$$\Delta x \cdot \Delta p \geq h / 4\pi m$$

- It is impossible to determine simultaneously the exact position and exact momentum of an electron.

Quantum Numbers

Quantum number	What it tells
Principal (n)	Main shell & approximate distance from nucleus; max electrons in shell = $2n^2$
Azimuthal (l)	Number of subshells in the shell (s, p, d, f); their shapes; for given n, $l = 0 \dots (n-1)$
Magnetic (m)	Preferred orientations of the electron in a subshell; for given l, $m = -l \dots +l$ ($2l+1$ values)
Spin (s)	Direction of spin; only two values, $-\frac{1}{2}$ and $+\frac{1}{2}$

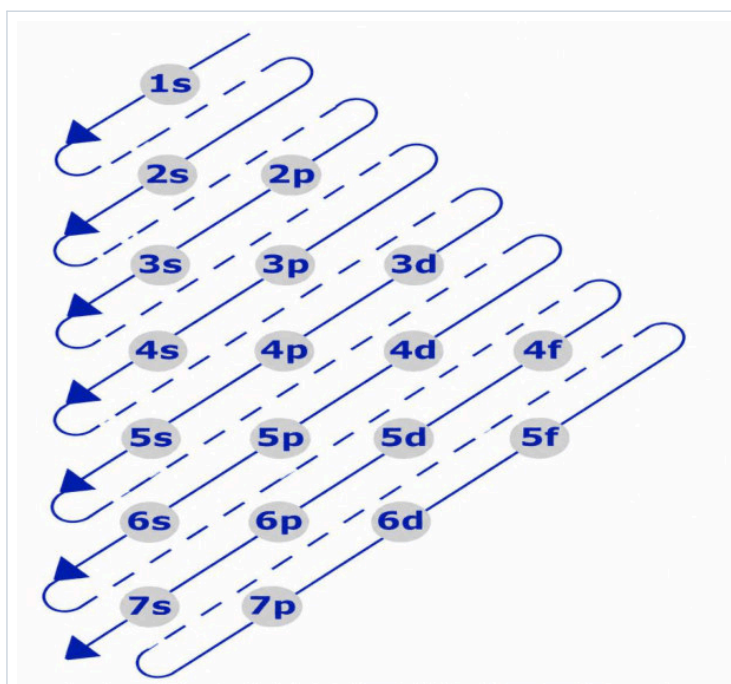
Value of l	0	1	2	3
Notation of sub shell	s	p	d	f
Values of m	0	-1,0,1	-2,-1,0,1,2	-3,-2,-1,0,1,2,3

➤ The spin quantum number (s)

Fig. 03-11 · de-Broglie relation & uncertainty principle

Rules for Filling of Electrons

- Aufbau principle** — electrons fill orbitals in order of increasing energy; lower (n+l) means lower energy.
- Pauli's exclusion principle** — an orbital holds at most two electrons, with opposite spins.
- Hund's rule of maximum multiplicity** — pairing in p, d, f orbitals cannot start until each orbital of the subshell is singly occupied.



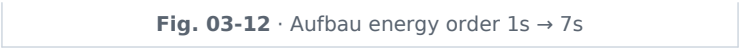


Fig. 03-12 · Aufbau energy order 1s → 7s

- Increasing energy order: $1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d...$

Exceptional Configurations — stability of half/full-filled orbitals

- Cu (29 electrons): expected $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^9$, but shifting one electron from 4s to 3d gives a symmetrical, more stable distribution → actual configuration **$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$** .
- Fully-filled and half-filled orbitals are extra stable — the same rule explains Cr and other exceptions.

UNIT 04

Periodic Table & Classification of Elements

Consolidated from: Periodic Table lecture (pp. 42–46) + periodic history merged from the Atomic Structure lecture (pp. 25–30) + trends revision (pp. 83–85)

4.1 Introduction

- Matter exists as elements, compounds and mixtures; **elements** contain only one type of atom (e.g. Na, Mg, Au). **118 elements** are known — all with different properties — so they are grouped such that elements of the same group have similar properties.

🔗 **De-dup note:** The history of the periodic table (Döbereiner → Newlands → Mendeleev → Modern) was taught twice — in the Atomic-Structure lecture (pp. 25–27) and in the Periodic-Table lecture (pp. 42–43). Both versions carried the same facts; the fuller version is kept here once.

4.2 Döbereiner's Triads (1829)

- By 1829, **30 elements** had been discovered. Döbereiner, a German scientist, grouped some elements into **Triads** of 3 with similar chemical properties — the atomic mass of the **middle element** $\approx$ **average of the other two**.

Triad	Elements	Atomic Mass	Arithmetic Mean
Triad-1	Lithium	7	$\frac{7+39}{2} = 23$
	Sodium	23	
	Potassium	39	
Triad-2	Chlorine	35.5	$\frac{35.5 + 126.5}{2} = 81$
	Bromine	80	
	Iodine	126.5	
Triad-3	Calcium	40	$\frac{40 + 137}{2} = 88.5$
	Strontium	87	
	Barium	137	

Limitations of Dobereiner's Triads

Fig. 04-1 · Döbereiner's triads with worked averages

Triad	Elements	Atomic mass	Arithmetic mean
Triad-1	Lithium, Sodium, Potassium	7, 23, 39	$(7+39)/2 = 23$
Triad-2	Chlorine, Bromine, Iodine	35.5, 80, 126.5	$(35.5+126.5)/2 = 81$
Triad-3	Calcium, Strontium, Barium	40, 87, 137	$(40+137)/2 = 88.5$

- Limitations:** only three triads were recognised; only **9 of the 30** known elements could be arranged as triads.

4.3 Newlands' Law of Octaves (1866)

- By 1866, **56 elements** were known. Newland, an English scientist, arranged them in increasing atomic mass and found every **8th element repeats** the properties of the 1st (like sa-re-ga-ma octaves in music) — e.g. properties of Li and Na were found similar.
- Limitations:**
 - Applicable only up to **calcium** (17 lighter elements).
 - Newly discovered elements did not fit the law.
 - Two unlike elements were sometimes put in one slot.
 - Placing of **iron (Fe)** far from **cobalt (Co)** and **nickel (Ni)** (which have similar properties) could not be explained.

4.4 Mendeleev's Periodic Table (1869)

- Dmitry Mendeleev, a Russian chemist — **63 elements** known by then. Arranged by **increasing atomic mass**; similar properties recur at regular intervals.

- **Mendeleev's periodic law:** "The properties of elements are a periodic function of their atomic masses."
- **Structure:** horizontal rows = **periods**; vertical columns = **groups** (eight, I to VIII). Groups I–VII subdivided into A and B; Group VIII has no subgroups. Properties in a period show regular gradation left → right; elements of a group show similar properties.

Group	I		II		III		IV		V		VI		VII		VIII		
Period	A	B	A	B	A	B	A	B	A	B	A	B	A	B	Transition Series		
Oxides:	R ₂ O		R ₂ O		R ₂ O ₃		R ₂ O		R ₂ O ₃		R ₂ O ₃		R ₂ O ₃		RO ₃		
Hydride	RH		RH ₂		RH ₃		RH ₂		RH ₃		RH ₃		RH				
1.	H 1.008																
2.	Li 6.939		Be 2.312		B 10.81		C 12.011		N 14.007		O 15.999		F 18.999				
3.	Na 22.99		Mg 22.99		Al 24.31		Si 28.09		P 30.504		S 32.06		Cl 35.453				
4. First Series	K 39.912		Ca 40.00		Sc 44.96		Ti 47.59		V 50.94		Cr 50.28		Mn 54.54		Fe 55.85	Co 56.93	Ni 58.71
Second Series	Ca 61.54		Zn 65.54		Ga 69.72		Ge 72.59		As 74.52		Se 73.66		Br 79.904				
5. First Series	Ba 85.47		Sr 87.62		Y 88.31		Zr 51.22		Nb 59.51		Mo 55.54		Tc 99		Ra 931.07	Rn 102.51	Pd 106.4
Second Series	Ag 107.77		Cd 112.48		In 114.37		Sn 118.69		Sb 121.60		Te 127.60		I 126.90				
6. First Series	Cs 132.30		Ba 137.34		La 138.91		Hf 178.40		Ta 108.55		W 183.85				Ra 190.2	Eh 152.2	Pd 155.09
Second Series	Au 156.50		Hg 200.55		Tl 204.37		Pb 207.19		Bi 233.98								

Fig. 04-2 · Mendeleev's periodic table (with oxide & hydride formulas)

Merits

- Predicted properties of elements from their positions.
- Inert gases, when discovered, fit into a **new group** without disturbing the existing order.
- Systematic study — knowing one element of a group lets us predict the others.
- Left gaps for undiscovered elements — **Gallium (Ga), Scandium (Sc), Germanium (Ge)** were discovered this way.
- **Corrected doubtful atomic masses** using expected positions and properties.

Limitations

- **Isotopes** (same element, different mass) violated the increasing-mass arrangement.
- **Hydrogen** resembles Group-I as well as Group-VII, but was placed in Group-I without explanation.
- Wrong (decreasing) order of atomic masses of some pairs could not be explained — e.g. **Co-Ni** and **Te-I**.

4.5 Modern Periodic Table

- In **1913, Moseley** proved that **atomic number** is the fundamental property of an element; after that **Neil Bohr** built the Modern Periodic Table on atomic number.
- **Modern periodic law:** properties of elements are a periodic function of their **atomic numbers**.
- Most properties depend on the number of **valence electrons**, and elements with the same number of valence electrons are grouped together.

Metals: Alkali metals, Alkaline earth metals, Lanthanides, Actinides, Transition metals, Poor metals, Metalloids.

Non-metals: Moderately active nonmetals, Highly active nonmetals, Noble gases.

Unknown chemical properties: Elements with unknown properties.

Group names:

- 1 Hydrogen & the alkali metals
- 2 Alkaline earth metals
- 11 Coinage metals (Cu, Ag & Au)
- 12 Volatile metals
- 13 Boron Group
- 14 Carbon Group
- 15 Pnictogens
- 16 Chalcogens
- 17 Halogens
- 18 Noble gases

Groups 3–10 are named after their first members i.e. Group 3 is the Scandium Group.

Fig. 04-3 · The Modern Periodic Table

Structure

- **Periods**: 7 horizontal rows (shells increase down a group). **Groups**: 18 vertical columns (same valence electrons).
- **Blocks**: s, p, d, f — by the orbital into which the last electron enters (s-block: groups 1-2; p-block: 13-18; d-block: 3-12, transition elements; f-block: inner transitions, lanthanides & actinides, placed at the bottom).

Period	Type	Elements
1st	Shortest	Only 2 — H and He
2nd & 3rd	Short periods	8 each (Li→Ne; Na→Ar)
4th & 5th	Long periods	18 each (K→Kr; Rb→Xe)
6th	Very long period	32
7th	Incomplete period—	

- **Group 1** (extreme left) = alkali metals (Li, Na, K, Rb, Cs, Fr); **Group 18** (extreme right) = noble gases (He, Ne, Ar, Kr, Xe, Rn); middle = metalloids/semi-metals showing mixed properties.
- **Exception**: group-18 elements have 8 valence electrons except **helium** (2) — yet He is rightly placed there because it has a stable configuration and similar properties.

Position & Electronic Configuration

- In **periods**: the n of the valence shell equals the period number; each period fills the next principal energy level ($n = 1, 2, \dots$); number of elements in a period = **twice the number of atomic orbitals** available in that level.
- In **groups**: same valence-shell configuration — e.g. group-1 alkali metals all show ns^1 configuration.

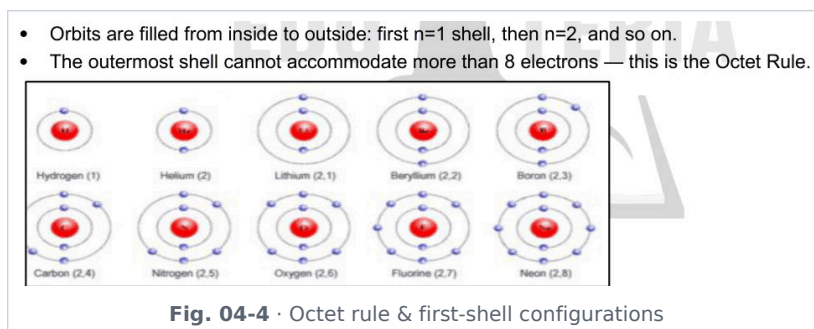


Fig. 04-4 · Octet rule & first-shell configurations

4.6 Periodic Trends — Master Table

🔗 **De-dup note:** Trends appeared three times in the source (a big table on p. 31, a summary on p. 46 and a revision on pp. 83–85). The three versions are identical in fact — one master table keeps them all.

Property	Across a period (left → right)	Down a group
Atomic radius	Decreases — increased nuclear charge pulls electrons inward	Increases — extra electron shells
Ionic radius	Decreases	Increases (anions larger than cations)
Ionization enthalpy / energy	Increases — stronger nuclear hold	Decreases — bigger size & shielding
Electron gain enthalpy	Increases (more negative) — tendency to gain e^- rises	Decreases — gain tendency falls
Electronegativity	Increases	Decreases
Metallic / electropositive character	Decreases	Increases
Non-metallic character	Increases	Decreases
Acidic nature of oxides	Increases	Decreases
Chemical reactivity	Metals: decreases; Non-metals: increases	Metals: increases; Non-metals: decreases
Valence electrons	Increase from 1 to 8	Remain the same

Factors Behind the Trends

- **Atomic radii are influenced by:** number of shells (+ → bigger); effective nuclear charge Z_{eff} (+ → smaller); screening effect (greater → larger).
- **Ionization energy factors:** atomic size, effective nuclear charge, screening effect, penetration effect, electronic configuration. Lower IE → atom readily forms a positive ion.
- **Electron gain enthalpy:** enthalpy change when an electron is added to a neutral gaseous atom forming an anion — positive = energy absorbed (less stable anion), negative = energy released (more stable). Depends on atomic size, Z_{eff} , screening effect, electronic configuration.
- **Electronegativity:** an atom's power, inside a molecule, to pull the shared pair towards itself — not fixed; depends on bonding partner. Factors: atomic size, Z_{eff} , magnitude of positive charge. It predicts the nature (ionic/covalent character) of the bond.

🔗 **De-dup note:** The ionization-energy factor list was printed twice on source p. 85 (under IE and again under electron gain enthalpy) — it is kept once above.

UNIT 05

Chemical Bonding & Molecular Structure

Consolidated from: Chemical Bonding & Molecular Structure (pp. 48–56) + Chemical Bond & Hydrogen Bond (pp. 58–64) — bond-type tables merged

5.1 What is Chemical Bonding & Why do Atoms Combine?

- **Chemical bond** — the attractive force that holds atoms together in a molecule, increasing stability and minimising energy.
- **Cause of combination:** atoms combine to achieve a **stable electronic configuration** — (i) to lower energy and (ii) to attain stable noble-gas configuration, usually a full **octet (8 valence electrons)**.



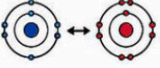
CHEMICAL BOND



What is a Chemical Bond?

A chemical bond is the attractive force that holds atoms or ions together in a molecule or compound. Bonds form to achieve stability, usually by completing the valence shell (octet rule).

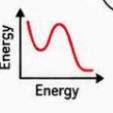
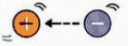
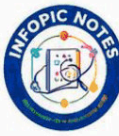
InfoPic Notes



stable ✓

Why Do Chemical Bonds Form?

- To attain **lower energy state**
- To achieve **stable electronic configuration**
- Due to **attractive forces** between atoms or ions

Types of Chemical Bonds

1 Ionic Bond (Electrovalent Bond)

- Formed by transfer of electrons
- Occurs between metal and non-metal
- Produces ions (cations & anions)
- Strong electrostatic attraction

$\text{Na} \cdot + \cdot \text{Cl} \rightarrow \text{Na}^+ + \text{Cl}^-$

★ Example: NaCl

★ Properties: High melting point, soluble in water, conducts electricity in molten/aqueous state

2 Covalent Bond

- Formed by sharing of electrons
- Occurs between non-metals
- Forms molecules

Single bond (–)	Double bond (=)	Triple bond (≡)
H–H	O=O	N≡N

★ Example: H₂O, O₂, N₂

★ Properties: Low melting point, poor electrical conductivity

3 Coordinate (Dative) Bond

- Shared electron pair contributed by one atom only
- Represented by an arrow (→)

$\text{H} \begin{array}{c} \text{H} \\ | \\ \text{H} \rightarrow \text{N}^+ \rightarrow \text{H}^+ \\ | \\ \text{H} \end{array}$

★ Example: NH₄⁺, CO

★ Special feature: Acts like covalent bond after formation

4 Hydrogen Bond

- Weak bond between H atom and highly electronegative atoms (F, O, N)

Intermolecular (between molecules)

 $\text{H}-\text{O} \cdots \text{O}-\text{H}$

Intramolecular (within molecule)

 $\text{H}-\text{O} \cdots \text{O}-\text{H}$

★ Importance: Responsible for high boiling point of water and DNA stability

5 Metallic Bond

- Attraction between positive metal ions and sea of delocalized electrons
- Found in metals



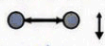
★ Properties explained: Electrical conductivity, malleability, ductility

Bond Parameters

• **Bond Length:** Distance between nuclei of bonded atoms



• **Bond Energy:** Energy required to break a bond



• **Bond Angle:** Angle between bonds in a molecule



• **Bond Order:** Number of bonds between two atoms (single = 1, double = 2)




Factors Affecting Bond Formation

• Atomic size ↓ • Electronegativity difference ↑ • Valency ↑ • Ionization energy ↓ • Electron affinity ↑

Significance of Chemical Bonds

• Determines structure and shape of molecules



• Explains physical and chemical properties

Melting point ↓

Boiling point ↑

• Essential for chemical reactions and life processes



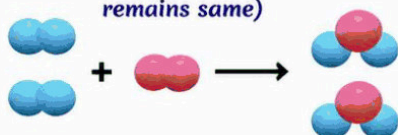

Fig. 05-1 · Chemical bond — complete summary chart (types, formation, significance)

Rules of Chemical Bonding (Chemical Combination)

Law of Conservation of Mass

Statement: Mass can neither be created nor destroyed in a chemical reaction.

Example: $\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ (Total mass remains same)

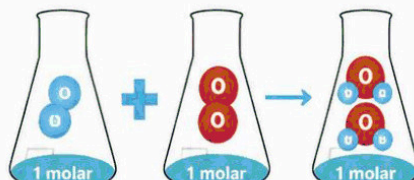


$2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}$
 Reactants Reaction Products

Gay-Lussac's Law of Gaseous Volumes

Statement: Gases react with each other in simple whole-number ratios by volume under similar conditions of temperature and pressure.

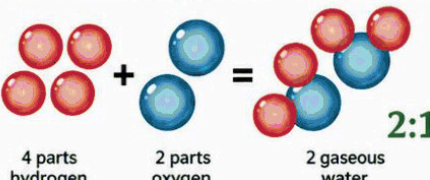
Example: $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$ (1:1:2 ratio)



Law of Definite Proportions (Proust's Law)

Statement: A chemical compound always contains the same elements in the same proportion by mass.

Example: Water (H_2O) always has hydrogen and oxygen in 2:16 mass ratio.



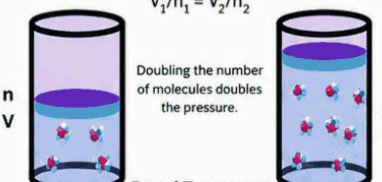
4 parts hydrogen 2 parts oxygen 2 gaseous water **2:1**

Avogadro's Law

Statement: Equal volumes of gases, under the same conditions of temperature and pressure, contain equal number of molecules.

Application: Basis of molar volume and molecular formula derivation.

$V/n = k$
 $V_1/n_1 = V_2/n_2$



Law of Multiple Proportions

Statement: If two elements combine to form more than one compound, the masses of one element that combine with a fixed mass of the other are in small whole-number ratios.

Example: CO and $\text{CO}_2 \rightarrow$ Oxygen: 16g in CO and 32g in CO_2 with 12g of Carbon



CO CO_2
 12:16 12:32
 1:1 1:2

Quick Study Quiz!

Qus: Gases react in simple volume ratios under same conditions. This is...

QUIZ!

- Avogadro's Law
- Law of Mass Conservation
- Law of Constant Composition
- Gay-Lussac's Law ☒

Fig. 05-2 · Laws of chemical combination — Gay-Lussac, Avogadro, Proust, Dalton (with quick quiz)

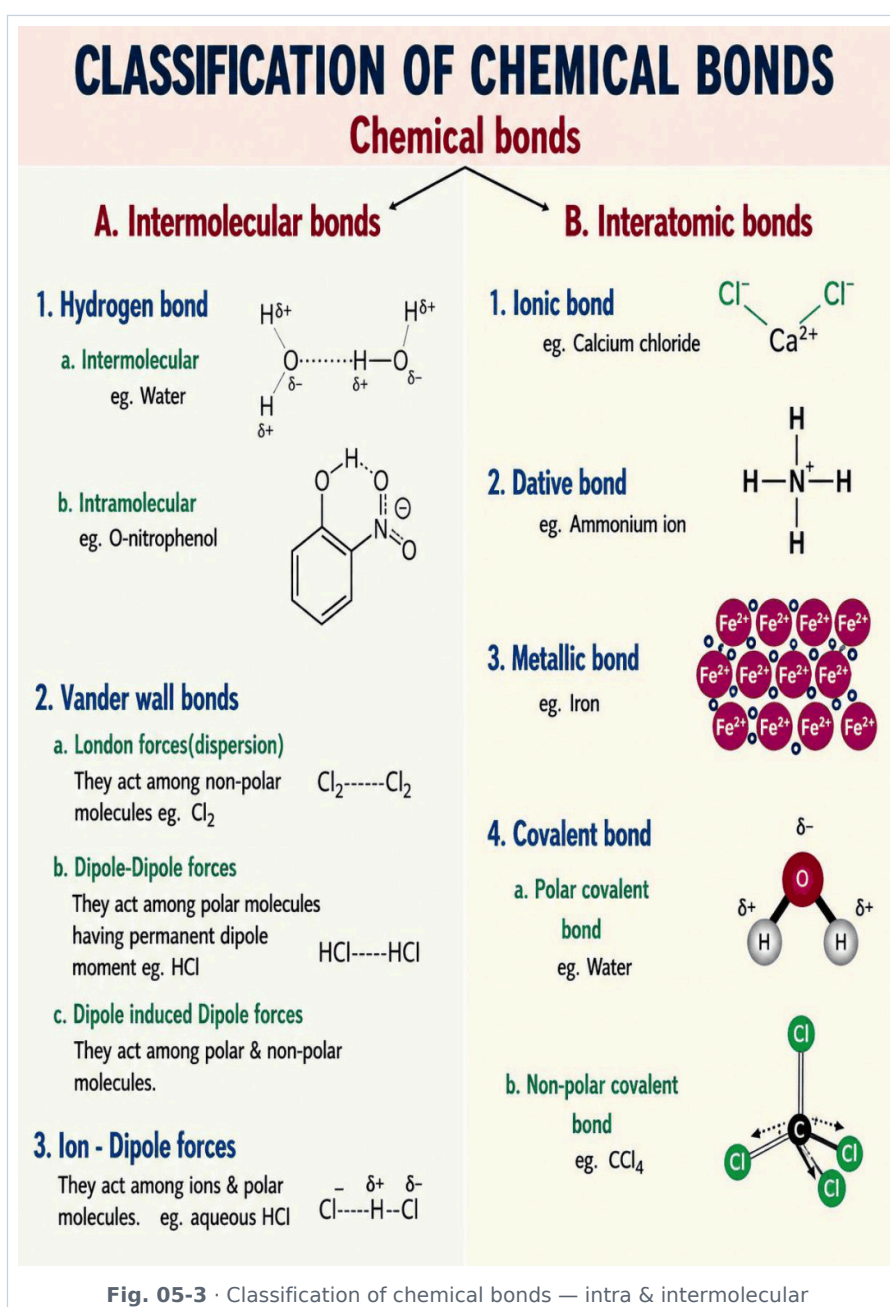
5.2 Types of Chemical Bonds — One Combined Picture

📌 **De-dup note:** Bond types and comparison tables were printed twice (Bonding lecture p. 48 and Chemical-Bond lecture pp. 58 & 63). The two sets carry the same facts — one combined set is kept.

Bond	Strength	Action / mechanism	Key feature	Typical participants	Example
Ionic (electrovalent)	Strong	Complete transfer of valence electrons	Electrostatic attraction between ions; high m.p./b.p.; water-soluble	Metal + non-metal	NaCl

Bond	Strength	Action / mechanism	Key feature	Typical participants	Example
Covalent	Strong	Sharing of electron pairs	Orbital overlap; polar (H ₂ O) or non-polar (O ₂) sharing	Non-metal + non-metal	H ₂ O, O ₂ , CH ₄
Metallic	Strong	Delocalisation	Positive metal ions in a "sea" of mobile electrons → conductivity	Metal + metal	Fe, Cu
Coordinate (dative)	Strong	Donor-acceptor	Both shared electrons come from the same atom	Donor + acceptor	NH ₄ ⁺ , BF ₃ ·NH ₃
Hydrogen	Weak	Dipole-dipole attraction	H bonded to N, O or F attracts another electronegative atom	Intermolecular	H ₂ O, DNA
Van der Waals	Very weak	Induced dipoles	Instantaneous uneven electron distribution (London forces)	Non-polar molecules	Noble gases, CH ₄

Relative strength: Ionic ≥ Covalent ≥ Metallic ≥ Hydrogen bond ≥ Van der Waals



TYPES OF CHEMICAL BONDS

1. IONIC BOND

- Transfer of electrons
- Metal + Non-metal

NaCl = Sodium Chloride (Table Salt)

2. COVALENT BOND

- Sharing of electrons
- Non-metal + Non-metal

H₂O = Water

3. METALLIC BOND

- Sea of delocalized electrons
- Between metal atoms

Fe = Iron • Cu = Copper

COORDINATION COVALENT BOND

- Shared pair donated by one atom
- Dative bond

HYDROGEN BONDING

- Weak attraction between H and F, O, or N
- Intermolecular

(In Water - H₂O)

COMPARISON TABLE

Property	Ionic	Covalent	Metallic
Formation	Electron Transfer	Electron Sharing	Electron Sea
Conductivity	In Water / Molten	Poor (Generally)	Excellent
Melting Point	High	Low to Moderate	High
Example	NaCl	H ₂ , O ₂ , CH ₄	Fe, Cu, Al

IMPORTANCE OF CHEMICAL BONDING

• Holds molecules together

• Basis of DNA structure

• Forms chemicals & compounds

• Explains material properties

✔ Tip: Chemical bonds = Foundation of all chemistry!

Fig. 05-4 · Types of chemical bonds with comparison table

- **Key differences:** ionic = transfer (metal + non-metal) vs covalent = sharing (non-metals); ionic/covalent/metallic bonds hold atoms **within** compounds, hydrogen/van der Waals act **between** molecules; ionic & metallic solids conduct, covalent generally do not.

5.3 Ionic (Electrovalent) Bond

- Formed by **electrostatic attraction between oppositely charged ions** (cation + anion), usually by complete transfer of electrons from a metal to a non-metal. e.g. **NaCl**.
- **Properties:** high melting/boiling points; soluble in water; conduct electricity when molten or dissolved.
- **Polarization (Fajan's rule)** — covalent character developed in ionic bonds: the power of an ion to distort the electron cloud of the other is its **polarising power**; the tendency to get polarised is **polarisability**. Higher either → higher covalent character.
- **Factors increasing polarisation:** small cation; large anion; higher charge on either ion. (A polar medium keeps ions apart and prevents polarisation.)

Classic example

LiCl is covalent whereas NaCl is ionic — Li^+ is very small (high polarising power), so it distorts Cl^- strongly → LiCl gains covalent character.

5.4 Covalent Bond & Its Parameters

- Formed by **sharing electron pairs** between non-metals to gain stability. Types: **non-polar** (equal sharing, H_2 , O_2) and **polar** (unequal sharing by electronegativity difference, H_2O); multiple bonds — single, double, triple — possible.

Bond Parameters

- Bond length** — equilibrium distance between nuclei of two bonded atoms; measured by spectroscopy, X-ray & electron diffraction. Varies as: single > double > triple.
- Bond angle** — angle between orbitals containing bonding pairs around the central atom (degrees); indicates orbital distribution and shape.
- Bond enthalpy** — energy to break 1 mole of a particular bond in gas phase, $\text{A-B(g)} \rightarrow \text{A(g)} + \text{B(g)}$, in kJ mol^{-1} . For polyatomic molecules (H_2O has two O-H bonds) an **average/mean bond enthalpy** is used.
- Bond order** — number of bonds between two atoms; bond strength $\propto$ bond order: triple > double > single.

Sigma & Pi bonds — orbital overlap

Point	Sigma (σ) bond	Pi (π) bond
Overlap	Head-on / end-to-end	Sideways / lateral
Formation	Along the internuclear axis	Above & below the axis
Strength	Stronger	Weaker
Rotation	Free rotation possible	Restricts rotation
Independence	Can exist alone	Only with a σ bond

DIFFERENCES BETWEEN SIGMA AND PI BONDS

Sigma Bond	Pi Bond
1. This bond is formed by the axial overlap of atomic orbitals.	1. The bond is formed by the sidewise overlap of atomic orbitals.
2. This bond can be formed by overlap of s - s , s - p or p - p orbitals.	2. It involves overlap of p -orbitals only.
3. The bond is stronger because overlapping can take place to a larger extent.	3. The bond is weaker because the overlapping occurs to a smaller extent.
4. The molecular orbital is symmetrical about the internuclear axis and consists of single charged cloud.	4. The molecular orbital is discontinuous and consists of two charged clouds above and below the plane of atoms.
5. There can be a free rotation of atoms around the σ bond.	5. Free rotation of atoms around π bond is not possible because it involves breaking of π bond.
6. The bond may be present between the two atoms either alone or alongwith π bond.	6. The bond is always present between the two atoms alongwith the sigma bond, <i>i.e.</i> , it is always superimposed on sigma bond.
7. The shape of molecule is determined by the sigma frame work around the central atom.	7. The π bonds do not contribute to the shape however they modify the dimensions of the molecule.

Fig. 05-5 · Differences between sigma and pi bonds

- Examples: CO_2 has two σ and two π bonds; N_2 has one σ and two π ; **benzene has 12 σ and 3 π bonds.**

VSEPR Theory (Valence Shell Electron Pair Repulsion)

- Predicts molecular **shape** by minimising repulsion between electron pairs around the central atom.
Repulsion order: **lone pair - lone pair > lone pair - bond pair > bond pair - bond pair.**

Steric number	Number of Bond pairs	Number of Lone pairs	Formula	Shape of molecule	Approximate Bond angles	Examples
1	1	0	AX	Linear	-	ClF, BrF, BrCl, HF, O ₂
2	2	0	AX ₂	Linear	180°	BeCl ₂ , HgCl ₂ , CO ₂
3	3	0	AX ₃	Trigonal planar	120°	BF ₃ , CO ₃ ²⁻ , NO ₃ ⁻ , SO ₃
	2	1	AX ₂ E	Angular	120°	SO ₂ , SnCl ₂ , O ₃ , NSF, NO ₂ ⁻
4	4	0	AX ₄	Tetrahedral	109°28'	CH ₄ , SiCl ₄ , NH ₄ ⁺ , PO ₄ ³⁻ , SO ₄ ²⁻ , ClO ₄ ⁻
	3	1	AX ₃ E	Trigonal pyramidal	around 109°28'	NH ₃ , PCl ₃ , XeO ₃
	2	2	AX ₂ E ₂	Angular	around 109°28'	H ₂ O, SCl ₂ , Cl ₂ O, OF ₂
5	5	0	AX ₅	Trigonal bipyramidal	120° & 90°	PCl ₅ , SOF ₄
	4	1	AX ₄ E	See saw or distorted tetrahedral	-	SF ₄ , TeCl ₄
	3	2	AX ₃ E ₂	T-Shape	90°	ClF ₃ , BrF ₃ , BrCl ₃
	2	3	AX ₂ E ₃	Linear	180°	XeF ₂ , I ₃ ⁻

Fig. 05-6 · VSEPR theory — summary card

Number of electron pairs around central atom		Full description of the molecule					
BONDING (B)	LONE (L)	Example	Bond angles (°)	Geometry of Electron Pairs	Geometry of Atoms	3D Shape	Type
2	0	BeCl ₂	180	Linear	Linear		AB ₂
3	0	BF ₃	120	Trigonal planar	Trigonal Planar		AB ₃
2	1	SO ₂	Slightly less than 120	Trigonal planar	Bent or V Shaped		AB ₃ E
4	0	CH ₄	109.5	Tetrahedral	Tetrahedral		AB ₄
3	1	NH ₃	107.5	Tetrahedral	Trigonal Pyramidal		AB ₃ E
2	2	H ₂ O	104.5	Tetrahedral	Bent or V Shaped		AB ₂ E ₂
5	0	PCl ₅	120 in plane, 90 perpendicular to plane	Trigonal bipyramidal	Trigonal Bipyramidal		AB ₅
4	1	SF ₄	Complex	Trigonal bipyramid	Seesaw		AB ₄ E
3	2	ClF ₃	Approx. 90	Trigonal bipyramidal	T-Shaped		AB ₃ E ₂
2	3	XeF ₂	180	Trigonal bipyramid	Linear		AB ₂ E ₃
6	0	SF ₆	90	Octahedral	Octahedral		AB ₆
5	1	BrF ₅	Approx. 90	Octahedral	Square Pyramidal		AB ₅ E
4	2	XeF ₄	90	Octahedral	Square Planar		AB ₄ E ₂

Fig. 05-7 · VSEPR molecular geometries with examples

- Postulates: electron pairs around the central atom arrange asymmetrically to minimise repulsion; lone pairs occupy more space; multiple bonds treated as one electron domain.

Hybridization

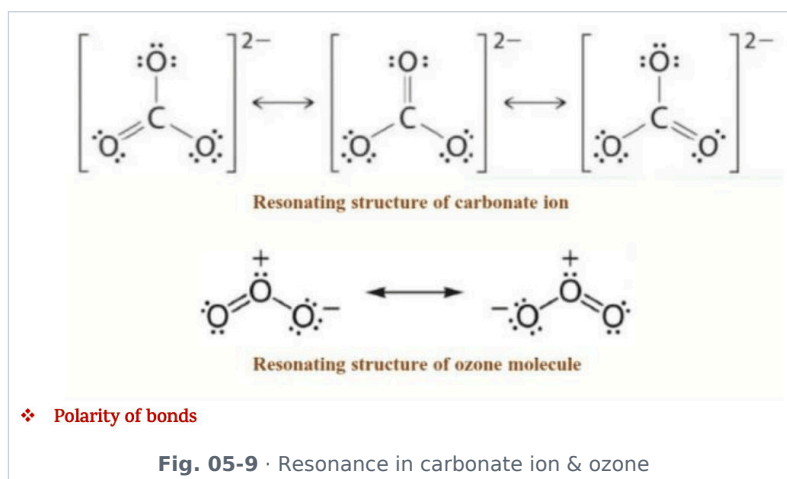
- Mixing of atomic orbitals (sp, sp², sp³ ...) to form equivalent hybrid orbitals that decide geometry.

Electron Groups	Bonding Groups	Lone Pairs	Electron Geometry	Molecular Geometry	Approximate Bond Angles	Example
2	2	0	Linear	Linear	180°	$\text{:}\ddot{\text{S}}=\text{C}=\ddot{\text{S}}\text{:}$
3	3	0	Trigonal planar	Trigonal planar	120°	$\text{:}\ddot{\text{Cl}}\text{:}$ $\text{:}\ddot{\text{Cl}}-\text{B}-\ddot{\text{Cl}}\text{:}$
3	2	1	Trigonal planar	Bent	<120°	$\text{:}\ddot{\text{O}}=\ddot{\text{S}}=\ddot{\text{O}}\text{:}$
4	4	0	Tetrahedral	Tetrahedral	109.5°	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H} \end{array}$
4	3	1	Tetrahedral	Trigonal pyramidal	<109.5°	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{P}-\text{H} \\ \\ \text{H} \end{array}$
4	2	2	Tetrahedral	Bent	<<109.5°	$\text{H}-\ddot{\text{S}}-\text{H}$
5	5	0	Trigonal bipyramidal	Trigonal bipyramidal	120° (equatorial) 90° (axial)	$\begin{array}{c} \text{:}\ddot{\text{Cl}}\text{:} \\ \\ \text{:}\ddot{\text{Cl}}-\text{P}-\ddot{\text{Cl}}\text{:} \\ \\ \text{:}\ddot{\text{Cl}}\text{:} \end{array}$
5	4	1	Trigonal bipyramidal	Seesaw	<120° (equatorial) <90° (axial)	$\begin{array}{c} \text{:}\ddot{\text{F}}\text{:} \\ \\ \text{:}\ddot{\text{F}}-\text{S}-\ddot{\text{F}}\text{:} \\ \\ \text{:}\ddot{\text{F}}\text{:} \end{array}$
5	3	2	Trigonal bipyramidal	T-shaped	<90°	$\begin{array}{c} \text{:}\ddot{\text{F}}\text{:} \\ \\ \text{:}\ddot{\text{F}}-\text{Cl}-\ddot{\text{F}}\text{:} \\ \\ \text{:}\ddot{\text{F}}\text{:} \end{array}$
5	2	3	Trigonal bipyramidal	Linear	180°	$\text{:}\ddot{\text{F}}-\ddot{\text{Xe}}-\ddot{\text{F}}\text{:}$
6	6	0	Octahedral	Octahedral	90°	$\begin{array}{c} \text{:}\ddot{\text{F}}\text{:} \\ \\ \text{:}\ddot{\text{F}}-\text{S}-\ddot{\text{F}}\text{:} \\ \\ \text{:}\ddot{\text{F}}\text{:} \end{array}$
6	5	1	Octahedral	Square pyramidal	<90°	$\begin{array}{c} \text{:}\ddot{\text{F}}\text{:} \\ \\ \text{:}\ddot{\text{F}}-\text{Br}-\ddot{\text{F}}\text{:} \\ \\ \text{:}\ddot{\text{F}}\text{:} \end{array}$
6	4	2	Octahedral	Square planar	90°	$\begin{array}{c} \text{:}\ddot{\text{F}}\text{:} \\ \\ \text{:}\ddot{\text{F}}-\text{Xe}-\ddot{\text{F}}\text{:} \\ \\ \text{:}\ddot{\text{F}}\text{:} \end{array}$

Fig. 05-8 · Orbital geometry table — localised & delocalised bonds

Resonance

- When a molecule/ion cannot be described by a single Lewis structure, several **resonating (canonical) structures** contribute to the true **resonance hybrid** — e.g. O_3 , CO_3^{2-} . Resonance stabilises the molecule/ion.



- **Resonance energy** = energy of actual molecule – energy of the most stable resonating structure.

Polarity & Dipole Moment

- Electronegativity difference creates partial charges → **polar covalent bonds**; measured by **dipole moment $\mu = \text{charge} \times \text{distance}$** — a vector directed from the positive to the negative end.
- **Applications:** (a) predicting polar/non-polar nature; (b) comparing polarities; (c) deciding shapes — CO_2 has polar $\text{C}=\text{O}$ bonds but $\mu = 0 \rightarrow$ linear; SO_2 has $\mu = 1.60 \text{ D} \rightarrow$ bent; (d) calculating % ionic character.

5.5 Molecular Orbital Theory (MOT)

- MOT (proposed by **F. Hund & R.S. Mulliken**, 1932): atomic orbitals combine to form **molecular orbitals** spread over the whole molecule; bonding MOs lower and antibonding MOs raise energy; electrons fill per Aufbau, Hund & Pauli.

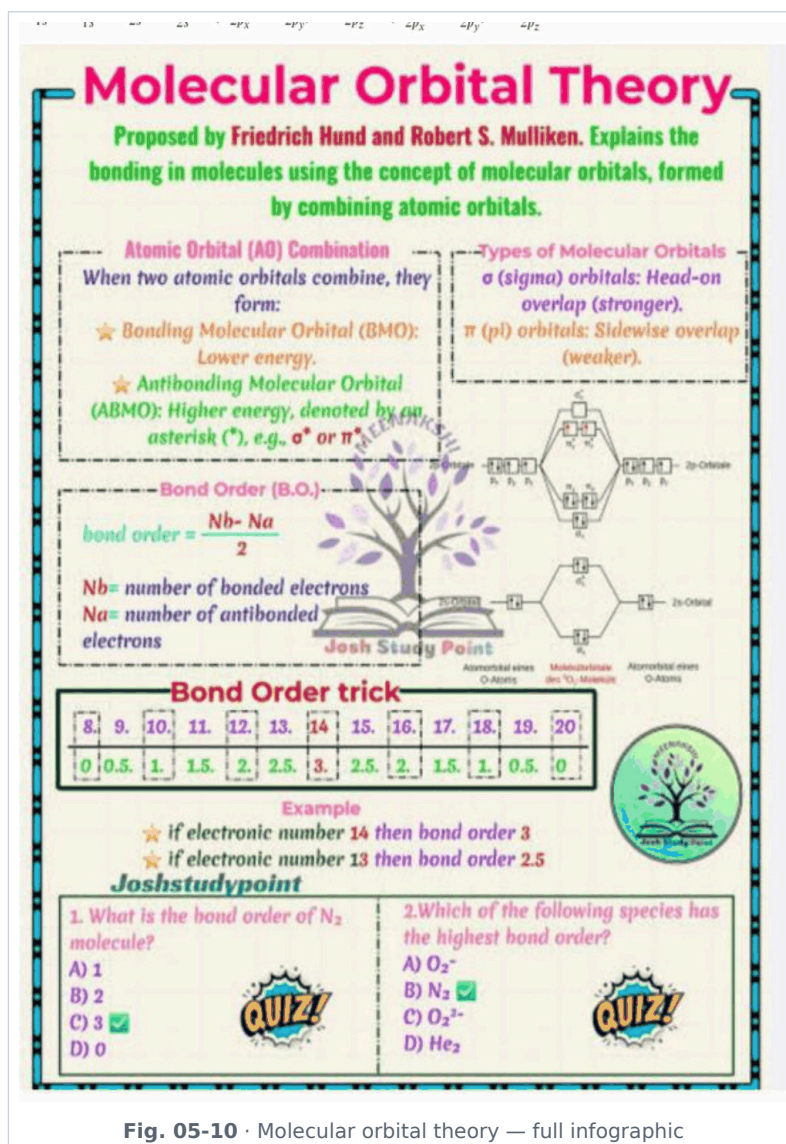
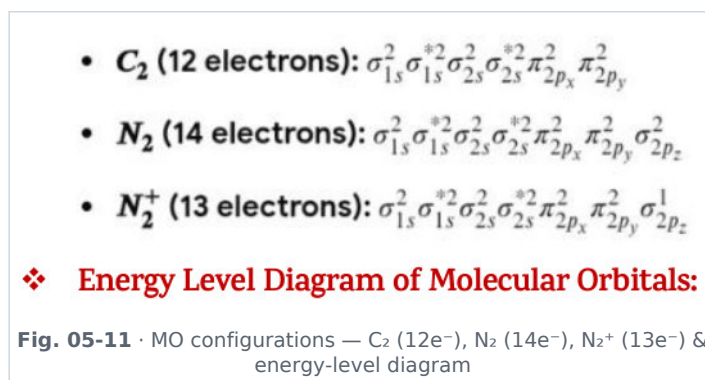
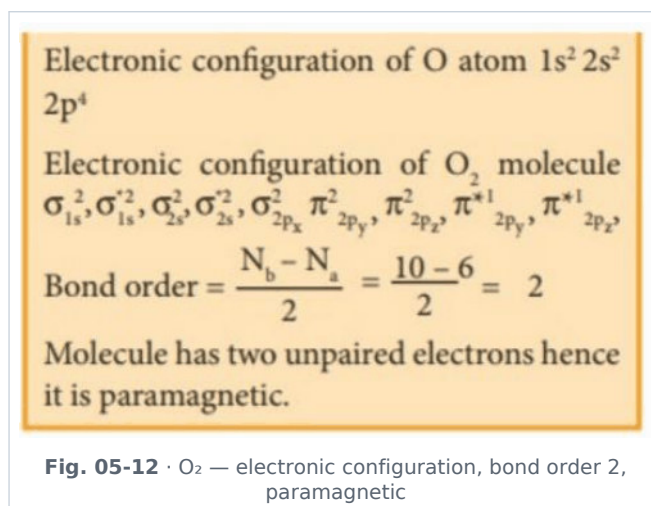


Fig. 05-10 · Molecular orbital theory — full infographic

- **Linear combination:** σ (from head-on) and π (side-on) MOs; bonding MO ($\psi_A + \psi_B$) has electron density between nuclei.
- **Filling order (1-14 electron systems):** $\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \pi_{2p_x} = \pi_{2p_y} < \sigma_{2p_z} < \pi_{2p_x}^* = \pi_{2p_y}^* < \sigma_{2p_z}^*$
- For 15-20 electron systems, σ_{2p_z} lies below $\pi_{2p_x} = \pi_{2p_y}$. Each σ/π MO holds 2 electrons; π_x, π_y are degenerate (4 total).
- **Bond order** = $\frac{1}{2} (N_b - N_a)$ — half the difference of bonding and antibonding electrons. Stability $\propto$ bond order; higher bond order $\rightarrow$ higher dissociation energy and shorter bond length. Molecules with unpaired electrons are **paramagnetic**.





⚠ **De-dup note:** Example from the source: Ne_2 has bond order 0 (16 bonding = 16 antibonding electrons) — the molecule is not stable; answer (c). The worked configuration is kept with the figures above.

Exam quick-checks (from the source infographic)

Bond order of $N_2 = 3$ (diamagnetic); $N_2^+ = 2.5$; $O_2 = 2$ (paramagnetic); $O_2^- = 1.5$; $He_2 = 0$ (does not exist).

Key point: 1s and 2s MOs cancel out in bond-order calculations of second-row diatomic molecules.

5.6 Hydrogen Bond (Short Notes)

- A **weak electrostatic dipole attraction** between an H atom covalently bonded to a highly electronegative atom (N, O or F) and another electronegative atom ($\delta^+ H \cdots \delta^-$). Bond energy $\approx$ up to 40 kJ/mol — stronger than van der Waals, weaker than covalent bonds.
- **Conditions:** H attached to N, O or F; the electronegative atom must be small and have lone pairs (highest in $F > O > N$).

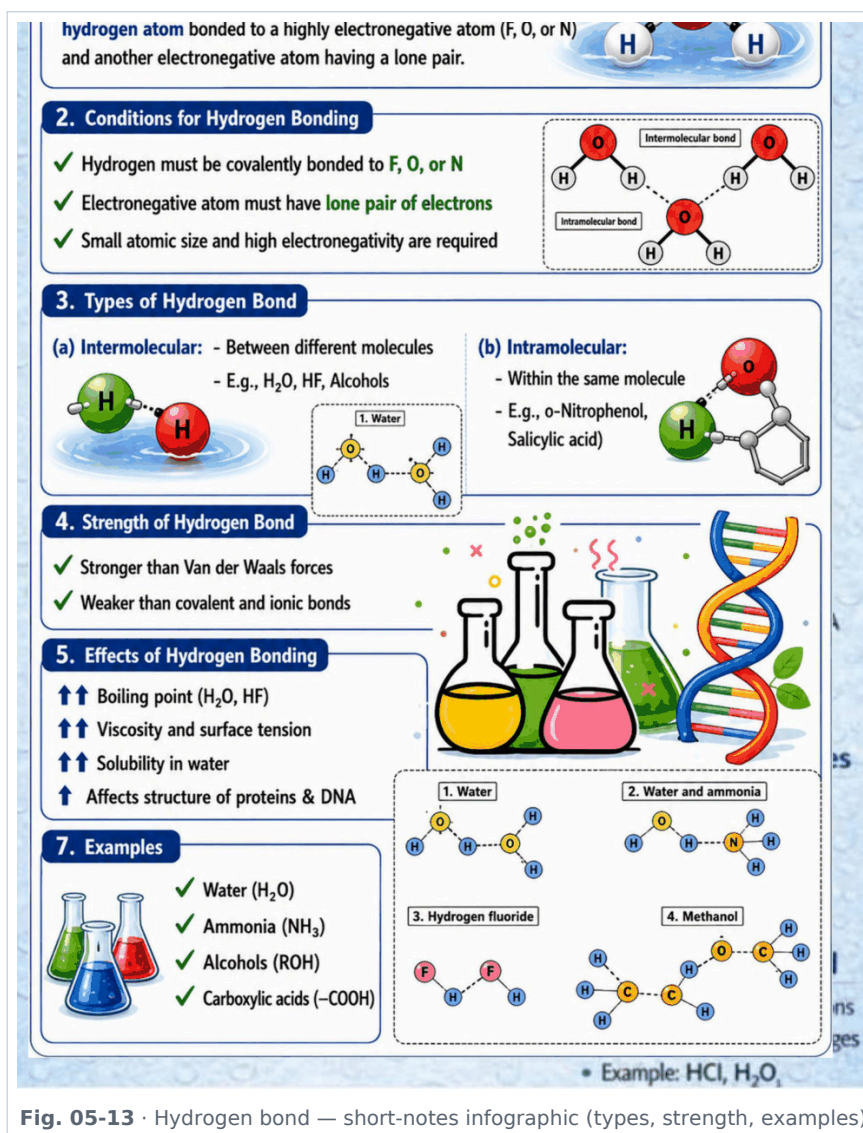


Fig. 05-13 · Hydrogen bond — short-notes infographic (types, strength, examples)

- **Types:**
 - **Intermolecular** — between different molecules: H_2O (water), HNO_3 , alcohols, carboxylic acids, sugar, NH_3 .
 - **Intramolecular** — within the same molecule: o-nitrophenol, o-hydroxybenzaldehyde (salicylaldehyde), o-hydroxybenzoic acid.
- **Strength comparison:** $\text{H-F} \cdots \text{H-F} > \text{H-O} \cdots \text{H-O} > \text{H-N} \cdots \text{H-N}$.
- **Effects:** high b.p. of H_2O (and H-F); ice less dense than water; H_2O a liquid while H_2S is a gas; alcohols / carboxylic acids associated; NH_3 , H_2O , HF have abnormally high b.p.; DNA double helix held by H-bonds.

UNIT 06

Chemical Reactions

Consolidated from: Chemical Reaction lecture (pp. 66–71) — foundation for UPSC/BPSC Science & Tech; explains rusting, digestion, photosynthesis

6.1 What is a Chemical Reaction?

- A chemical reaction is a **rearrangement of atoms** — one or more substances (reactants) are converted into new substances (products) with different properties. The five primary types are combination, decomposition, single displacement, double displacement and combustion/redox.

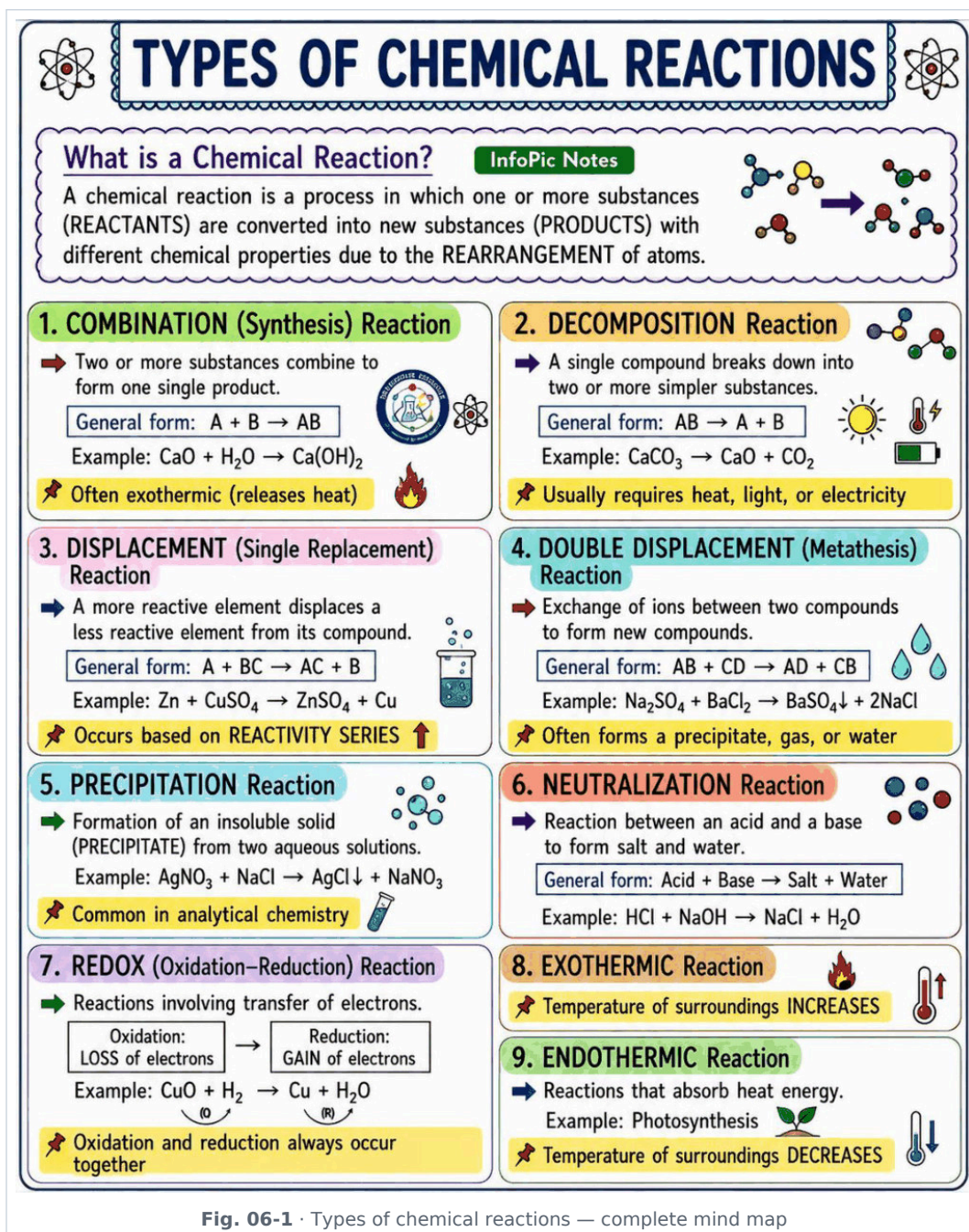


Fig. 06-1 · Types of chemical reactions — complete mind map

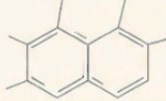
1. Combination Reaction $A + B \rightarrow AB$ $2H_2 + O_2 \rightarrow 2H_2O$ 	3. Displacement Reaction $A + BC \rightarrow AC + B$ $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ 
2. Decomposition Reaction $AB \rightarrow A + B$ $CaCO_3 \rightarrow CaO + CO_2$ 	4. Double Displacement Reaction $AB + CD \rightarrow AD + CB$ $NaCl + AgNO_3 \rightarrow NaNO_3 + AgCl$ 
5. Oxidation Reaction $2Mg + O_2 \rightarrow 2MgO$ $Fe + O_2 \rightarrow Fe_2O_3$ 	6. Reduction Reaction $CuO + H_2 \rightarrow Cu + H_2O$ $ZnO + C \rightarrow Zn + CO$ 
7. Redox Reaction $Zn + CuO \rightarrow ZnO + Cu$ 	8. Neutralization Reaction $Acid + Base \rightarrow Salt + Water$ $HCl + NaOH \rightarrow NaCl + H_2O$ 
9. Exothermic Reaction $C + O_2 \rightarrow CO_2 + Heat$ $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + Heat$ 	10. Endothermic Reaction $CaCO_3 + Heat \rightarrow CaO + CO_2$ 
11. Corrosion <i>Rusting:</i> $Fe + O_2 + H_2O \rightarrow Fe_2O_3 \times H_2O$ 	
12. Rancidity <i>Oxidation of oils / fats causing bad smell and taste.</i> 	

Fig. 06-2 · Reaction types at a glance with examples

6.2 Core Types of Chemical Reactions

ChemistryCorner

1 COMBINATION REACTION

Two or more reactants form a **SINGLE** product

$A + B \rightarrow AB$

- Building up
- Synthesis

Example: Formation of Water

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

Hydrogen + Oxygen → water

2 DECOMPOSITION REACTION

A single reactant **breaks down** into **TWO OR MORE** products

$AB \rightarrow A + B$

- Breaking down
- Requires energy

Types

- Thermal decomposition of $CaCO_3$: $CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)$
- Decomposition of Water: $2H_2O(l) \xrightarrow{\text{energy}} 2H_2(g) + O_2(g)$

3 SINGLE DISPLACEMENT REACTION

A more reactive element **replaces** a less reactive element in a compound

$A + BC \rightarrow AC + B$

- Dependent on Reactivity Series

Ex. Zinc & Copper Sulfate

$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$$

4 DOUBLE DISPLACEMENT REACTION

Exchange of **ions** between two compounds to form new ones

$AB + CD \rightarrow AD + CB$

- Ionic exchange, often forms a precipitate

Example: Precipitation

$$BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$$

(White ppt) + 2NaCl(aq)

5 REDOX REACTIONS (OXIDATION-REDUCTION)

Reactions involving simultaneous transfer of **electrons**

Definition in terms of electron transfer

- Concept of reducing and oxidizing agents
- Importance in various chemical processes

Simple example:

Concept of reducing and oxidizing agents

Ex: $Zn(s) \rightarrow Zn^{2+} + 2e^-$ (oxidized)

$Cu^{2+} + 2e^- \rightarrow Cu(s)$ (reduced)

Fig. 06-3 · Ten reaction types — "Chemistry Corner" chart

Reaction Type	Pattern	Key Example
Combination	$A + B \rightarrow AB$	$2Mg + O_2 \rightarrow 2MgO$
Decomposition	$AB \rightarrow A + B$	$2KClO_3 \rightarrow 2KCl + 3O_2$
Single Displacement	$A + BC \rightarrow AC + B$	$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
Double Displacement	$AB + CD \rightarrow AD + CB$	$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$
Redox	Ox/Loss & Red/Gain (e^-)	$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

Fig. 06-4 · Twelve reaction types with formulas and examples

The five primary types

Type	General form	Example
Combination (synthesis)	$A + B \rightarrow AB$	$C + O_2 \rightarrow CO_2$; $2H_2 + O_2 \rightarrow 2H_2O$

Type	General form	Example
Decomposition	$AB \rightarrow A + B$	$\text{CaCO}_3 \xrightarrow{\text{heat}} \text{CaO} + \text{CO}_2$
Single displacement	$A + BC \rightarrow AC + B$	$\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$
Double displacement	$AB + CD \rightarrow AD + CB$	$\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} \downarrow + \text{NaNO}_3$
Redox (oxidation-reduction)	electron transfer	one substance oxidised (loses e^-), other reduced (gains e^-)

- **Combustion** — a substance (usually a hydrocarbon) reacts with oxygen releasing heat & light: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$.

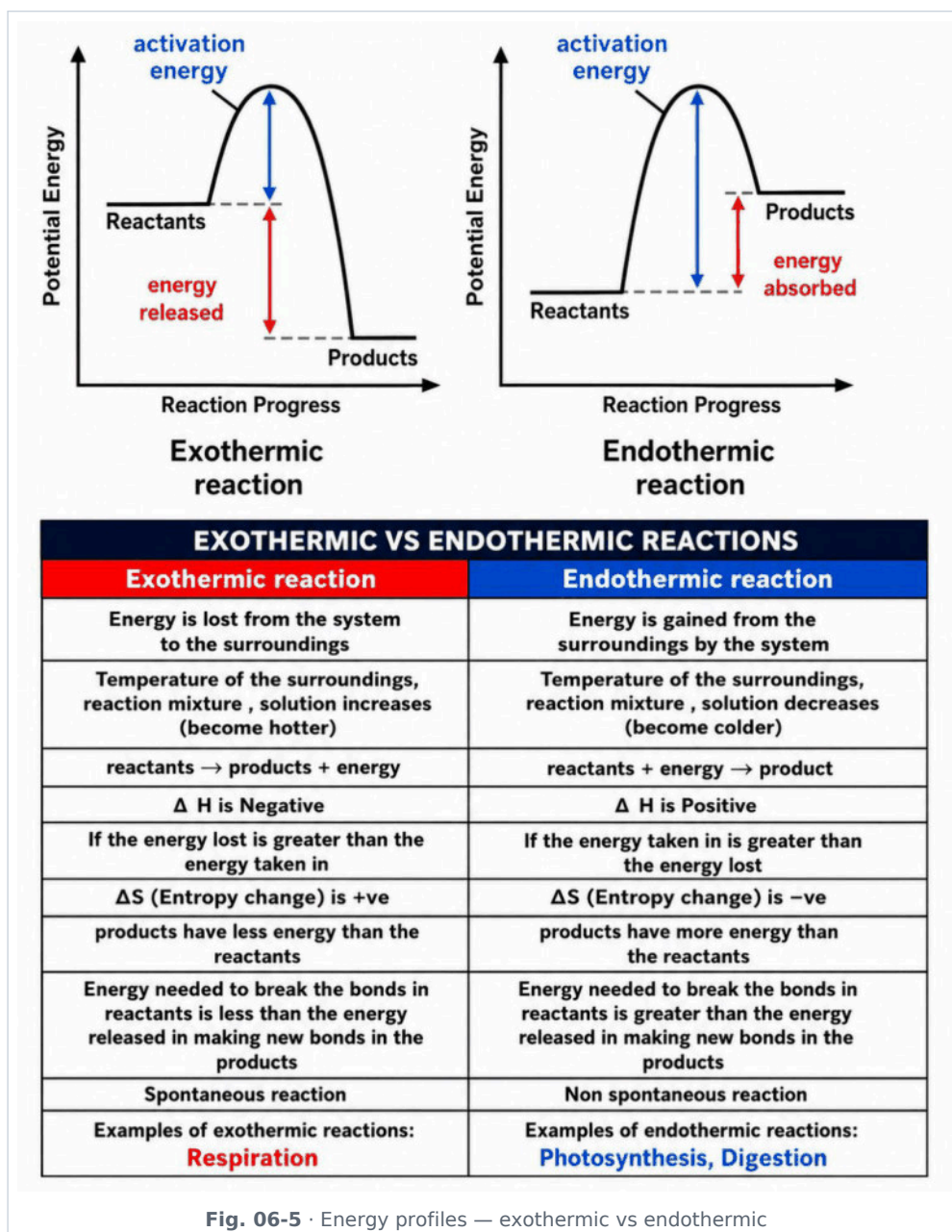
Other important classifications

- **Neutralisation** — double displacement of acid + base $\rightarrow$ salt + water: $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$.
- **Precipitation** — a double displacement producing an insoluble solid (precipitate).
- **Endothermic vs Exothermic** — endothermic absorbs energy; exothermic releases it (see 6.4).

6.3 Effects of Oxidation in Everyday Life

- **Corrosion** — a metal is attacked by substances around it (moisture, acids etc.):
 - Iron articles get a reddish-brown coating — **rusting of iron**.
 - Black coating on **silver** and green coating on **copper** — other examples of corrosion.
- **Rancidity** — fats and oils get oxidised; smell and taste change. Antioxidants are added to fatty/oily foods; air-tight containers slow oxidation; chip bags are flushed with **nitrogen** to prevent oxidation.

6.4 Exothermic vs Endothermic Reactions



Point	Exothermic	Endothermic
Energy change	Releases energy (heat, light, sound) to surroundings	Absorbs energy from surroundings
Temperature	Surroundings become hot	Surroundings become cold
Enthalpy	Negative ($\Delta H < 0$)	Positive ($\Delta H > 0$)
Bonding	Forms stronger bonds; releases excess energy	Absorbs energy to break bonds
Examples	Combustion of fuels, rusting of iron, nuclear fission, mixing acids with bases, freezing water	Photosynthesis, melting ice, evaporation, thermal decomposition, dissolving ammonium nitrate in water

Exam pointers

Exothermic feels hot, endothermic feels cold.

Respiration = redox + decomposition; baking with baking soda = decomposition + displacement; vinegar + baking soda = double displacement + gas evolution; burning candle = combustion.

Rusting = redox reaction; cooking = decomposition.

UNIT 07

Acids, Bases & Salts

Consolidated from: Acids, Bases and Salts lecture (pp. 73–79); the carbon intro of p. 81 is consolidated with organic chemistry in Unit 10

7.1 Acids — Definitions

Concept	Definition	Examples
Arrhenius acid	Compound that increases H^+ (hydronium) concentration when dissolved in water—	
Brønsted-Lowry acid	Hydrogen-containing substance that can donate a proton to another substance—	
Lewis acid	Substance capable of accepting a pair of electrons	$AlCl_3$, BF_3



Fig. 07-1 · Acid & base strengths in the flask

Strong & Weak Acids

- **Strong acids** — sulphuric acid (H_2SO_4), hydrochloric acid (HCl), nitric acid (HNO_3), chloric acid ($HClO_3$), hydrobromic acid (HBr).
- **Weak acids** — hydrofluoric acid (HF), sulfurous acid (H_2SO_3), carbonic acid (H_2CO_3), formic/methanoic acid ($HCOOH$), acetic/ethanoic acid (CH_3COOH) etc.

Physical Properties of Acids

- Sour in taste; sticky in touch; good conductors of electricity; strong acids are corrosive on skin and cause painful burns.

Action with Indicators

Indicator	With acid	With base
Blue litmus	Turns red—	
Methyl orange	Turns pink	Turns yellow
Phenolphthalein	Remains colourless	Turns pink
Turmeric	No change	Turns deep red
Red cabbage	Turns pink	Turns dark blue
Hibiscus	No change	Turns blue (gudhal)
Olfactory (onion, vanilla, clove oil)	No change in smell	Smell changes in basic medium

Classification of Acids

- **By source:** organic (natural sources) / inorganic (H_2SO_4 , HCl , HNO_3 , HBr ...). **Hydracids** — acids without oxygen.
- **By concentration:** dilute (more water) / concentrated (less or no water).
- **By H^+ released:** strong (HCl , H_2SO_4 , HNO_3) / weak (all organic acids, H_2CO_3 ...).

- **By basicity** (hydronium ions producible per molecule): **monobasic** — HCl, HBr, HF; **dibasic** — H₂SO₄, H₂SO₃, H₂CO₃; **tribasic** — H₃PO₄, citric acid.

S.N.	Name of Acid	Source
1	Tartaric Acid	Grapes, Tamarind, Citrus, Bananas
2	Acetic Acid (Ethanoic Acid)	Vinegar
3	Citric Acid	Lemon, Orange, and commonly in all citrus fruits
4	Oxalic Acid	Tomato, Spinach, Broccoli
5	Lactic Acid	Curd
6	Malic Acid	Apple
7	Ascorbic Acid (Vitamin C)	Amla, Guava, Papaya, Orange
8	Formic Acid	Stings of ants and Stinging Nettle (a plant)

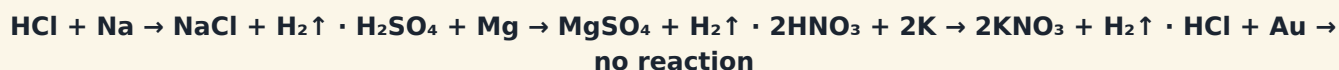
Depending on their concentration:

Fig. 07-2 · Common acids and their uses

- Uses (from source table): acetic acid — vinegar; boric acid — eyewash; carbonic acid — soft drinks; citric acid — flavouring; formic acid — preservation; lactic acid — curd/milk products; maleic acid — phenolic resins; nitric acid — explosives/fertiliser; oxalic acid — polishing & cleaning; benzoic/sorbic acids — food preservatives; tartaric acid — baking; sulphuric acid — batteries.

Chemical Properties of Acids

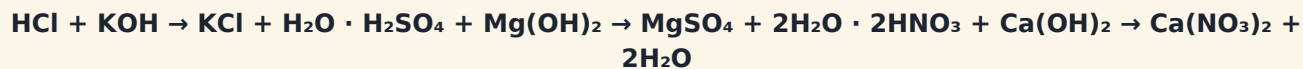
- **With metals:** Acid + Metal → Salt + H₂ ↑



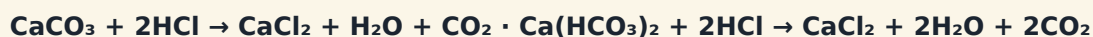
- **With metal oxides:** Acid + Metal oxide → Salt + Water



- **With bases (neutralisation):** Acid + Base → Salt + Water



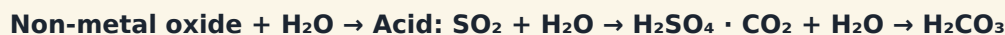
- **With metal carbonates/bicarbonates:** Carbonate/Bicarbonate + Acid → Salt + Water + CO₂



- **General use:** Baking soda + tartaric acid (NaHCO₃) → **baking powder**.

7.2 Acid Rain

- Rain water mixed with pollutant gases (non-metal oxides) forms dilute acid — acid rain.



- **Effects:** skin problems in humans & animals; damages plants; damages monuments; soil degradation; harms aquatic life.

7.3 Bases

- **Arrhenius base** — increases hydroxide (OH⁻) concentration in water. **Brønsted-Lowry base** — accepts a proton.
- **Examples:** NaOH, KOH, NH₄OH, Ca(OH)₂, Mg(OH)₂ etc.

Physical Properties & Indicators

- Bitter in taste; slippery in touch; good conductors; mild corrosive action on skin.
- With indicators (see table in 7.1): methyl orange → yellow; phenolphthalein → pink; red cabbage → dark blue; turmeric → deep red; hibiscus → blue; olfactory indicators change smell in basic medium.

Classification of Bases

- **By strength:** strong (NaOH , KOH , $\text{Mg}(\text{OH})_2$, $\text{Ca}(\text{OH})_2$) / weak (NH_4OH , CH_3NH_2) — weak bases do not produce a large number of OH^- ions.
- **By acidity** (OH^- ions producible per molecule): **monoacidic** — NaOH , LiOH , KOH ; **diacidic** — $\text{Mg}(\text{OH})_2$, $\text{Ca}(\text{OH})_2$; **triacidic** — $\text{Al}(\text{OH})_3$, $\text{Fe}(\text{OH})_3$.

♣ **De-dup note:** The base definitions and examples were printed twice on source p. 75 — kept once here.

Chemical Properties of Bases

- **With acids:** $\text{Base} + \text{Acid} \rightarrow \text{Salt} + \text{Water}$ (neutralisation)



- **With non-metal oxides:** $\text{Base} + \text{Non-metal oxide} \rightarrow \text{Salt} + \text{Water}$



7.4 pH — Strength of Acids & Bases

Acid	Use
1. Boric acid	Eye-wash/antiseptic.
2. Citric acid	Food preservation. Vitamin C preparation.
3. Oxalic acid	Ink stain remover.
4. Carbonic acid	Flavoured drinks.
5. Tartaric acid	Baking powder.
6. Acetic acid	Table vinegar and cooking.
7. Hydrochloric acid	Cleaning of metal items
8. Benzoic acid	Preservation of food and making of perfumes and medicines.
9. Nitric acid	Explosives.
10. Phosphoric acid	Fertilizers.

Fig. 07-3 · The pH scale

- **pH = Potenz (potential) of Hydrogen** — negative logarithm (base 10) of H^+ ion concentration (mol/L): **$\text{pH} = -\log_{10}[\text{H}^+]$** . SI unit of concentration: mol/L; 1 mole = 6.022×10^{23} particles.
- **Log recap (from the notes):** $2^3 = 8 \Rightarrow \log_2 8 = 3$; $\log_{10} 1000 = \log_{10} 10^3 = 3$; $\log_m m = 1$; $\log_{10} m^n = n \cdot \log m$.

Importance of pH in Everyday Life

- Our body works within pH **7.0 – 7.8**; plants and animals survive in narrow ranges.
- **Agriculture:** rice prefers slightly acidic soil; sugarcane neutral; citrus fruits alkaline.
- **Medicine:** diseases diagnosed from blood/urine pH.
- **Digestion:** stomach HCl helps digestion; excess acid $\rightarrow$ pain; **antacids** like milk of magnesia $\text{Mg}(\text{OH})_2$ neutralise.
- **Tooth decay:** mouth bacteria degrade sweets; below pH 5.5 enamel (calcium phosphate — hardest substance in the body) corrodes; saliva is slightly alkaline; toothpaste neutralises excess acid.
- **Bee sting:** leaves acid; baking soda (a base) applied on the sting gives relief.

7.5 Salts

- Salts are formed by displacement of H^+ ions of an acid by a metallic ion: $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$. e.g. NaCl , MgCl_2 , MgSO_4 , Na_2CO_3 .
- **General properties:** salts are ionic compounds; conduct electricity in aqueous/molten state; mostly soluble in water.

Types of Salts

Type	Definition	Examples
Neutral salt	Complete neutralisation of acid & base	NaCl, KCl, K ₂ SO ₄
Acidic salt	Still contains H ⁺ ; increases H ⁺ in water (pH < 7); behave as acids	NaHSO ₄ (from H ₂ SO ₄ +NaOH); sodium bisulphite NaHSO ₃ ; sodium dihydrogen phosphate NaH ₂ PO ₄
Basic salt	Contains OH ⁻ ; increases OH ⁻ in water (pH > 7); behave as bases; e.g. Mg(OH)Cl from the polyacidic base Mg(OH) ₂ + HCl	Basic lead chloride Pb(OH)Cl; basic copper chloride Cu(OH)Cl; basic copper nitrate Cu(OH)NO ₃
Double salt	Salt crystals containing 2 salts	Potash alum K ₂ SO ₄ ·Al ₂ (SO ₄) ₃ ·24H ₂ O; Mohr's salt FeSO ₄ ·(NH ₄) ₂ SO ₄ ·6H ₂ O; Dolomite CaCO ₃ ·MgCO ₃
Mixed salt	More than one cation or anion	NaKCO ₃ ; CaOCl ₂ (bleaching powder)

🔗 **De-dup note:** The source wrote the potash-alum formula as K₂SO₄·Al(SO₄)₃·24H₂O; the standard reading Al₂(SO₄)₃ is shown — everything else unchanged.

Why does blue coloured copper sulphate crystals turn to white powder on heating above 100 degrees Celsius?

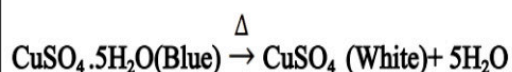


Fig. 07-4 · Salts — quick recap strip

Water of Crystallisation & Related Terms

Term	Meaning	Examples
Water of crystallisation	Number of water molecules present with one formula unit of a salt crystal—	
Efflorescence	Losing water of crystallisation fully/partially on exposure to air	Glauber's salt Na ₂ SO ₄ ·10H ₂ O; Epsom salt MgSO ₄ ·7H ₂ O
Hygroscopy	Absorbing moisture from air without dissolving	Silica gel; quick lime (CaO); phosphorus pentoxide P ₂ O ₅
Deliquescence	Absorbing moisture, becoming wet and finally dissolving	Caustic soda NaOH; caustic potash KOH; MgCl ₂
Drying / desiccating agents	Absorb moisture from other substances	Conc. H ₂ SO ₄ , silica gel
Dehydrating agents	Remove water molecules from other substances	Blue vitriol CuSO ₄ ·5H ₂ O

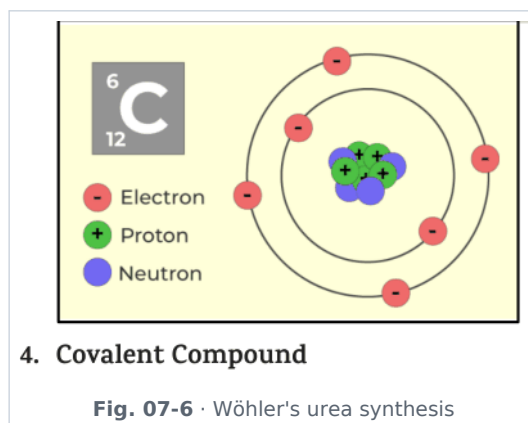
Common name	Chemical name	Formula
Washing soda crystals	Sodium carbonate decahydrate	$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
Epsom salt	Magnesium sulphate heptahydrate	$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
Potash alum	Hydrated potassium aluminium sulphate	$\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
Glauber's salt	Sodium sulphate decahydrate	$\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$
Blue vitriol	Copper (II) sulphate pentahydrate	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
Plaster of Paris	Calcium sulphate semihydrate	$\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$
White vitriol	Zinc sulphate heptahydrate	$\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$
Gypsum	Hydrated calcium sulphate	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

Fig. 07-5 · Hygroscopic vs deliquescent substances

7.6 Carbon — The Gateway to Organic Chemistry

🔗 **De-dup note:** This page closed the acids-bases lecture but actually introduces organic chemistry; it is consolidated with the organic units (10) — repeated there only once.

- **Vital Force theory** — "organic compounds can only come from natural sources, not be made by humans" — proved wrong by **Friedrich Wöhler (1828)** by converting **ammonium cyanate** → **urea**.



- **Basics about carbon:** atomic number 6; mass number 12; 6 e⁻, 6 protons, 6 neutrons; **valence electrons & valency = 4**.
- **Covalent compounds** — formed by sharing of electrons. **Covalent bond** — bond between atoms due to sharing; types: single, double, triple.
- σ/π counts (from the notes): CO_2 — two σ + two π ; N_2 — one σ + two π ; **benzene** — **12 σ + 3 π** .
- **Organic compounds** — compounds containing carbon (excluding oxides of carbon, metallic carbonates, metal cyanides and metal carbides). **Unique nature of carbon: tetravalency & catenation** — full treatment in Unit 10.

UNIT 08

Metals, Non-metals & Metallurgy

Consolidated from: Metal & Nonmetal lecture (pp. 112-124) + element classes & alloys merged from Part-II (pp. 13-17, 22)

8.1 Metals

- Elements with a tendency to **lose electrons** (electropositive); position — left side and centre of the periodic table.

Fig. 08-1 · Metals in the periodic table — block map

Physical Properties

- Malleability** — can be hammered into thin sheets. **Ductility** — can be drawn into wires. **Sonorosity** — produce a singing sound when struck. **Lustre** — shiny surface.
- Good conductors of heat & electricity (good for pots, pans, wires); high density; very hard and non-brittle; high melting & boiling points.

Fig. 08-2 · Lanthanides & actinides strip

15. Invar	Iron (Fe) 64% + Nickel (Ni) 36%	Precision instruments like clocks, measuring tapes, scientific instruments (due to very low thermal expansion)
16. Monel Metal	Nickel (Ni) ~70% + Copper (Cu) ~30% (+ small Fe, Mn)	Marine engineering parts, chemical plant equipment, turbine blades
17. Ferromanganese	Iron (Fe) + Manganese (Mn)	Steel manufacturing to remove oxygen and sulfur impurities
18. Pewter	Tin (Sn) 85–99% + Copper (Cu) + Antimony (Sb)	Decorative items, tableware, figurines
19. Coinage Alloy	Copper (Cu) + Nickel (Ni)	Coins, medals
20. Type Metal	Lead (Pb) + Antimony (Sb) + Tin (Sn)	Printing press movable type (sharp casting)
21. Rose Metal	Bismuth (Bi) + Lead (Pb) + Tin (Sn)	Low-melting-point applications, fire sprinkler triggers
22. Tumbaga	Gold (Au) + Copper (Cu)	Jewellery, ornaments

Fig. 08-3 · Types of metals — commercial classification]

(Highest reactive)→	क K	नि Ni	औ Au
	न Na	सन Sn	पट Pt
	बा Ba	प० Pb	(Least reactive)
	का Ca	H	
	मग Mg	Cu	
	अल Al	Hg	
	जन Ze	अग Ag	
	फे Fe		

- Reaction of metals with non-metals:

Fig. 08-4 · d-, f-, p-block element families chart

- Exceptions (from the notes):**

- Brittle metals — beryllium, arsenic, zinc fracture instead of deforming.
- Alkali metals (Na, K) are soft — cut with a knife — and have low density.
- Lead, bismuth and tungsten conduct less than other metals.

- **Mercury** is liquid at room temperature; **gallium & caesium** melt at very low temperatures.
- Most metals are silver-grey, but copper is reddish-brown and gold is yellow.
- Magnetic metals: iron, cobalt, nickel.

Chemical Properties

- **With oxygen:** Metal + O₂ → Metal oxide (basic)



- With Fe the oxide depends on reactivity/heat: FeO (ferrous), Fe₂O₃ (ferric) — Fe(II) & Fe(III) oxides form under different conditions.
- **With water:** Metal + H₂O → Metal hydroxide + H₂↑



- **With other metal salts (displacement):** P + AB → PB + A — possible only if P is more reactive than A.



- **Double displacement:** AB + PQ → PB + AQ; e.g. Na₂SO₄ + CuCl₂ → NaCl + CuSO₄.
- **With non-metals:** Na + Cl₂ → NaCl; 2K + I₂ → 2KI; Mg + H₂ → MgH₂ — metals + non-metals form **ionic compounds**.

METALS REACTIVITY SERIES			
Metal Name		Symbol	Reactivity
Cesium		Cs	HIGHLY REACTIVE Cs, Fr, Rb & K Violent Reaction with Cold Water Na, Li, Ba, Ra, Sr & Ca Moderate reaction with Cold water, Strong reaction with hot water or steam
Francium		Fr	
Rubidium		Rb	
Potassium		K	
Sodium		Na	
Lithium		Li	
Barium		Ba	
Radium		Ra	
Strontium		Sr	
Calcium		Ca	
Magnesium		Mg	MODERATELY REACTIVE Mg Very Slow Reaction with Cold Water, Strong reaction with hot water or steam Al, Ti, Mn, Zn & Fe Moderate reaction with hot water or steam and dilute acid Ni, Sn & Pb Slow reaction with dilute acid
Aluminum		Al	
Titanium		Ti	
Manganese		Mn	
Zinc		Zn	
Chromium		Cr	
Iron		Fe	
Cadmium		Cd	
Cobalt		Co	
Nickel		Ni	
Tin		Sn	
Lead		Pb	
Hydrogen		H	Nonmetal (For Comparison)
Antimony		Sb	LEAST REACTIVE Sb, Bi, Cu, W, Hg, Ag, Au & Pt Do not react with cold water, hot water, steam or dilute acids
Bismuth		Bi	
Copper		Cu	
Tungsten		W	
Mercury		Hg	
Silver		Ag	
Gold		Au	
Platinum		Pt	

Fig. 08-5 · Metals reactivity series — most to least reactive

Types of Metals

Type	Where	Character
Alkali metals (Group 1)	s-block	Highly reactive; always form +1 cations; hydroxides are water-soluble alkalis
Alkaline earth metals (Group 2)	s-block	Form +2 cations (Mg → Mg ²⁺ + 2e ⁻ , Ca → Ca ²⁺ + 2e ⁻); hydroxides soluble but less than group 1
Transition metals	d-block, groups 3-12	Variable valency; colourful compounds; hard, strong, high m.p.; catalysts; Fe/Co/Ni magnetic

Type	Where	Character
Inner transition metals	f-block	Lanthanides (Ce–Lu) & actinides (Th–Lr); heavy, high m.p., variable valency, many radioactive

Mnemonic (as in the source)

Group-2 order "Be Mg Ca Sr Ba Ra" = Be, Mg, Ca, Sr, Ba, Ra.

Alkali metals: most reactive metal family; soft & light; low m.p./b.p.; single valence electron (highly electropositive); give alkaline solutions with water ($\text{Na} \rightarrow \text{NaOH}$, a strong base); used in batteries (Li-ion), pharma (Na compounds), organic synthesis (K compounds).

Catalyst

Substance that changes the rate of a reaction without participating in it. Transition metals and their compounds are widely used as catalysts.

8.2 Non-metals

- Elements having a tendency to **gain or share electrons**; more electronegative than metals; 19 in number.

Elements having tendency to gain or share electrons.

Solid (7)	Liquid (1)	Gas (11)	
Sulphur	Bromine	Oxygen	Helium
Carbon		Chlorine	Neon
Phosphorus		Fluorine	Argon
Selenium		Nitrogen	Argon
Iodine		Hydrogen	Radon
Boron			Xenon
Astatine			

Fig. 08-6 · Non-metals at room temperature — 11 gases, 1 liquid, 7 solids

Physical Properties

- Brittle — non-malleable & non-ductile; non-lustrous & non-sonorous; bad conductors (**exception: graphite** conducts); low density, low m.p./b.p.; generally solids & gases (**exception: bromine**, the only liquid non-metal — red & volatile).

Chemical Properties

- Accept or share electrons → covalent compounds (e.g. $\text{N} \equiv \text{N}$).

These elements accept or share electron.

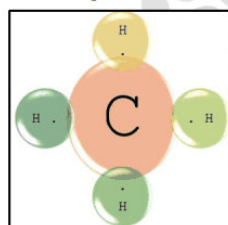
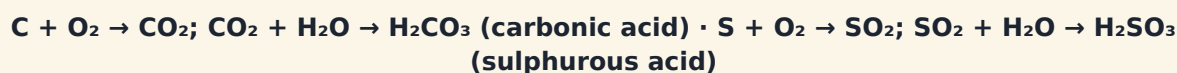


Fig. 08-7 · Carbon shares four electrons — covalent compound formation

- With oxygen:** Non-metal + $\text{O}_2 \rightarrow$ Non-metal oxide (acidic)



- Electronegativity** — tendency to attract the shared pair in a covalent compound; non-metals are more electronegative than metals.

Metals vs Non-metals — Summary Tables

🔗 **De-dup note:** The metals/non-metals/metalloids comparison appeared in Part-II (pp. 13-14) and again in the Metal & Nonmetal lecture (pp. 112, 116-117). The two versions are merged below.

Property	Metals	Non-metals	Metalloids
State	Mostly solid (exception Hg)	Solid, liquid & gas	Solid
Appearance	Shiny (lustrous)	Dull	Variable (metallic look, brittle)
Conductivity	Good	Poor (insulators); graphite excepted	Semiconductor (rises with temperature)
Malleability	High	Brittle — not malleable	Brittle
Ion type	Cations (+)	Anions (–)	Variable
Oxides	Basic / amphoteric (Na ₂ O, Al ₂ O ₃)	Acidic / neutral (CO ₂ , SO ₂ , N ₂ O)	Amphoteric / weakly acidic
Ionization enthalpy & EN	Low	High	Intermediate
Position	Left & centre	Upper right	Zig-zag line between the two

- **Metalloids:** B, Si, Ge, As, Sb, Te (+ Po) — intermediate properties; shiny like metals but brittle like non-metals; **semiconductors**; backbone of the electronics industry (Si, Ge).
- **Semiconductor doping (kept from Part-II):**

Feature	P-type	N-type
Majority carriers	Holes (positive)	Electrons (negative)
Doping elements	Trivalent (3 valence e [–]): B, Al, Ga, In	Pentavalent (5 valence e [–]): P, As, Sb
Common base	Si, Ge	Si, Ge
Common materials	Si+B, Si+Ga, Ge+In	Si+P, Si+As, Ge+Sb
Usage	Diodes, transistors, solar cells	Diodes, transistors, solar cells

Metal Alloys and Their Uses

- An **alloy** is a homogeneous mixture of two or more metals (and non-metals) — pure iron rusts easily but alloying gives desirable properties. Steel and brass are common alloys; some steels also contain non-metals.

Some metal alloys, their composition and uses		
Name of alloy	Composition of alloy	Uses
solder	50% tin + 50% lead	soldering wire
type metal	75% lead + 25% antimony and tin	lead printing machines
bronze	10% tin and zinc + 90% copper 5% tin and zinc + 95% copper	machinery, decorative metalwork, coins
stainless steel	chromium + nickel + iron (in different proportions)	pans, cutlery, chemical and food-processing industries
armoured steel	nickel + chromium + iron (in different proportions)	armour plates, gears and wires
tool steel	chromium + molybdenum + iron (in different proportions)	tools (hammers, chisels, etc.)

Fig. 08-8 · Metal alloys — composition & uses (source chart)

Non-metal	Uses
nitrogen (N)	fertilisers, cleansers, medication, food-processing, explosives
oxygen (O)	in many foods, welding
fluorine (F)	etching, refrigerant, toothpaste
chlorine (Cl)	bleach, disinfectant
iodine (I)	dye, medicine, antiseptic
sulfur (S)	sulfuric acid, matches, fungicide
phosphorus (P)	fertiliser, matches, detergents
neon (Ne)	illuminated advertising signs
carbon (C)	graphite in pencils, carbon fibre, diamonds in jewellery

Fig. 08-9 · Uses of metals & non-metals (iodine, sulphur, neon, carbon...)

Alloy oddities (from the notes)

Iodine is a non-metal but shiny; graphite (carbon) conducts and has a high m.p.; mercury is a liquid metal.

8.3 Hydrogen & Water (from Metal & Nonmetal lecture)

🔗 **De-dup note:** Hydrogen sits with the element chemistry of this lecture in the source; kept here — facts unchanged.

General Facts about Hydrogen

- **Most abundant element in the universe;** on Earth it exists in combined form (water, organic matter, minerals); diatomic gas (H_2).
- Shows **dual similarity** — with alkali metals ($H \rightarrow H^+ + e^-$) and halogens ($H + e^- \rightarrow H^-$).

• Isotopes of Hydrogen:

	Protium	Deuterium	Tritium
no of e-	1	1	1
no. of protone	1	1	1
no of neutron	0	1	2

Fig. 08-10 · Isotopes of hydrogen

Isotope	Protons	Neutrons	Electrons	Symbol
Protium	1	0	1	${}_1H^1$
Deuterium	1	1	1	${}_1H^2$
Tritium	1	2	1	${}_1H^3$

Heavy Water (D_2O)

- Contains deuterium instead of protium; discovered in **1932 by H. C. Urey & E. W. Washburn.**
- **Uses:** nuclear **moderator & coolant** (slows neutrons); **tracer** in chemistry/metabolism studies; production of deuterium compounds.

Hard & Soft Water

- **Soft water** lathers with soap easily; **hard water** does not — due to salts of **Ca^{2+} and Mg^{2+}** (bicarbonates, chlorides, sulphates).

Basis	Temporary Hard Water	Permanent Hard Water
Cause	Presence of bicarbonates of calcium and magnesium $\text{Ca}(\text{HCO}_3)_2$, $\text{Mg}(\text{HCO}_3)_2$.	Presence of sulphates and chlorides of calcium and magnesium CaSO_4 , MgSO_4 , CaCl_2 , MgCl_2 .
Nature	Can be removed easily by simple methods.	Cannot be removed by simple methods like boiling.

Fig. 08-11 · Temporary vs permanent hardness

Point	Temporary hard water	Permanent hard water
Cause	Bicarbonates of Ca & Mg — $\text{Ca}(\text{HCO}_3)_2$, $\text{Mg}(\text{HCO}_3)_2$	Chlorides & sulphates of Ca & Mg — CaCl_2 , CaSO_4 , MgCl_2 , MgSO_4
Removal	Can be removed simply by boiling	Cannot be removed by simple methods; needs chemical treatment

- **Effects of hard water:** soap wastage; scale (boiler) formation; problems in dyeing & laundry.

Hydrogen Peroxide (H_2O_2)

- Discovered **1818 by Thenard**; pale blue liquid; m.p. -0.43°C ; b.p. 150.2°C ; density 1.47 g/cm^3 .
- Strong **oxidising agent** (also a reducing agent in some reactions); unstable $\rightarrow$ decomposes to water + oxygen.
- **Uses:** germicide & antiseptic; bleaching of textiles, paper, hair, leather, oils & fats; manufacture of sodium perborate & sodium carbonate; milk/food preservation; oxygen source in submarines.

8.4 Sulphur & Its Important Compounds

- **Occurrence:** 0.03–0.1 % of Earth's crust; sources — gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), Epsom salt ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$), barite (BaSO_4), galena (PbS), zinc blende (ZnS), copper pyrites (CuFeS_2); found in volcanic H_2S ; present in proteins of egg, onion, garlic, mustard, pulses.
- Oxidised forms: H_2SO_4 , H_2SO_3 , peroxodisulphuric acid ($\text{H}_2\text{S}_2\text{O}_8$), pyrosulphuric acid ($\text{H}_2\text{S}_2\text{O}_7$).

Compound	Facts
Sulphur dioxide (SO_2)	Colourless gas, pungent smell; liquefies at 263 K under pressure; highly water-soluble. Uses — petroleum refining; bleaching wool & silk; disinfectant & preservative; refrigerant
Sulphur hexafluoride (SF_6)	Colourless, odourless, non-toxic, octahedral; excellent electrical insulator
Sulphuric acid (H_2SO_4)	"King of chemicals"; colourless, oily, hygroscopic; high b.p.; very strong acid. Uses — petroleum refining; fertilisers; explosives, detergents, plastics; storage batteries; medicines (nitroglycerine)

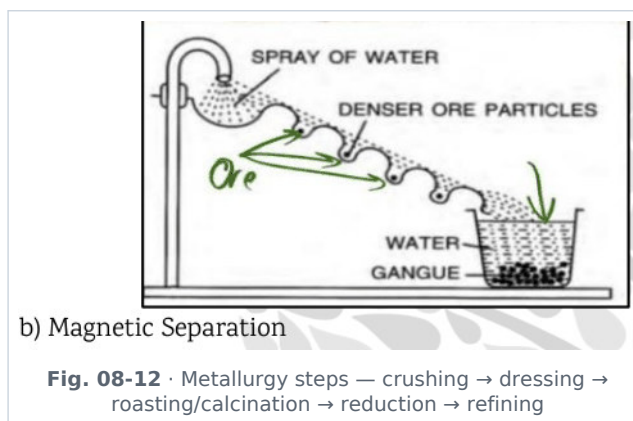
Inert (Noble) Gases

- He, Ne, Ar, Kr, Xe, Rn — rare/noble gases; filled outer shells $\rightarrow$ very low reactivity; colourless, odourless, non-flammable.

Gas	Uses
Helium	Balloons & airships; O_2 -mix for deep-sea divers (avoids nitrogen narcosis); asthma treatment; inflating aeroplane tyres; inert atmosphere for welding reactive metals
Neon	Discharge/advertising tubes; beacon lights for air navigation (fog-piercing)
Argon	Fills electric lamps with nitrogen (more inert than N_2); fluorescent lamps
Krypton	High-intensity airport runway lamps; photographic flashes
Xenon	Powerful lamps; camera flashes; some medical anaesthesia
Radon	Cancer radiotherapy (radioactive — used carefully)

8.5 Metallurgy

- **Metallurgy** — the branch of science dealing with extraction of metals from ores and their purification.



Basic Terms

Term	Meaning
Minerals	Naturally occurring compounds of metals mixed with soil, sand, limestone, rocks
Ore	Mineral from which metal is extracted commercially at low cost & minimum effort
Gangue (matrix)	Earthy impurities (mud, silica, sand) in the ore
Flux	Substance added in smelting to promote fluidity and remove impurities as slag
Slag	Product formed when flux reacts with the gangue; e.g. $\text{CaO}(\text{flux}) + \text{SiO}_2(\text{gangue}) \rightarrow \text{CaSiO}_3(\text{slag})$

Metal	Mineral/Ores	Composition of Mineral/Ores
Sodium (Na)	Rock Salt	NaCl
	Trona	$\text{Na}_2(\text{HCO}_3)(\text{CO}_3) \cdot 2\text{H}_2\text{O}$
	Chilli Saltpetre	NaNO_3
	Borax	$\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$
	Glauberite	$\text{Na}_2\text{Ca}(\text{SO}_4)_2$
Potassium (K)	Glauber's Salt	$\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$
	Sylvine	KCl
	Carnallite	$\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
Magnesium (Mg)	Schoenite (Picromerite)	$\text{K}_2\text{Mg}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$
	Magnesite	MgCO_3
	Dolomite	$\text{MgCO}_3 \cdot \text{CaCO}_3$
	Carnallite	$\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
	Kieserite	$\text{MgSO}_4 \cdot \text{H}_2\text{O}$
Calcium (Ca)	Epsom Salt (Epsomite)	$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
	Limestone	CaCO_3
	Calcite	CaCO_3
	Gypsum	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
	Fluorspar	CaF_2

Fig. 08-13 · Important minerals & ores (carbonates & sulphides)

Metal	Mineral/Ores	Composition of Mineral/Ores
Silver (Ag)	Native Silver	Ag
	Argentite	Ag_2S (Silver Glance)
	Cerargyrite	AgCl (Horn Silver)
Zinc (Zn)	Zinc Blende (Black Jack)	ZnS
	Franklinite	ZnFe_2O_4
	Calamine	ZnCO_3
	Zincite	ZnO (Red Zinc)
Mercury (Hg)	Cinnabar	HgS
Manganese (Mn)	Pyrolusite	MnO_2
	Manganite	$\text{MnO}(\text{OH})$
	Hausmannite	Mn_3O_4
Iron (Fe)	Magnetite (Load stone or Magnetic oxide of iron)	Fe_3O_4
	Haematite	Fe_2O_3
	Limonite	$\text{FeO}(\text{OH}) \cdot n\text{H}_2\text{O}$
	Siderite	FeCO_3 (Spathic Iron)
	Iron Pyrites	FeS_2

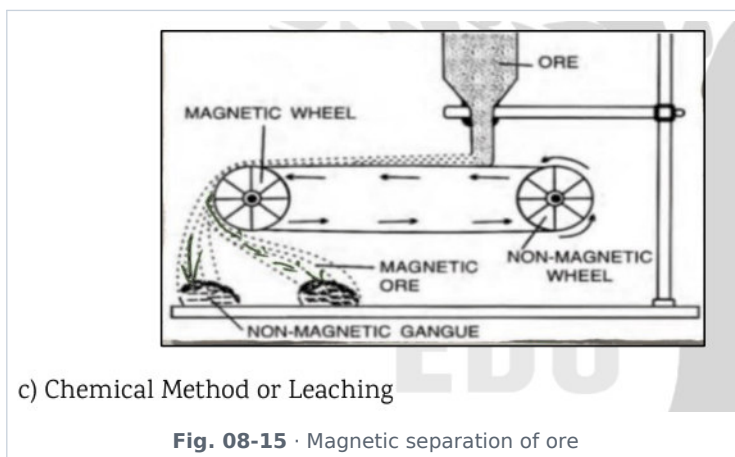
Fig. 08-14 · Mineral/ore composition table (Al, Zn, Fe, Mg, Na, Ca...)

Step 1 — Crushing & Grinding

- Ores are crushed to fine powder in **big jaw crushers & ball mills**; **pulverisation** = crushing & grinding of ore to powder.

Step 2 — Dressing (Concentration) of Ores

- Removal of impurities (sand, soil, rock particles) from powdered ore. Methods:
 - Gravity / hydrolytic separation** — needs density difference between ore and impurities.
 - Magnetic separation** — for ores attracted by a magnet.
 - Leaching (chemical)** — a suitable solvent dissolves the ore; insoluble impurities are filtered. e.g. bauxite + hot conc. NaOH → soluble sodium aluminate Na[Al(OH)₄]; CO₂ then precipitates Al₂O₃.



Step 3 — Roasting & Calcination

Roasting	Calcination
I. The ore is heated in the excess of air. II. Generally, sulphide ores are roasted, so SO ₂ is given off $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$ (at 800-900degree) III. Volatile impurities are removed as oxides (SO ₂ , P ₂ O ₅ , As ₂ O ₃) and the ore becomes porous and more reactive.	I. The ore is heated in the absence of air II. Carbonate and hydrated ores are calcined and CO ₂ or water vapours are given off $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$ III. Moisture and organic impurities are removed and the ore becomes porous and more reactive.

Fig. 08-16 · Roasting vs calcination

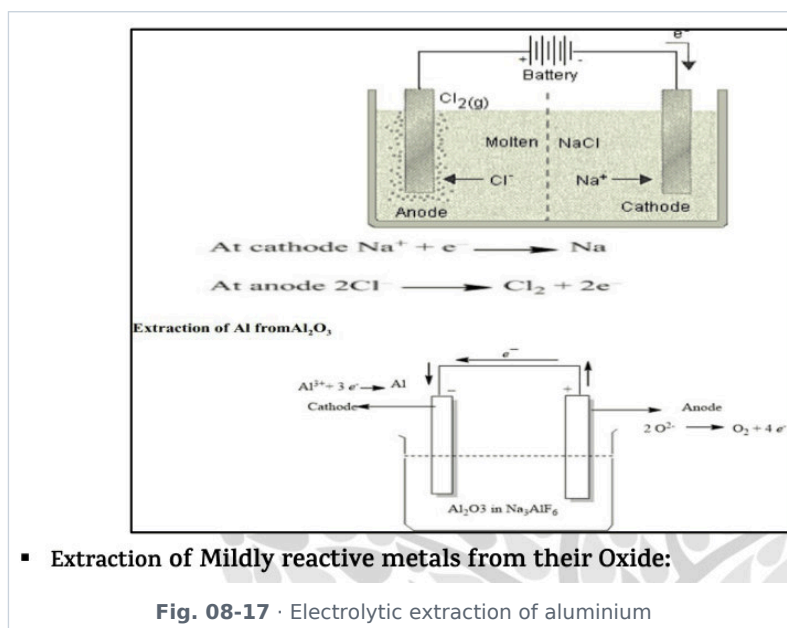
Point	Roasting	Calcination
Done	In the presence of excess air (O ₂)	In the absence/restricted supply of air
Ores	Sulphide ores	Carbonate & hydrated oxide ores
Examples	$2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$; $2\text{PbS} + 3\text{O}_2 \rightarrow 2\text{PbO} + 2\text{SO}_2$	$\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$; $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O} \rightarrow \text{Al}_2\text{O}_3 + 2\text{H}_2\text{O}$
Moisture & organic matter	Removed	Removed

Step 4 — Reduction of Metal Oxides

- Highly reactive metals (K, Na, Ca, Mg, Al)** — oxides cannot be reduced by carbon; extracted by **electrolytic reduction**.
- Mildly reactive metals** — oxide + reducing agent (C, CO, H₂ or a more reactive metal):



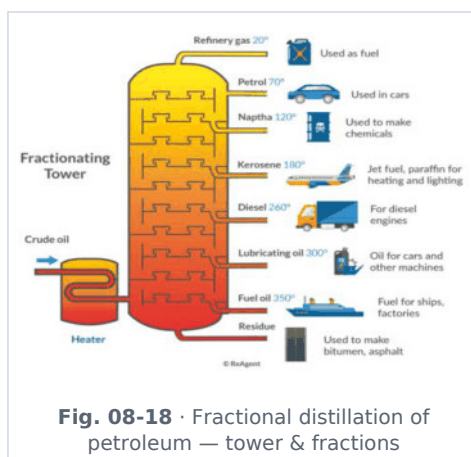
- Least reactive metals** — oxides decompose to metal on heating alone.



Step 5 — Refining of Metals

♻ **De-dup note:** Refining methods were printed twice on source pp. 123-124 — kept once.

Method	Used for	Idea
Distillation	Volatile metals — Hg, Zn	Impure metal heated above b.p.; vapour separated from non-volatile impurities and condensed
Liquation	Low-melting metals — Pb, Sn	Metal melts and flows away from higher-melting impurities
Electrolytic refining	Cu, Al, Zn etc.	Impure anode dissolves; pure metal deposits on cathode



UNIT 09

Important s, p, d & f Block Elements

Consolidated from: Important S,P,D,F Elements lecture (pp. 87-110) — group-wise chemistry, compounds & uses

9.1 s-Block Elements

The periodic table shows the following elements in the s-block (Groups 1 and 2):

- Group 1 (Alkali Metals):** H (1), Li (3), Na (11), K (19), Rb (37), Cs (55), Fr (87).
- Group 2 (Alkaline Earth Metals):** He (2), Be (4), Mg (12), Ca (20), Sr (38), Ba (56), Ra (88).

The table also includes the Lanthanide and Actinide series at the bottom.

Fig. 09-1 · s-block — physical properties of alkali & alkaline-earth metals

Electronic Configuration of s-block Elements		
Element	Symbol	Electronic configuration
Lithium	Li	$1s^2s^1$
Sodium	Na	$1s^22s^22p^63s^1$
Potassium	K	$1s^22s^22p^63s^23p^4s^1$
Rubidium	Rb	$1s^22s^22p^63s^23p^63d^{10}4s^24p^65s^1$
Caesium	Cs	$1s^22s^22p^63s^23p^63d^{10}4s^24p^64d^{10}5s^25p^66s^1$ Or [Xe] $6s^1$
Francium	Fr	[Rn] $7s^1$

Fig. 09-2 · Atomic/chemical behaviour — alkali vs alkaline earth metals

- **Group 1 — alkali metals:** Li, Na, K, Rb, Cs, Fr — named for the **strongly alkaline hydroxides** they form with water.
- **Group 2 — alkaline earth metals:** Be, Mg, Ca, Sr, Ba, Ra (Be excluded from the name) — oxides & hydroxides are alkaline, commonly found in the earth's crust.
- Both are s-block groups but differ in properties & reactivity.

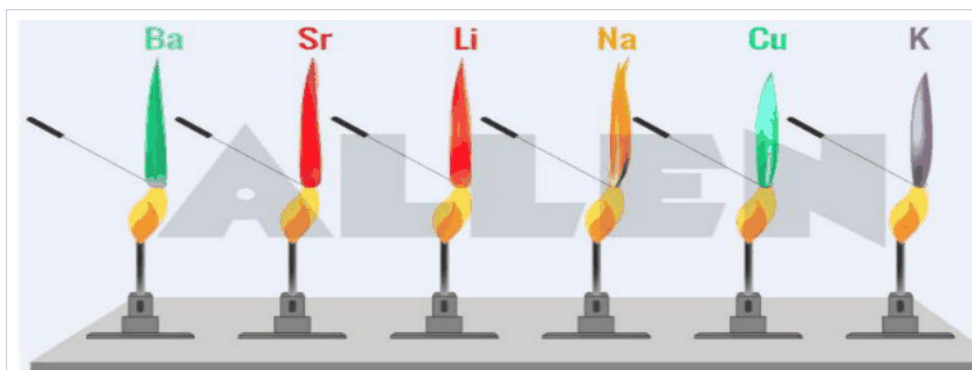


Fig. 09-3 · Melting & boiling points — alkali vs alkaline earth

Flame Test

Physical Properties of s-block Elements	
Alkali Metals	Alkali Earth Metals
One electron in outermost shell & General formula- ns^1	Two electrons in outermost shell & General formula- ns^2
Francium is a radioactive element.	Radium is a radioactive element.
All are silvery white	All are silvery white.
Light soft, malleable and ductile metals with metallic lustre.	These metals are harder than alkali metals.
Atomic size	
Alkali Metals	Alkali Earth Metals
Largest in their respective period (except noble gas element)	Smaller than IA group elements, since extra charge on nucleus attracts the electron cloud.
Size increases from Li to Cs due to addition of an extra shell. $Li < Na < K < Rb < Cs$	Size increases gradually from Be to Ba $Be < Mg < Ca < Sr < Ba$
Note- In s-block elements Beryllium (Be) has the smallest size, while Caesium (Cs) has the largest size.	
Melting and Boiling Point	

Fig. 09-4 · Flame colours of metal ions

- The **flame test** identifies metal ions by the characteristic flame colour on heating (electronic transitions) — a qualitative-analysis method.

Important Compounds of s-block

Alkali Metals	Alkali Earth Metals
Alkali metals and their salts give characteristic colour to Bunsen flame. The flame energy causes an excitation of the outer most electron which on dropping back to ground state emits absorbed energy as a visible light.	Due to small size of Be & Mg outer most n electrons are tightly bounded. So not excited to higher level, hence they do not give flame test.
Li - Crimson red; Na - Golden yellow K - Violet; Rb - Red violet; Cs - Blue	Ca - Brick red; Sr - Crimson red; Ba - apple green

Fig. 09-5 · Compounds of s-block — properties & uses ($NaCl$, $NaOH$, $NaHCO_3$)

2. Reactions with air-	
Alkali Metals	Alkali Earth Metals
Alkali metals gets tarnish in air due to the formation of oxide at their surface hence they are kept in kerosene or paraffin wax.	Except Be, these metals easily tarnished in air. Beryllium in powdered form, burns brilliantly on ignition in air.
These elements reacts with moist air to form carbonates $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ $\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$ (moist) $2\text{NaOH} + \text{CO}_2 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$ (In air)	In moist air, except Be all the elements converts into carbonates.
In dry air only Li gives nitride and oxide both while other elements gives only oxides.	In dry air all elements of II A give nitride and oxide both.

Fig. 09-6 · Compounds of s-block — Na_2CO_3 , $\text{Ca}(\text{OH})_2$, CaSO_4 , Plaster of Paris

Compound	Properties	Uses
NaCl — sodium chloride	White crystalline solid; high m.p.; very water-soluble	Table salt; seasoning & preservative; chemical manufacture; food processing; water softening
NaOH — sodium hydroxide	White solid; strongly alkaline solution	Pulp & paper, textiles, soaps & detergents, cleaners; biodiesel; pH regulation in water treatment
NaHCO_3 — sodium bicarbonate	White crystalline; slightly alkaline taste	Baking soda (leavening); antacid for heartburn; cleaning & deodorising
Na_2CO_3 — sodium carbonate	White, odourless, alkaline	Soda ash / washing soda; glassmaking; detergents; water treatment; pool pH control
$\text{Ca}(\text{OH})_2$ — calcium hydroxide	White powder; low solubility; strong alkaline	Slaked lime — mortar & plaster; wastewater treatment; agriculture; pickling; toothpaste & antacids
CaSO_4 — calcium sulphate	Gypsum $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ and anhydrous form	Plaster of Paris, plasterboard, cement; soil conditioner; water conditioner in brewing
Plaster of Paris	White powder; hardens with water (from heating gypsum)	Casts, moulds, sculptures; limb casts in medicine; dental impressions; art & crafts

9.2 p-Block Elements — General Properties

p block Elements

Fig. 09-7 · General properties of p-block elements

General electronic configurations and oxidation states for each group of the p-block elements-				
Group	General Electronic Configuration	First Member	Group Oxidation State	Other Oxidation States
13	ns^2np^1	B (Boron)	+3	+1
14	ns^2np^2	C (Carbon)	+4	+2, -4
15	ns^2np^3	N (Nitrogen)	+5	+3, -3
16	ns^2np^4	O (Oxygen)	+6	+4, +2, -2
17	ns^2np^5	F (Fluorine)	+7	+5, +3, +1, -1
18	ns^2np^6 (for He)	He (Helium)	+8 (theoretical)	+6, +4, +2

Fig. 09-8 · General properties (continued)

- **Configuration:** $ns^2 np^{1-6}$ — the last electron enters a p-orbital (groups 13–18).
- **Variety:** metals, non-metals and metalloids together — N, O, F gaseous; Al, Sn solid with high m.p.
- **Oxidation states:** often –3 to +5+ from ns & np availability; non-metals negative, metals positive.
- **Reactivity:** F most reactive (non-metals); Ne, Ar notably inert.
- **Allotropy:** C — diamond, graphite, fullerenes; also P and S show allotropy.

9.3 Group 13 — Boron Family (B, Al, Ga, In, Tl)

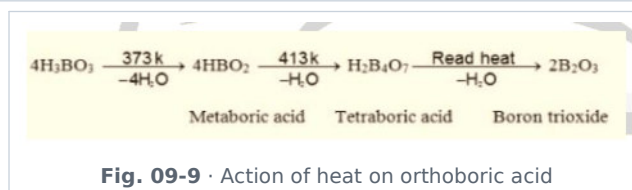


Fig. 09-9 · Action of heat on orthoboric acid

- **Trends:** m.p. decreases B→Ga then rises to Tl; ionization energies 1st $\ll$ 2nd $\ll$ 3rd; metallic character increases B→Tl (**B is a non-metal**). Al — light, corrosion-resistant — used in packaging & construction.

Preparation of Boron

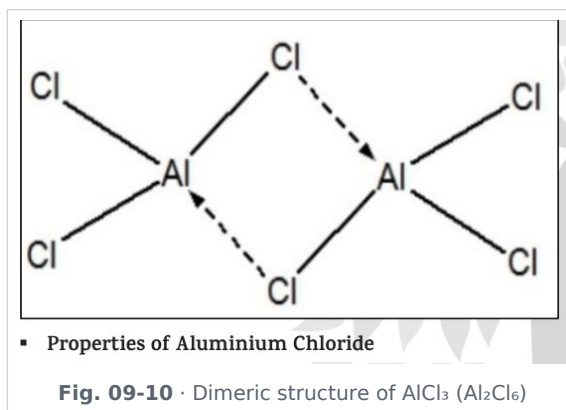
- From boric acid route: $B_2O_3(s) + 3Mg(s) \rightarrow 2B(s) + 3MgO(s)$
- From boron trichloride: $2BCl_3 + 3H_2(g) \xrightarrow{1270\text{ K}} 2B(s) + 6HCl(g)$; $2BCl_3(g) + 3Zn(s) \xrightarrow{900\text{ }^\circ\text{C}} 2B(s) + 3ZnCl_2(s)$
- Electrolysis of fused boric anhydride (B_2O_3) + MgO + MgF_2 at $1100\text{ }^\circ\text{C}$: $2MgO \rightarrow 2Mg + O_2$; $B_2O_3 + 3Mg \rightarrow 2B + 3MgO$
- Thermal decomposition of boron hydrides: $B_2H_6(g) \xrightarrow{\Delta} 2B(s) + 3H_2(g)$

Compounds of Boron

Compound	Preparation	Properties
Orthoboric acid H_3BO_3	From borax: $Na_2B_4O_7 + H_2SO_4 + 5H_2O \rightarrow Na_2SO_4 + 4H_3BO_3$; from colemanite: $Ca_2B_6O_{11} + 2SO_2 + 11H_2O \rightarrow 2Ca(HSO_3)_2 + 6H_3BO_3$	Heating: $2H_3BO_3 \xrightarrow{\Delta} H_2B_4O_7 \xrightarrow{\Delta} H_2O + 2B_2O_3$; weak monobasic acid — $H_3BO_3 + NaOH \rightarrow NaBO_2 + 2H_2O$ (sodium metaborate)
Borax $Na_2B_4O_7 \cdot 10H_2O$	$4H_3BO_3 + Na_2CO_3 \rightarrow Na_2B_4O_7 + 6H_2O + CO_2$	Aqueous solution alkaline by hydrolysis: $Na_2B_4O_7 + 7H_2O \rightarrow 2NaOH + 4H_3BO_3$

Compounds of Aluminium

- **Alumina Al_2O_3 :** $2Al(OH)_3 \xrightarrow{\text{heat}} Al_2O_3 + 3H_2O$ (source shows $Al_2O_3 + 2H_2O$); $(NH_4)_2Al_2(SO_4)_3 \cdot 24H_2O \xrightarrow{\Delta} 2NH_3 + Al_2O_3 + 4SO_3 + 25H_2O$.
- **Aluminium chloride $AlCl_3$:** white hygroscopic solid; sublimes at $183\text{ }^\circ\text{C}$; addition compounds with NH_3 , PH_3 , $COCl_2$; hydrolysis $AlCl_3 + 3H_2O \rightarrow Al(OH)_3 + 3HCl$; heating $2AlCl_3 \cdot 6H_2O \rightarrow 2Al(OH)_3 \rightarrow Al_2O_3 + 6HCl + 3H_2O$.



Why AlCl_3 dimerises

Electron-deficient Al completes its octet by bridging through chlorine atoms — the dimer Al_2Cl_6 .

9.4 Group 14 — Carbon Family (C, Si, Ge, Sn, Pb)

- C — heart of organic chemistry; Si — semiconductor technology; Pb — historical pipes & paints.
- **Trends:** IE decreases $\text{C} \rightarrow \text{Sn}$ then rises to Pb; C & Si non-metals, Ge metalloid, Sn & Pb metals; **catenation** — C (and Si) form long chains.
- **Compounds of carbon:** CO burns with blue flame, turns platinum/palladium-chloride paper pink-green. CO_2 — from acids on carbonates ($\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$) or burning carbon; turns lime water milky, $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$, milkiness disappears in excess CO_2 ($\rightarrow \text{Ca(HCO}_3)_2$); dry ice sublimates.
- **Carbides:** salt-like (C_2^{2-} acetylide / C^{4-} methanide — CaC_2 , Al_4C_3 , Be_2C); covalent (SiC — carborundum); interstitial (transition-metal lattices — WC, VC).
- **Compounds of silicon:** sodium silicate Na_2SiO_3 (fused soda ash + sand); **silicones** — Si-O-Si polymers from chlorosilane hydrolysis; **silicates** — salts of silicic acid H_4SiO_4 with sp^3 tetrahedral SiO_4^{4-} units.

9.5 Group 15 — Nitrogen Family (N, P, As, Sb, Bi)

- Atomic radii increase down the group (small rise $\text{As} \rightarrow \text{Bi}$); oxidation states -3 to $+5$, $+3$ stability increases down; IE decreases $\text{N} \rightarrow \text{Bi}$.

Ammonia (NH_3)

- **Preparation:** $\text{NH}_4\text{Cl} + \text{NaOH} \rightarrow \text{NH}_3 + \text{NaCl} + \text{H}_2\text{O}$; $\text{Mg}_3\text{N}_2 + 6\text{H}_2\text{O} \rightarrow 3\text{Mg(OH)}_2 + 2\text{NH}_3$; **Haber's process** $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$.
- **Properties:** aqueous solution basic — turns red litmus blue ($\text{NH}_3 + \text{H}_2\text{O} \rightarrow \text{NH}_4^+ + \text{OH}^-$).
- **Complex formation:** $\text{Ag}^+ + \text{NH}_3 \rightarrow [\text{Ag(NH}_3)_2]^+$; $\text{Cu}^{2+} + 4\text{NH}_3 \rightarrow [\text{Cu(NH}_3)_4]^{2+}$; $\text{Cd}^{2+} + 4\text{NH}_3 \rightarrow [\text{Cd(NH}_3)_4]^{2+}$.
- **Precipitation of metal ions:** $\text{FeCl}_3 + 3\text{NH}_4\text{OH} \rightarrow \text{Fe(OH)}_3 \downarrow$ (brown) + $3\text{NH}_4\text{Cl}$; $\text{AlCl}_3 + 3\text{NH}_4\text{OH} \rightarrow \text{Al(OH)}_3 \downarrow$ (white); $\text{CrCl}_3 + 3\text{NH}_4\text{OH} \rightarrow \text{Cr(OH)}_3 \downarrow$ (green).

Phosphorus

Oxy Acids	Name of oxy – acid
1. $\text{H}_2\text{N}_2\text{O}_2$	Hyponitrous acid
2. H_2NO_2	Hydronitrous acid
3. HNO_2	Nitrous acid
4. HNO_3	Nitric acid
5. HNO_4	Per nitric acid

Fig. 09-11 · Phosphine & phosphorus halides

- **Allotropy: white P** — translucent waxy solid, extremely reactive, poisonous, water-insoluble; **red P** — from heating white P without air; does not burn spontaneously at room temperature; **black P** — from further heating of red P.
- **Oxy-acids of phosphorus** and **oxy-acids of nitrogen** — charts kept with the tables below.

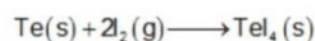
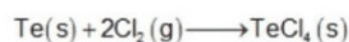
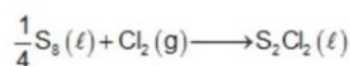
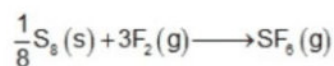
Formation of volatile Hydrides:**Formation of Halides:**

Fig. 09-12 · Oxyacids of nitrogen

Sr. No.	Property	Oxygen	Sulfur	Selenium	Tellurium	Polonium
1.	Configuration	$[\text{He}]2s^22p_4$	$[\text{Ne}]3s^23p_4$	$[\text{Ar}]4s^24p_4$	$[\text{Kr}]5s^25p_4$	$[\text{Xe}]6s^26p_4$
2.	Common oxidation state	-2	-2, +4, +6	+4, +6	+4, +6	
3.	Atomic radius (pm)	66	104	116	143	167
4.	First ionization energy (kJ/mol)	1314	1000	941	869	812
5.	Electronegativity	3.5	2.5	2.4	2.1	2.0

Fig. 09-13 · Oxyacids of phosphorus

9.6 Group 16 — Oxygen Family (O, S, Se, Te, Po)

Oxo acid	Name
H_3PO_2	Hypophosphorus acid
H_3PO_3	Phosphorus acid
$\text{H}_4\text{P}_2\text{O}_5$	Hypophosphoric acid
H_3PO_4	Orthophosphoric acid
$\text{H}_4\text{P}_2\text{O}_7$	Pyrophosphoric acid
HPO_3	Metaphosphoric acid

Fig. 09-14 · Group-16 chemical properties card

- **Oxides:** all elements form dioxides MO_2 — SO_2 gas, SeO_2 volatile solid, TeO_2 & PoO_2 non-volatile crystalline solids. **Ozone** — unstable, decomposes to O_2 ; strong oxidising agent (liberates nascent

oxygen).

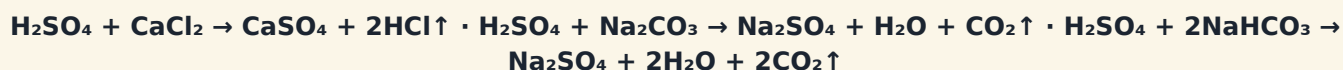
• **Allotropes of sulphur:**

- **Rhombic (α):** bright yellow; insoluble in water & CS₂; density 2.07 g/cm³; exists as S₈ puckered rings.
- **Monoclinic (β):** stable above 369 K; dull yellow; soluble in CS₂; slowly changes to rhombic; also S₈ rings, different crystal symmetry.
- **Plastic sulphur:** molten sulphur poured into cold water; amorphous; insoluble in water & CS₂.

Sulphur	Selenium	Tellurium
Sulphurous acid H ₂ SO ₃ . Sulphuric acid H ₂ SO ₄ Peroxomonosulphuric acid H ₂ SO ₅ (Caro's acid) Peroxodisulphuric acid H ₂ S ₂ O ₈ (Marshall's acid) Thio sulphuric acid H ₂ S ₂ O ₃ Dithionic acid H ₂ S ₂ O ₆ Pyrosulphuric acid H ₂ S ₂ O ₇	Selenious acid H ₂ SeO ₃ Selenic acid H ₂ SeO ₄	Tellurous acid H ₂ TeO ₃ . Telluric acid H ₂ TeO ₄ .

Fig. 09-15 · Allotropes of sulphur & sulphuric acid properties

- **Sulphuric acid — properties:** powerful **dehydrating agent** (strong affinity for water); **charring** of sugar/wood/paper → black carbon mass; moderately strong oxidising agent; decomposes carbonates, bicarbonates, sulphides, sulphites, thiosulphates, nitrites at room temperature:



- Salts like **chlorides, fluorides, nitrates, acetates, oxalates** are decomposed by hot conc. H₂SO₄, liberating their corresponding acids.

9.7 Group 17 — Halogens (F, Cl, Br, I)

Type XX' ₁ (n = 1) (with linear shape)	Type XX' ₃ (n = 3) (with T-shape)	XX' ₅ (n = 5) (with square pyramidal shape)	XX' ₇ (n = 7) with pentagonal bipyramidal shape)
ClF	ClF ₃	ClF ₅	
BrF BrCl	BrF ₃	BrF ₅	
ICl, IBr, IF	ICl ₃ , IF ₃	IF ₅	IF ₇

Fig. 09-16 · Hydrogen halides — comparative properties

- F and Cl — water purification & disinfectants; I — essential in medical treatment.
- **Reducing behaviour of HX:** HCl is not attacked by H₂SO₄; 2HBr + H₂SO₄ → 2H₂O + SO₂ + Br₂; 2HI + H₂SO₄ → 2H₂O + SO₂ + I₂.
- All three react with KMnO₄/K₂Cr₂O₇: 2KMnO₄ + 16HCl → 2KCl + 2MnCl₂ + 8H₂O + 5Cl₂; K₂Cr₂O₇ + 14HBr → 2KBr + 2CrBr₃ + 7H₂O + 3Br₂.

Halide comparison	Order
Dipole moment	HI < HBr < HCl < HF
Bond length	HF < HCl < HBr < HI
Bond strength	HI < HBr < HCl < HF

Halide comparison	Order
Thermal stability	$\text{HI} < \text{HBr} < \text{HCl} < \text{HF}$
Acid strength	$\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$
Reducing power	$\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$

- **Pseudohalides:** best known — CN^- . **Oxyacids of chlorine:** acidity rises with oxidation number — $\text{HClO}_4 > \text{HClO}_3 > \text{HClO}_2 > \text{HOCl}$.

9.8 Group 18 — Noble Gases

- He (balloons, cryogenics), Ar (lighting, welding). Inert because of: complete octet (stable configuration), high ionization energies, low electron affinity.

Molecule	Total electron pairs (BP + LP)	Hybridisation	Shape
XeF_2	5	sp^3d	Linear
XeF_4	6	sp^3d^2	Square planar
XeF_6	7	sp^3d^3	Distorted octahedral

Fig. 09-17 · Xenon compounds & their shapes

Xenon compound	Shape
XeO_3	Pyramidal
XeO_4	Tetrahedral
XeOF_4	Square pyramidal
XeO_2F_2	Distorted octahedral

Pseudohalide ions	Name
CN^-	Cyanide ion
OCN^-	Cyanate ion
SCN^-	Thiocyanate ion
SeCN^-	Selenocyanate ion
NCN^{2-}	Cyanamide ion
N_3^-	Azide ion
OMC^-	Fulminate ion

- Some important stable compound of Xenon

Fig. 09-18 · Uses of noble gases

- (Uses table consolidated with Unit 8.4 to avoid repetition.)
- **p-block trends recap:** electronegativity ↓ (size & shielding — $\text{F} > \text{I}$); atomic size ↑ ($\text{C} < \text{Pb}$); ionization energy ↓ ($\text{B} > \text{Ti}$).

Fig. 09-19 · Important compounds of the p-block — chart

Compound (group)	Uses	Properties
Ammonia NH_3 (15)	Fertilizers (ammonium nitrate), cleaning agents, nitric-acid production	Colourless pungent gas; highly water-soluble; strongly basic
Silicon dioxide SiO_2 (14)	Glass manufacture, cement, semiconductor material	Hard crystalline (quartz); water-insoluble
Sulfuric acid H_2SO_4 (16)	Phosphate fertilizers, battery acid, chemical synthesis	Strong, highly corrosive; strong affinity for water
Phosphoric acid H_3PO_4 (15)	Fertilizers (superphosphates), food additive (acidifier in soft drinks), rust removal	Weak, non-volatile; solidifies to crystals
Aluminium oxide Al_2O_3 (13)	Aluminium manufacture, abrasives, refractory materials	Very high m.p.; very hard; electrical insulator
Chlorine Cl_2 (17)	Water disinfection, PVC production, bleaching	Yellow-green gas; strong odour; reactive; toxic
Hydrofluoric acid HF (17)	Etching glass, cleaning metals, fluorine compounds	Weak but highly corrosive; dangerous to handle
Boric acid H_3BO_3 (13)	Antiseptic, insecticide, borosilicate glass	Weak acid; water-soluble; mild antiseptic
Dichloromethane CH_2Cl_2 (14)	Paint removers, degreasing, pharmaceutical industry	Volatile liquid; sweet odour; moderately toxic; non-flammable

9.9 d-Block Elements (Transition Metals)

(a) 3d-Series (First Transition Series)			
Atomic No.	Element	Symbol	Electron Configuration
21	Scandium	Sc	$[\text{Ar}]3d^14s^2$
22	Titanium	Ti	$[\text{Ar}]3d^24s^2$
23	Vanadium	V	$[\text{Ar}]3d^34s^2$
24	Chromium	Cr	$[\text{Ar}]3d^54s^1$
25	Manganese	Mn	$[\text{Ar}]3d^54s^2$
26	Iron	Fe	$[\text{Ar}]3d^64s^2$
27	Cobalt	Co	$[\text{Ar}]3d^74s^2$
28	Nickel	Ni	$[\text{Ar}]3d^84s^2$
29	Copper	Cu	$[\text{Ar}]3d^{10}4s^1$
30	Zinc	Zn	$[\text{Ar}]3d^{10}4s^2$

Fig. 09-20 · d-block — four transition series

- **Series:** 3d — Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn; 4d — Y, Zr, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Cd; 5d — La, Hf, Ta, W, Re, Os, Ir, Pt, Au, Hg; 6d — incomplete.

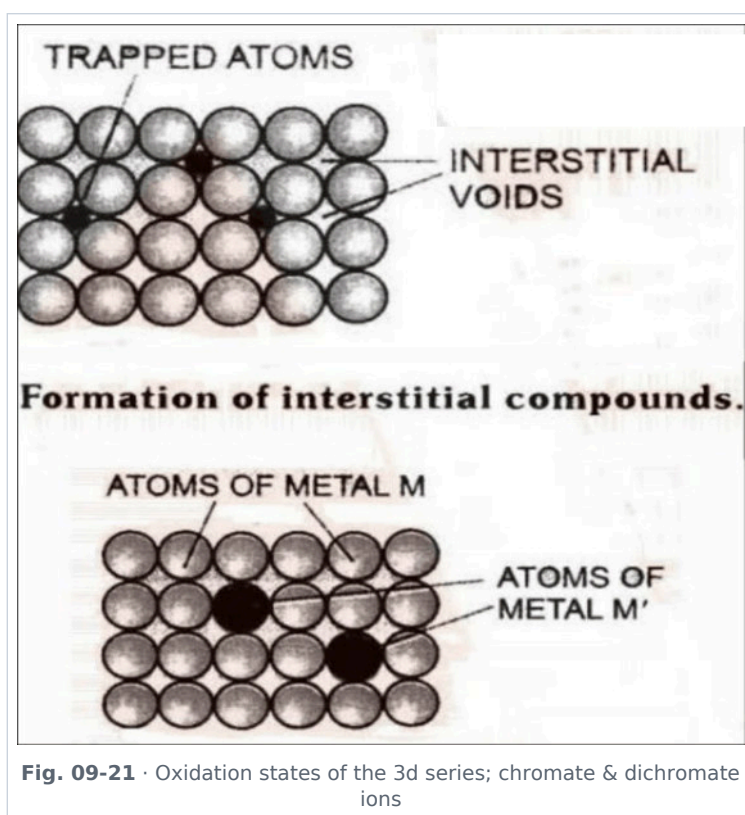
Electronic Configuration

- General: $(n-1)d^{1-10} ns^{1-2}$. Exceptions arise because (i) energy gap between $(n-1)d$ and ns is small, (ii) half/full-filled sets are extra stable — seen in 3d: **Cr & Cu**; 4d: Nb, Mo, Tc, Ru, Rh, Pd, Ag; 5d: Pt & Au; 6d: Rg.

Atomic Radii & Density

- Across 3d series: Sc→Mn radii decrease (nucleus–valence attraction dominates); Fe/Co/Ni nearly equal; Cu→Zn increase.
- Group trend: $4d > 3d$; $4d \approx 5d$ (**lanthanide contraction**).
- Densities $\sim 2\text{--}22\text{ g/cm}^3$; $Sc < Ti < V < Cr < Mn < Fe < Co < Ni = Cu > Zn$ (Zn low — large atomic volume); minimum density in 3d = Sc, maximum = Ni & Cu; $3d < 4d < 5d$; heaviest — **Os 22.51 and Ir 22.61 g/cm³**.

Oxidation States & Coloured Ions



- Transition metals show **variable oxidation states** (+1 to +8 typical) as ns and $(n-1)d$ electrons participate; zero state also occurs — Ni in $Ni(CO)_4$, Fe in $Fe(CO)_5$.
- **Colour:** partially-filled d -orbitals $\rightarrow$ d - d electronic transitions absorb visible light. **Cu^{2+} — blue; Cr^{3+} — green.**

Magnetic Properties

- **Paramagnetic** — unpaired d electrons (d^1 – d^9 ions); **diamagnetic** — all paired; **ferromagnetic** — strongest paramagnetism (Fe, Co, Ni). Magnetic moment: $\mu = \sqrt{n(n+2)}$ B.M. (n = unpaired electrons).

Interstitial Compounds & Alloys

Some important alloys-		
(a)	Bronze	Cu + Sn
(b)	Brass	Cu + Zn
(c)	Gun metal	Cu + Zn + Sn
(d)	German Silver	Cu + Zn + Ni
(e)	Stainless Steel	Cr + Ni
(f)	Invar	Ni + Fe
(g)	Alnico	Al + Ni + Co
(h)	Duralumin	Cu + Al + Mn
(i)	22 Carat gold	Au + Ag
(j)	18 Carat gold	Au + Ag + Cu

Fig. 09-22 · Formation of interstitial compounds — "trapped atoms"

- Small atoms (H, C, N) fit into lattice gaps of the d-block metal: **TiC** (extreme hardness — cutting tools), **MnN** (magnetic, electronics), **FeH** (hydrogen storage, fuel cells).

♣ **De-dup note:** The alloy-formation block was printed twice on source p. 104 — kept once.

- Alloys:** **brass** (Cu + Zn — malleable, acoustic); **alnico magnets** (Al + Ni + Co — strong magnets); **titanium alloys** (Ti + Al/V — aerospace, medical implants).

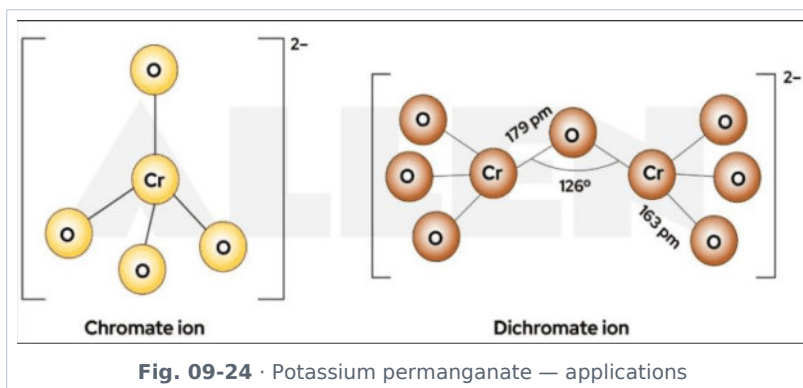
Complexes & Catalysis

- Ligands (lone-pair donors) form **coordinate covalent bonds** with the central metal ion. **Coordination number** 2-12; geometry — octahedral, square planar, tetrahedral etc.
- Examples: $[\text{Fe}(\text{CN})_6]^{3-}$ hexacyanoferrate(III); $[\text{Cu}(\text{NH}_3)_4]^{2+}$ tetraamminecopper(II); $[\text{PtCl}_4]^{2-}$ tetrachloridoplatinate(II).
- Complexes are coloured and catalytically active; some act as enzyme cofactors or therapeutic agents.
- Transition metals are exceptional **catalysts** — they activate molecules and enable hydrogenation to CO_2 conversion; vital in petroleum refining, chemical production and renewable energy.

Important Compounds of d-block

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
								+1	
	+2	+2	+2	<u>+2</u>	<u>+2</u>	<u>+2</u>	<u>+2</u>	<u>+2</u>	<u>+2</u>
<u>+3</u>	+3	<u>+3</u>	<u>+3</u>	+3	+3	+3	+3		
	<u>+4</u>	+4	+4	<u>+4</u>	+4	+4	+4		
		<u>+5</u>	+5	+5					
			<u>+6</u>	+6	+6				
				<u>+7</u>					

Fig. 09-23 · Potassium dichromate — properties & uses



♣ **De-dup note:** In the source, the $K_2Cr_2O_7$ property block was accidentally repeated under the $KMnO_4$ heading on p. 107; the two compound blocks are separated correctly here — content unchanged.

Compound	Highlights
$K_2Cr_2O_7$ — potassium dichromate	Bright orange-red crystals; powerful oxidising agent; water-soluble (orange solution); stable normally but decomposes on heating/with reducing agents; uses — qualitative analysis & titrations, old photographic processes, pigments & dyes, wood preservation (biocide); highly toxic & carcinogenic — inhalation or ingestion causes severe health issues; restricted in many countries
$KMnO_4$ — potassium permanganate	Dark purple crystals/powder; strong oxidising agent; uses — water treatment (removes impurities & odour for drinking/industrial water), dilute antiseptic for wounds/dermatitis/fungal infections, organic synthesis & industrial oxidiser, analytical reagent for identifying reducing agents

9.10 f-Block Elements (Inner Transition Metals)

1 H Hydrogen																	2 He Helium				
3 Li Lithium	4 Be Beryllium															5 B Boron	6 C Carbon	7 N Nitrogen	8 O Oxygen	9 F Fluorine	10 Ne Neon
11 Na Sodium	12 Mg Magnesium															13 Al Aluminium	14 Si Silicon	15 P Phosphorus	16 S Sulfur	17 Cl Chlorine	18 Ar Argon
19 K Potassium	20 Ca Calcium	21 Sc Scandium	22 Ti Titanium	23 V Vanadium	24 Cr Chromium	25 Mn Manganese	26 Fe Iron	27 Co Cobalt	28 Ni Nickel	29 Cu Copper	30 Zn Zinc	31 Ga Gallium	32 Ge Germanium	33 As Arsenic	34 Se Selenium	35 Br Bromine	36 Kr Krypton				
37 Rb Rubidium	38 Sr Strontium	39 Y Yttrium	40 Zr Zirconium	41 Nb Niobium	42 Mo Molybdenum	43 Tc Technetium	44 Ru Ruthenium	45 Rh Rhodium	46 Pd Palladium	47 Ag Silver	48 Cd Cadmium	49 In Indium	50 Sn Tin	51 Sb Antimony	52 Te Tellurium	53 I Iodine	54 Xe Xenon				
55 Cs Cesium	56 Ba Barium	57 La Lanthanum	72 Hf Hafnium	73 Ta Tantalum	74 W Tungsten	75 Re Rhenium	76 Os Osmium	77 Ir Iridium	78 Pt Platinum	79 Au Gold	80 Hg Mercury	81 Tl Thallium	82 Pb Lead	83 Bi Bismuth	84 Po Polonium	85 At Astatine	86 Rn Radon				
87 Fr Francium	88 Ra Radium	89 Ac Actinium	104 Rf Rutherfordium	105 Db Dubnium	106 Sg Seaborgium	107 Bh Bohrium	108 Hs Hassium	109 Mt Meitnerium	110 Ds Darmstadtium	111 Rg Roentgenium	112 Cn Copernicium	113 Nh Nihonium	114 Fl Flerovium	115 Mc Moscovium	116 Lv Livermorium	117 Ts Tennessine	118 Og Oganesson				

Firstly let's discuss the general differences between Lanthanides and Actinides in brief-

Lanthanides	Actinides
These elements range from atomic number 58 (Cerium) to 71 (lutetium).	These elements range from atomic number 90 (Thorium) to 103 (Lawrencium).
They are typically shiny metals, often silvery-white in appearance.	Most actinides are synthetic and highly radioactive.
Lanthanides share common properties due to similar electron configurations within the f-orbital.	Uranium and plutonium are the most well-known actinides due to their nuclear applications.

Fig. 09-26 · Lanthanides vs actinides — properties (continued)

Here's a summary table to highlight the differences between lanthanides and actinides:

Aspect	Lanthanides	Actinides
Electronic Configuration	4f sublevel partially filled	5f sublevel partially filled
Occurrence	More abundant, naturally found in Earth's crust	Predominantly synthetic, found in trace amounts due to radioactivity
Stability	Generally more stable, longer half-lives	Often radioactive, less stable
Applications	Electronics, catalysts, magnets, lighting	Nuclear reactors, research, nuclear weaponry
Position in Periodic Table	Sixth period	Seventh period
Properties	Relatively uniform properties and chemical behaviors	Greater variability in properties and behaviors

Fig. 09-27 · Periodic table with f-block placement

- Lanthanoids = **4f** series; actinoids = **5f** series; both are inner transition metals placed below the main table.

Lanthanides — properties

- Silvery-white, malleable, high conductivity; high density (varies across series); high m.p./b.p. (exceptions among lighter ones); unique magnetism (para/ferromagnetic at low T); tarnish slowly (oxide layer); **+3 most common** oxidation state; similar properties throughout.

Actinides — properties

- Mostly radioactive** (α , β , γ decay); very dense (U, Pu); high m.p./b.p.; **variable valency** from 5f filling — highest +7 (unstable) for Np & Pu; highest stable **+6 for U**; magnetism at low T; allotropy.
- Similarities:** inner transition metals; similar configurations & chemistry; many actinides radioactive (heavier lanthanides slightly so); applications — lanthanides in lighting, catalysts, magnets, electronics; actinides in nuclear reactors, research, weaponry.

Lanthanide	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
	Lanthanum	Cerium	Praseodymium	Niodymium	Promethium	Samarium	Europium	Gadolinium	Terbium	Dysprosium	Holmium	Erbium	Thulium	Ytterbium	Lutetium
Actinides	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
	Actinium	Thorium	Protactinium	Uranium	Neptunium	Plutonium	Americium	Curium	Berkelium	Californium	Einsteinium	Fermium	Mendelevium	Nobelium	Lawrencium

Fig. 09-28 · Lanthanide & actinide series strip

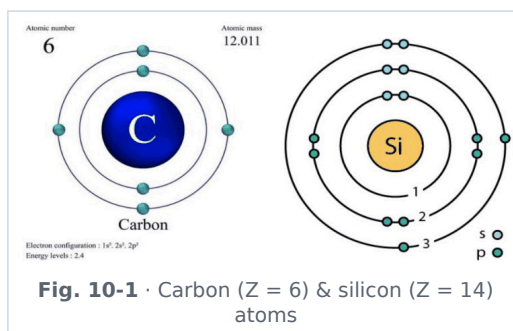
UNIT 10

Organic Chemistry — Basics & Hydrocarbons

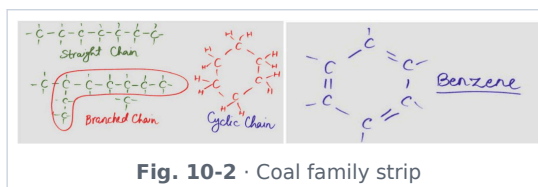
Consolidated from: Organic Chemistry Part-1 (pp. 126–137) + carbon intro merged from Acids-Bases lecture (p. 81)

10.1 Allotropy & Allotropes of Carbon

- Allotropy** — the property of elements to exist in different physical forms. e.g. coal, diamond & graphite (natural carbon allotropes); oxygen (O_2 & O_3), phosphorus (red & white), sulphur (rhombic & monoclinic).



- Coal types:** (a) Peat → (b) Lignite → (c) Bituminous → (d) Anthracite.



Graphite

- The **only non-metal that conducts electricity**.
- Uses:** electrodes; lubricant (slippery); **moderator** in nuclear reactors; pencil lead.

10.2 Organic Compounds & the Uniqueness of Carbon

- Organic compounds** — compounds containing carbon (excluding oxides of carbon, metallic carbonates, metal cyanides, metal carbides etc.).

🔗 **De-dup note:** The definition, Wöhler's urea synthesis and "nature of carbon" appeared both on source p. 81 (Acids-Bases lecture) and pp. 126–127 (Organic lecture) — kept once (see also Unit 7.6).

- Unique nature of carbon:**
 - Tetravalency** — valency 4: one carbon atom forms 4 covalent bonds.
 - Catenation** — "self-linking ability of carbon": bonds with other carbon atoms to form straight, branched and cyclic chains. C, Si, Ge, S, P, N also catenate.

Carbon Counting

Carbons	1	2	3	4	5	6	7	8	9	10
Root	Meth	Eth	Prop	But	Pent	Hex	Hept	Oct	Non	Dec

Structural Representation

- Complete (all bonds shown), condensed and bond-line structural formulae:

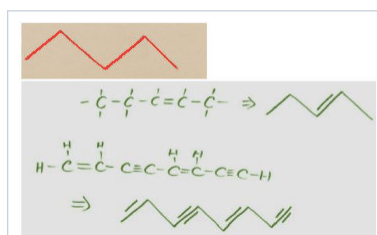
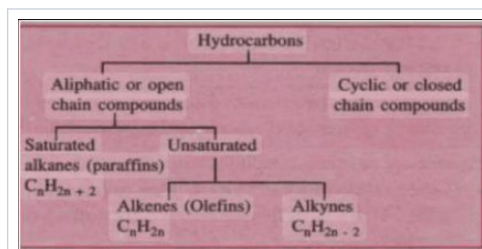
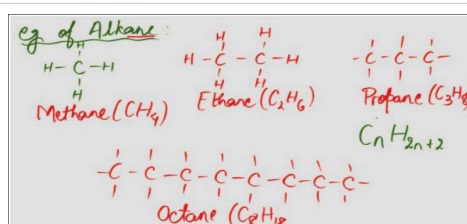
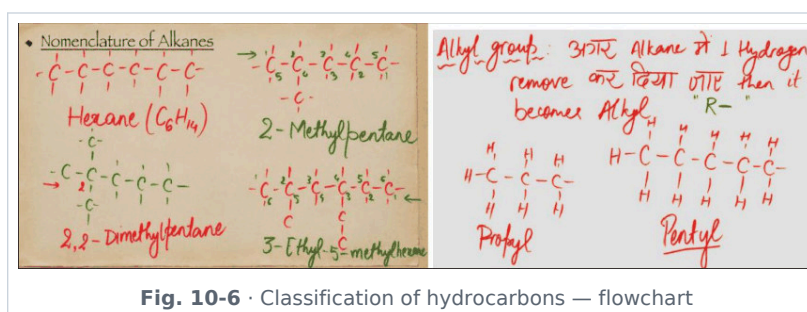


Fig. 10-3 • Methane — complete structural formula**Fig. 10-4** • Pentane — complete & condensed form**Fig. 10-5** • Pentane — bond-line form

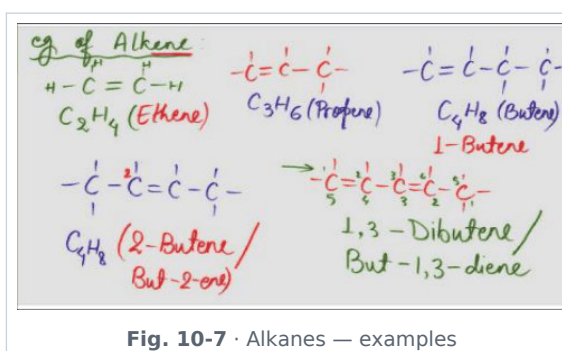
10.3 Hydrocarbons

- Compounds containing only **carbon & hydrogen** — CH_4 (methane), C_2H_4 (ethene), C_2H_2 (ethyne).

**Fig. 10-6** • Classification of hydrocarbons — flowchart

Saturated Hydrocarbons (Alkanes, $\text{C}_n\text{H}_{2n+2}$)

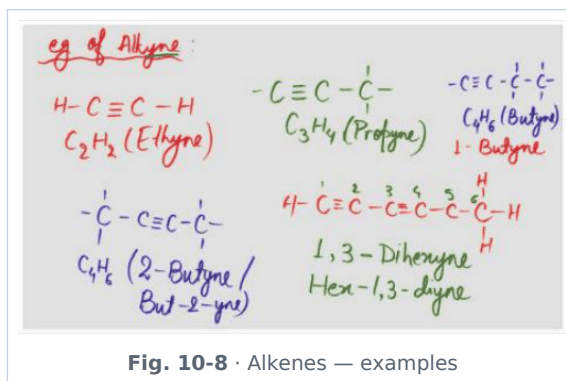
- Only **single covalent bonds**; also called **paraffins** (less reactive); first member — **methane**.

**Fig. 10-7** • Alkanes — examples

- Homologous series:** compounds with similar physical & chemical properties, successive members differing by a $-\text{CH}_2-$ unit (alkanes, alkenes, alkynes, alcohols, aldehydes...).
- Properties of alkanes:** non-polar; weak van der Waals forces; first four members gases, C_5 – C_{17} liquids, C_{18} onwards solids; low reactivity.

Alkenes (C_nH_{2n})

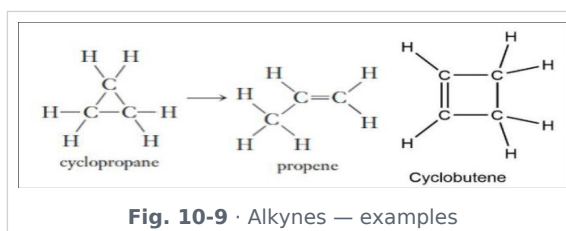
- At least one **double bond**; e.g. ethene, propene, butene, pentene — first member **ethene (ethylene)**.



- **Properties:** non-polar; first 3 members gases, C₅–C₁₈ liquids, C₁₉+ gases (source wording); colourless & odourless (exception ethene); insoluble in water.

Alkynes (C_nH_{2n-2})

- At least one **triple bond**; e.g. ethyne, propyne, butyne, pentyne — first member **ethyne (acetylene)**.



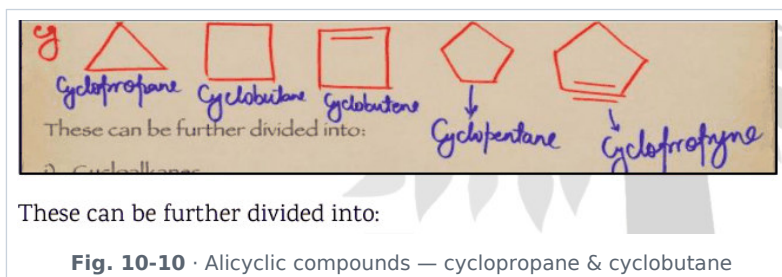
- **Properties:** non-polar; first 3 gases, C₅–C₁₃ liquids, C₁₄+ solids; odourless & colourless (exception ethyne); insoluble in water.

10.4 Cyclic (Closed-Chain) Compounds

- Ring of carbon atoms (≥ 3) instead of a chain; divided into **alicyclic** and **aromatic** compounds.

Alicyclic Hydrocarbons (non-aromatic)

- Cyclic hydrocarbons with single/double/triple bonds arranged in any (non-alternate) fashion; e.g. cyclopropane, cyclobutane.

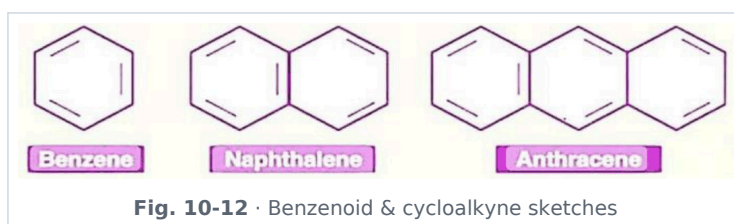


- **Homocyclic / carbocyclic** — ring of only carbon atoms → cycloalkanes (single bonds), cycloalkenes (double), cycloalkynes (triple).
- **Heterocyclic** — ring contains atoms other than carbon.



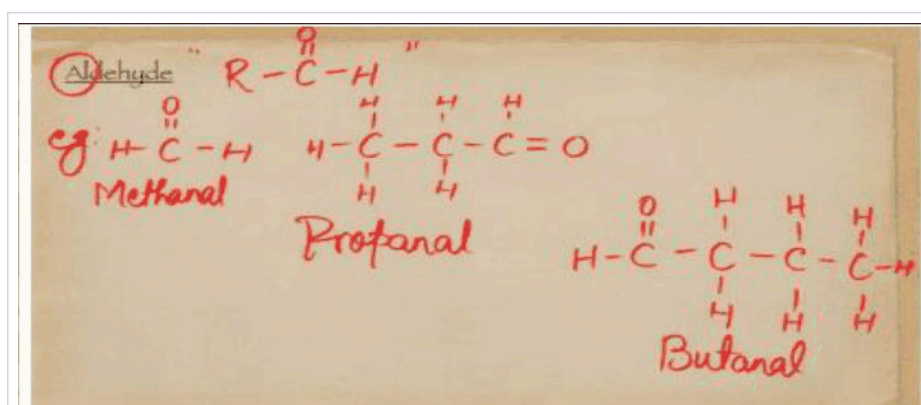
Aromatic Hydrocarbons (Arenes)

- Cyclic with **alternating single & double bonds**; most have **pleasant smell**; divided into **benzenoids** (contain benzene ring), **non-benzenoids** (no benzene ring) and **heterocyclic aromatics** (other atoms in ring).



10.5 Functional Groups

- An atom/group that, when added to an organic compound, changes its properties — e.g. R-OH (alcohol), aldehydes, ketones, carboxylic acids.

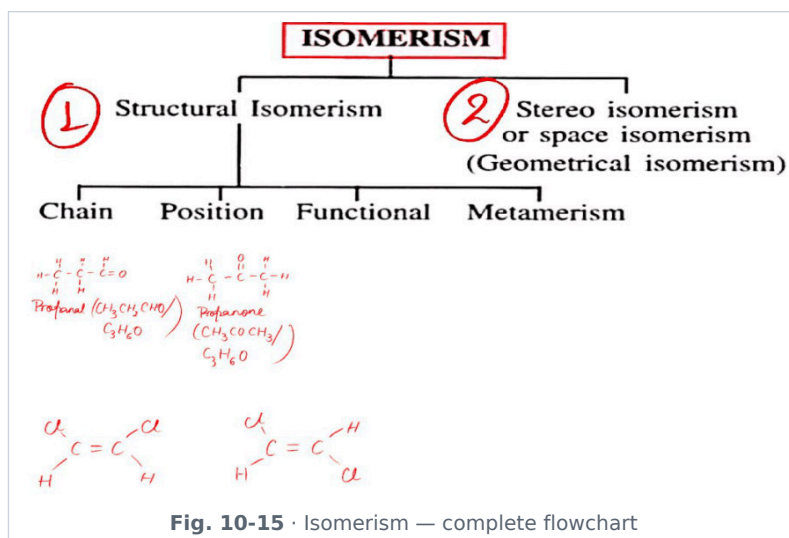


Functional group	General formulae	Types of Organic compounds	Suffix
Halide - X (F, Cl, Br, I)	R-X	Haloalkanes	ane
Hydroxyl - OH	R-OH	Alcohols	ol
Aldehyde -CHO	$\text{H} \diagup \text{C}=\text{O}$ $\text{R} \diagdown$	Aldehydes	al
Carboxyl - COOH	$\text{R}-\text{C} \begin{matrix} \text{O} \\ \parallel \\ \text{O}-\text{H} \end{matrix}$	Carboxylic acids	oic acid
Keto $\begin{matrix} \text{O} \\ \parallel \\ -\text{C}- \end{matrix}$	$\text{R}-\text{C}(=\text{O})-\text{R}'$	Ketones	one
Ethers $\begin{matrix} & & \\ -\text{C}-\text{O}-\text{C}- \\ & & \end{matrix}$	R-O-R'	Ethers	oxy

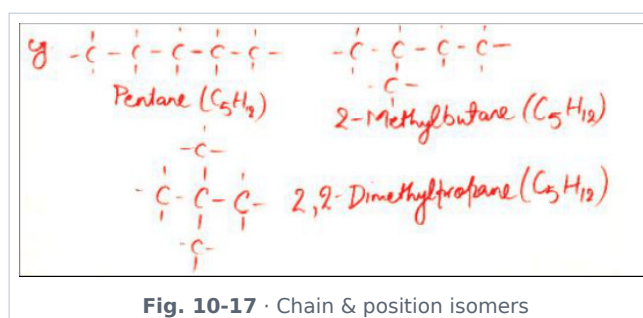
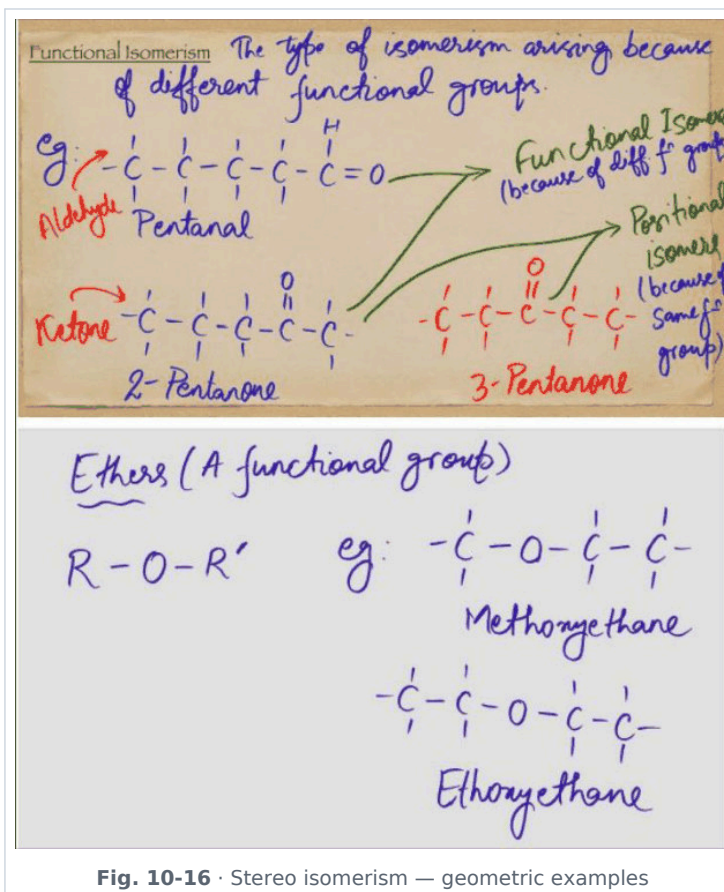
Fig. 10-14 · Functional groups — structural sketches

10.6 Isomerism

- Compounds with the **same molecular formula but different arrangement of atoms** show isomerism.



- **Chain isomerism** — difference in parent & branched chains.
- **Position isomerism** — different position of the substituent/double bond.
- **Functional isomerism** — same formula, different functional group.
- **Stereo isomerism** — same formula & structure, different **geometrical orientation** (e.g. cis-trans).



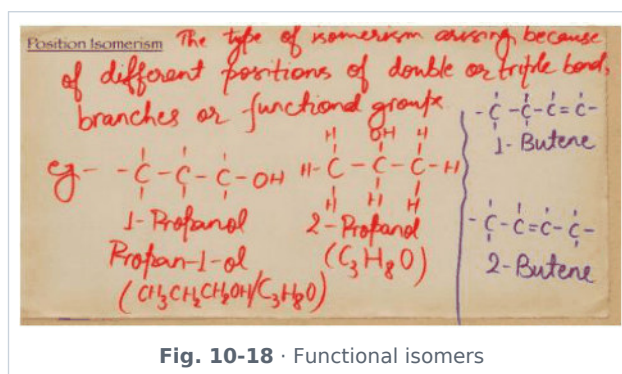


Fig. 10-18 · Functional isomers

10.7 Chemical Properties of Carbon Compounds

Combustion

- Burning in presence of $\text{O}_2 \rightarrow \text{CO}_2$ + water + heat & light energy:

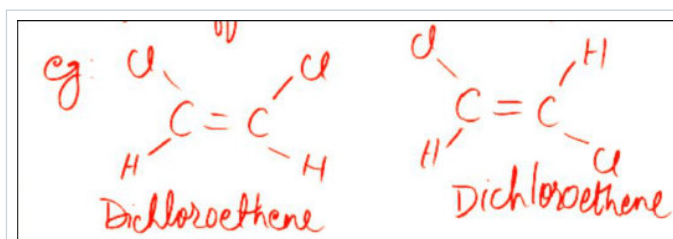
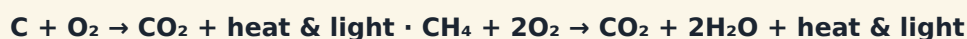


Fig. 10-19 · Combustion equations

Oxidation (addition of oxygen)

- Ethanol $\rightarrow$ ethanoic acid with **acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or alkaline KMnO_4** — both strong oxidising agents.

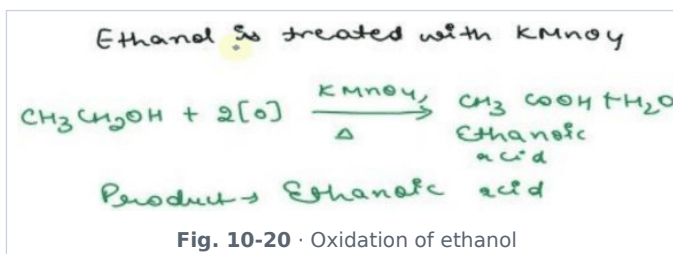


Fig. 10-20 · Oxidation of ethanol

Addition Reaction / Hydrogenation

- Addition of hydrogen (in presence of **Ni or Pd catalyst**) to unsaturated molecules — used for **hydrogenation of vegetable oils**.

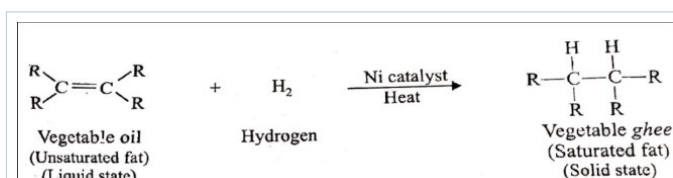


Fig. 10-21 · Addition (reduction) reaction

Substitution Reaction

- One atom/group replaced by another (characteristic of saturated compounds — e.g. methane chlorination).

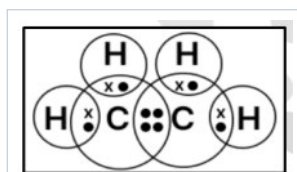


Fig. 10-22 · Methane — shared electron pairs (electron-dot structure)

10.8 Methane, Ethene, Ethyne, Ethanol & Ethanoic Acid

Methane (CH₄) — 1st alkane

- Also called **marsh gas**; $\approx$ 80–85 % of natural gas; colourless, odourless; produced in paddy fields.
- Preparation:** $\text{Al}_4\text{C}_3 + \text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3 + \text{CH}_4 \uparrow$; $\text{CH}_3\text{COONa} + \text{NaOH} \xrightarrow{\Delta} \text{Na}_2\text{CO}_3 + \text{CH}_4$ (sodium ethanoate route).

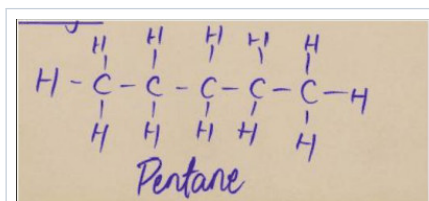


Fig. 10-23 · Methane — Lewis structure

Ethene (C₂H₄) — 1st alkene

- Plant hormone — responsible for **ripening of fruits**; colourless with sweet smell.
- Preparation:** ethanol $\xrightarrow{\text{conc. H}_2\text{SO}_4, \Delta}$ ethene + H₂O — conc. H₂SO₄ acts as a **dehydrating agent**.

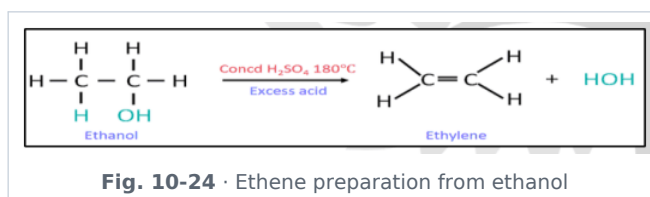


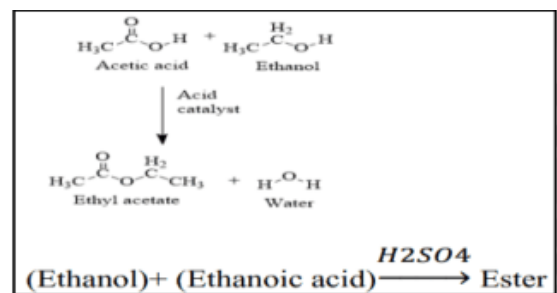
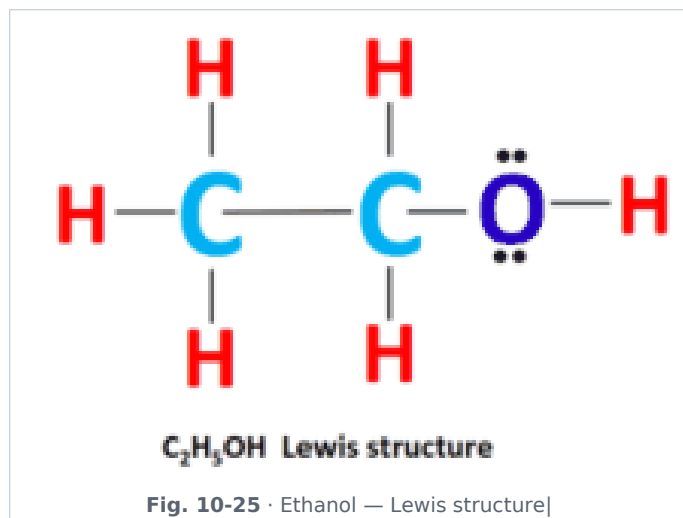
Fig. 10-24 · Ethene preparation from ethanol

Ethyne (C₂H₂) — 1st alkyne

- Structure H-C $\equiv$ C-H; colourless gas with a smell; lighter than air.
- Preparation:** $\text{CaC}_2 + 2\text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2 + \text{C}_2\text{H}_2 \uparrow$.

Ethanol (C₂H₅OH)

- Liquid at room temperature; common organic solvent; consumption causes drunkenness — 100 % (absolute) ethanol is lethal.
- Reactions:**
 - With sodium: $2\text{C}_2\text{H}_5\text{OH} + 2\text{Na} \rightarrow 2\text{C}_2\text{H}_5\text{ONa}$ (sodium ethoxide) + H₂.
 - Neutral (pH $\approx$ 7).
 - Dehydration with hot conc. H₂SO₄ gives unsaturated ethene.

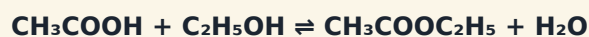


Note:-

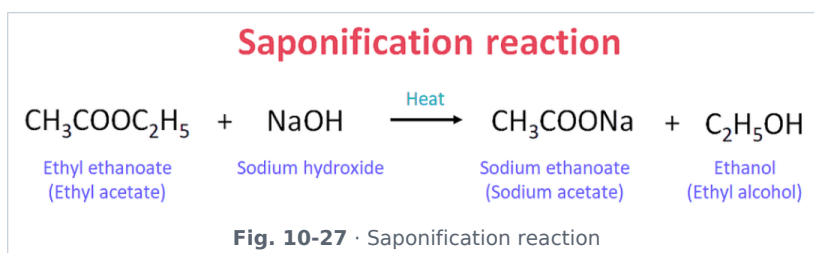
Fig. 10-26 · Ethanol + sodium → sodium ethoxide + H_2

Ethanoic Acid (CH_3COOH) — carboxylic acid

- Called **glacial acetic acid** — low melting point ($\sim 15\text{--}17^\circ\text{C}$); 5–8 % solution = **vinegar**.
- **Esterification** — alcohol + carboxylic acid $\xrightarrow{-\text{H}_2\text{SO}_4}$ **ester** (sweet-smelling) + water:



- **Saponification** — opposite process: ester + base → alcohol + metallic salt of carboxylic acid (**soap formation reaction**).



- **With bases:** $\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa}$ (sodium ethanoate) + H_2O .
- **With carbonates:** $2\text{CH}_3\text{COOH} + \text{K}_2\text{CO}_3 \rightarrow 2\text{CH}_3\text{COOK}$ (potassium ethanoate) + $\text{H}_2\text{O} + \text{CO}_2$.

UNIT 11

Polymers

Consolidated from: Polymers & Chemistry in Everyday Life lecture — polymer part (pp. 139–147)

11.1 Polymers & Polymerisation

- Polymer** (Greek — "many units"): a chemical substance of **high molecular mass** formed by combining a large number of simple molecules (**monomers**).

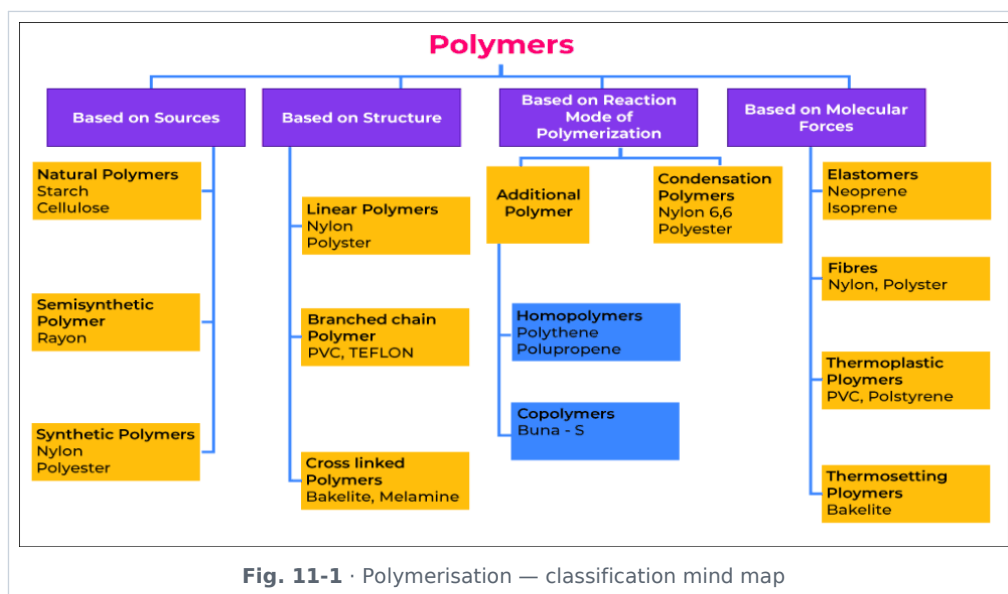
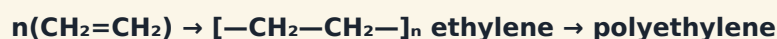


Fig. 11-1 · Polymerisation — classification mind map

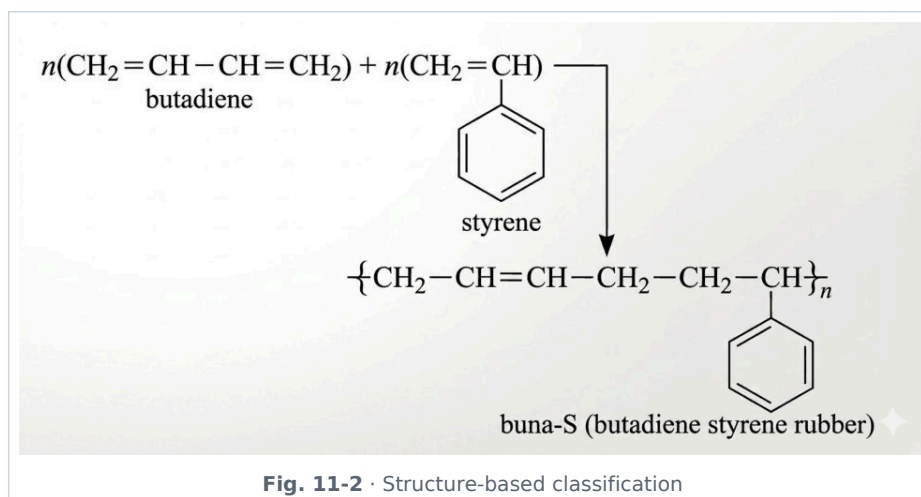
- Polymerisation** — the process by which monomers combine & transform into polymers: $n [\text{Monomer}] \rightarrow \text{Polymer}$.
- Polymers vs macromolecules:** all polymers are macromolecules, but not conversely — chlorophyll ($\text{C}_{55}\text{H}_{72}\text{O}_5\text{N}_4\text{Mg}$) is a macromolecule **without monomer units**, so not a polymer.

11.2 Classification of Polymers

Based on Source

Type	Idea	Examples
Natural	Occur in plants/animals	Starch (reserve food of plants); cellulose (structural material of plants); proteins (building blocks of animals); natural rubber (latex of *Hevea brasiliensis*)
Synthetic	Prepared in the laboratory	Polythene, synthetic rubber, PVC, nylon-66, teflon, orlon
Semisynthetic	Natural polymers modified artificially	Cellulose acetate (rayon), vulcanised rubber

Based on Structure



- **Linear** — long close-packed chains → high density, tensile strength, m.p.: HDPE, nylon, polyesters.
- **Branched** — (source figure) chain branches lower packing: LDPE.
- **Cross-linked / network** — 3-D network, hard, rigid, brittle: bakelite, glyptal, melamine-formaldehyde.

Based on Molecular Forces

Class	Forces	Character	Examples
Elastomers	Weakest	Rubber-like; stretchable; few cross-links let them retract	Natural rubber, buna-S, buna-N
Fibres	Strong (H-bonding)	Thread-like, woven; high tensile strength; crystalline; sharp m.p.	Nylon-66, terylene, polyacrylonitrile
Thermoplastics	Intermediate (van der Waals)	Melt on heating, set on cooling — castable into moulds; plasticizers (high-boiling esters/haloalkanes) soften them	Polythene, polystyrene
Thermosetting	Cross-linked on heating	Semi-fluid low-mass resins that harden irreversibly into 3-D infusible mass	Bakelite, melamine-formaldehyde, urea-formaldehyde

Based on Mode of Polymerisation

- **Addition polymers** — monomers with double/triple bonds add up; empirical formula same as monomer.
Homopolymers — single monomer (polythene from ethene); *copolymers* — two or more monomers (buna-S).
- **Condensation polymers** — monomers combine with elimination of small molecules (H_2O , alcohol, HCl) — e.g. **nylon-6,6** from hexamethylenediamine + adipic acid.

11.3 Types of Polymerisation

Chain-Growth (Addition) Polymerisation

- Reactive intermediates — **free radical, carbocation or carbanion**; monomers are unsaturated (alkenes, alkadienes & derivatives).
 - **Free-radical** — monosubstituted alkenes; catalysts benzoyl peroxide, H_2O_2 , t-butyl peroxide.
 - **Cationic** — carbocation intermediates via Lewis acids (BF_3 , AlCl_3 , SnCl_4) or protonic acids (H_2SO_4 , HF); more stable carbocation → more reactive monomer.
 - **Anionic** — carbanion intermediate; **strong bases** as initiators.

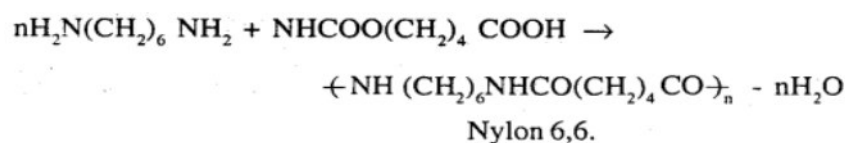


Fig. 11-3 · Nylon-66 — condensation of hexamethylenediamine with adipic acid

Step-Growth (Condensation) Polymerisation

- Stepwise condensation eliminating small molecules (H_2O , NH_3 , HCl , ROH) — e.g. adipic acid + hexamethylenediamine; phenol + formaldehyde.

	Chain growth polymerisation	Step growth polymerisation
1.	In this type of polymerisation, repeated addition of monomer units to a growing polymer chain occurs.	Polymerisation occurs by a series of condensation reactions with the elimination of simple molecules such as H_2O , HCl , NH_3 , etc.
2.	The monomers involved are unsaturated compounds such as alkenes, alkadiene, etc. e.g., ethene, vinyl chloride, etc.	Monomers involved are saturated compound having two or more functional groups. e.g., terephthalic acid, ethylene glycol, etc.

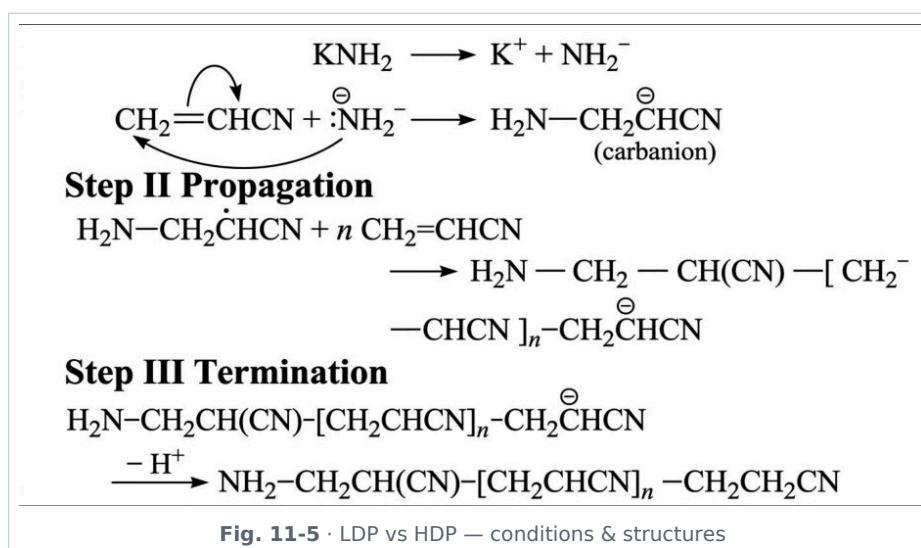
1. Polythene

Fig. 11-4 · Step-growth polymerisation

11.4 Important Addition Polymers

Polythene

- LDP (low density)** — tough, flexible, transparent, chemically inert, poor electrical conductor; moderate tensile but good tearing strength; used for insulation of electric wires, squeeze bottles, toys, flexible pipes.
- HDP (high density)** — close packing → high density; chemically inert, tougher & harder; containers, housewares, bottles, toys, electric insulation.



Polystyrene

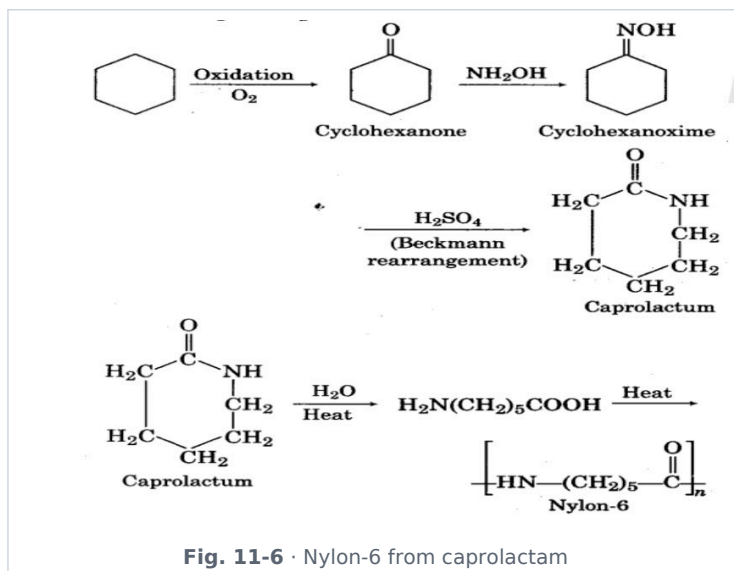
- Monomer styrene; thermoplastic; toys, radio & TV cabinets; rain coats, electrical insulation; hard, resistant to heat & chemicals. (Foamed → thermocol, see Unit 12 polymers list.)

Polypropylene (PP)

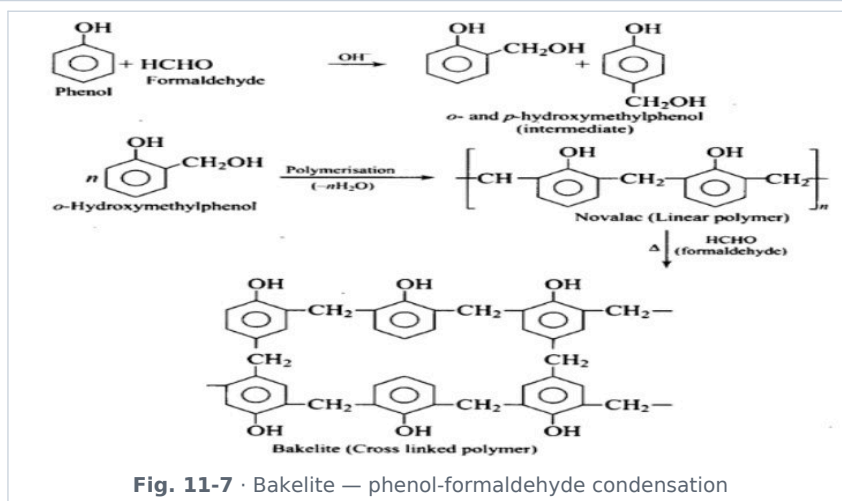
- Propylene polymerised with **Ziegler-Natta catalyst**.

Polyamides — Nylon-66 & Nylon-6

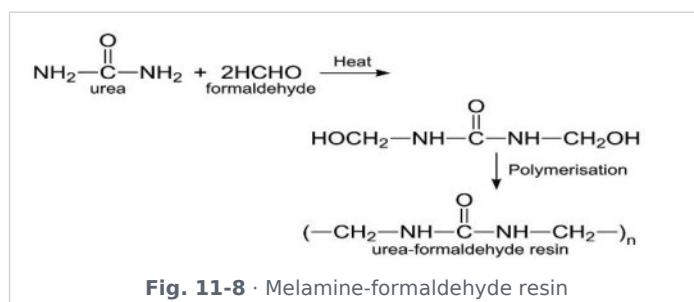
- Numbers = carbon counts of diamine & dibasic acid (6 & 6 → **nylon-66**).
 - Nylon-66** — linear; very high tensile strength; abrasion-resistant; sheets, brush bristles, textiles.
 - Nylon-6** — from caprolactam heated with water at high temperature.



11.5 Resins (Condensation Polymers)



- **Phenol-formaldehyde (Bakelite)** — condensation in acid/base; methylene bridges at ortho/para positions; linear or cross-linked. Uses — combs, photo records, electrical switches; soft bakelite as binding glue for laminated wood, varnishes, lacquers.



- **Melamine-formaldehyde** — copolymer of melamine (heterocyclic triamine) + formaldehyde; very hard & tough; **crockery** that survives drops.
- **Urea-formaldehyde resin** — (structure in figure) used as adhesive/moulding material.

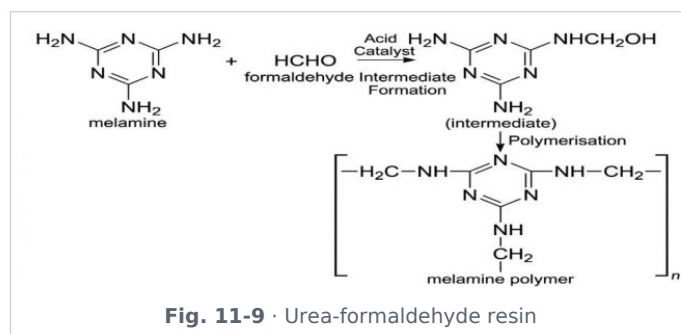


Fig. 11-9 · Urea-formaldehyde resin

11.6 Natural & Synthetic Rubber

- **Natural rubber** — coiled linear **cis-1,4-polyisoprene**; double bonds between C2-C3 of isoprene (cis configuration) → coiled chain that stretches and recoils; only van der Waals forces (no polar groups).
- Soft & sticky; usable only 10-50 °C (softens hot, brittle cold); high water absorption; attacked by oxidising agents & solvents — limited commercial use as-is.

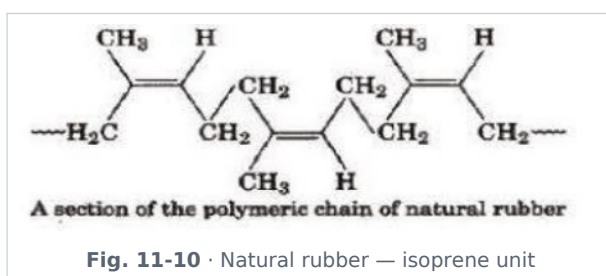


Fig. 11-10 · Natural rubber — isoprene unit

- **Vulcanisation** — heating rubber with **sulphur (3-5 %)** + **zinc oxide at ~110 °C for 20-30 min** (ZnO accelerates) introduces **-S-S- polysulphide crosslinks** between chains — limits chain motion → stronger, elastic material.

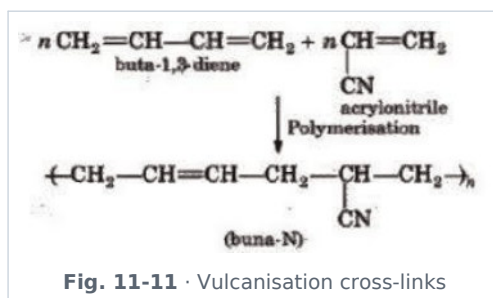


Fig. 11-11 · Vulcanisation cross-links

- **Neoprene** — polymer of chloroprene (synthetic rubber); conveyor belts, gaskets, hoses.
- **Buna-N** — copolymer of buta-1,3-diene + acrylonitrile; oil-resistant.

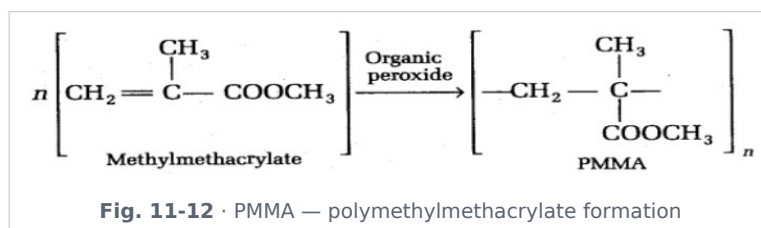
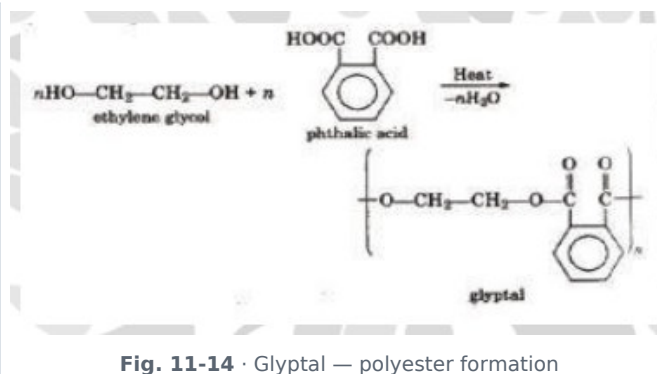
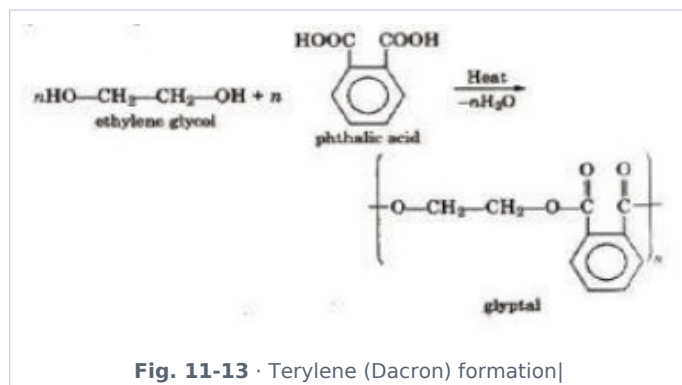


Fig. 11-12 · PMMA — polymethylmethacrylate formation

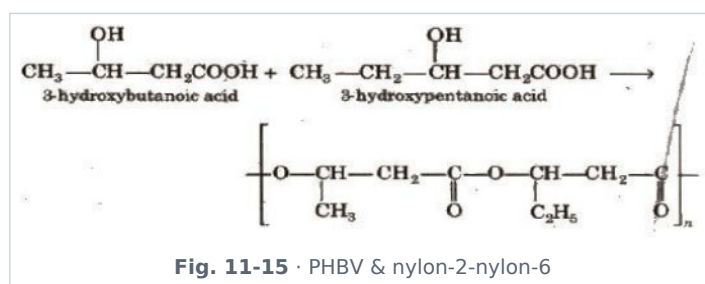
11.7 Polyesters & Biodegradable Polymers

- **Polyesters** — polymers with ester linkages (e.g. dacron).
 - **PMMA (polymethylmethacrylate)** — from methylmethacrylate + organic peroxide; sold as lucite, acrylite, plexiglass, perspex; hard, transparent, resists light/heat/ageing; unbreakable lights, protective coatings, dentures, aircraft windows.
 - **Glyptal** — cross-linked polyester (ethylene glycol/glycerol + phthalic acid); thermosetting; tough non-flexible film → paints & lacquers.
 - **Terylene (Dacron)** — ethylene glycol + terephthalic acid, 420-460 K, zinc acetate + antimony trioxide catalyst; resistant to chemicals & biology; blended with wool/cotton (terycot, terywool, terysilk); magnetic

tapes, conveyor belts, aprons.



- **Biopolymers & biodegradable polymers:** synthetic polymers are mostly non-biodegradable (polythene bags!). Biopolymers from biological systems — **polysaccharides, proteins, nucleic acids** — decompose/hydrolyse by enzymes → biodegradable. **Aliphatic polyesters** are common biodegradable polymers.
 - **PHBV** — copolymer of 3-hydroxybutanoic acid + 3-hydroxypentanoic acid.
 - **Nylon-2-nylon-6** — alternating polyamide copolymer of glycine ($\text{H}_2\text{N}-\text{CH}_2-\text{COOH}$) + aminocaproic acid ($\text{H}_2\text{N}(\text{CH}_2)_5\text{COOH}$); biodegradable.



11.8 Some More Important Polymers

Polymer	Composition / fact	Use
Saran	Copolymer of vinyl chloride	Wrapping food materials
ABS rubber	Acrylonitrile + buta-1,3-diene + styrene—	
Bubble gum	Styrene-butadiene rubber—	
Dynel	Vinyl chloride + acrylonitrile	Human hair wigs
Silk	Natural thread-like polyamide fibre from silkworm cocoons—	
Thermocol	Foamed plastic — air blown through molten polystyrene/polyurethane—	
Superglue	Methyl- α -cyanoacrylate polymer (anionic polymerisation)—	

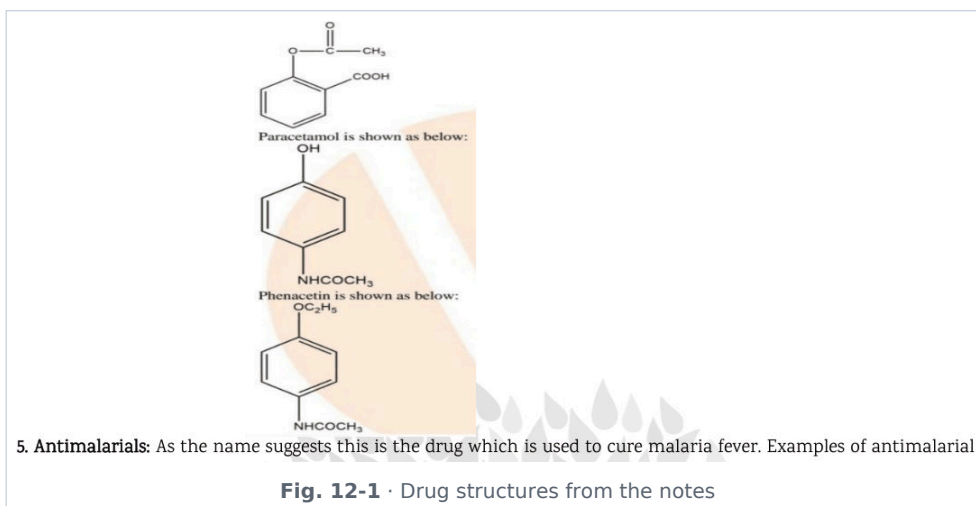
UNIT 12

Chemistry in Everyday Life & Fuels

Consolidated from: Everyday-life half of the lecture — medicines, dyes, food chemicals, soaps & detergents, fuels (pp. 147–157)

12.1 Medicine & Drugs

- **Medicine/drug** — chemical substances that treat diseases or reduce pain. Treatment with compounds that destroy microorganisms without harming the body = **chemotherapy**; the compounds = **chemotherapeutic agents**.



Class	Action	Examples
Antiseptics	Prevent/destroy growth of harmful microorganisms	Dettol (chloroxylenol + terpineol), Savlon, acriflavine, iodine, methylene blue, mercurochrome; bithional added to soaps
Disinfectants	Destroy microorganisms completely; toxic to living tissues	Floor/sanitary/cloth sterilisation; 1 % phenol = disinfectant, 0.2 % = antiseptic
Analgesics	Painkillers	Narcotic — morphine (opium alkaloid, habit-forming, causes sleep/unconsciousness at high doses); non-narcotic — aspirin, ibuprofen, naproxen (also antipyretic)
Antipyretics	Control fever	Aspirin
Antimalarials	Cure malaria fever	Quinine, chloroquine, paraquine, primaquine
Tranquilizers	Act on CNS; calming; psychotherapeutic drugs	Sedatives/hypnotics (reduce nervous tension) — reserpine, barbituric acid & derivatives luminal, seconal; mood elevators/antidepressants (for depressed patients) — benzedrine (amphetamine)
Anaesthetics	Relieve pain / local insensibility; general = unconsciousness for major surgery, local = small body portion for minor operations	General — nitrous oxide, ethylene, cyclopropane (gas); chloroform, divinyl ether, sodium pentothal (liquid); local — xylocaine (jelly), ethyl chloride (spray), procaine (injection)
Antibiotics	Produced by microorganisms (fungi, bacteria, mould); stop growth of others	Penicillin (pneumonia, bronchitis, abscesses, sore throat); streptomycin (tuberculosis); chloromycetin (typhoid, meningitis, pneumonia, diarrhoea, dysentery); tetracycline (acute fever, trachoma, dysentery, urinary-tract infection)
Sulpha drugs	Great antibacterial power; sulfanilamide derivatives	Sulphathiazole (severe infections); sulpha pyridine (pneumonia); sulfadiazine (dysentery, urinary & respiratory infections)

12.2 Dyes

- **Dyes** — coloured substances used in solution or dispersion on substrates (cotton, wool, silk, polyester, nylon, paper, leather, hair, fur, plastics) to give a coloured, shiny appearance.
- **Colour & theory:** colour depends on light absorption — a compound absorbing in the visible region shows the complementary colour of what it reflects (a dye absorbing blue looks yellow).
 - **Chromophores** — groups that absorb light and produce colour (nitro, azo groups etc.).
 - **Auxochromes** — groups that do not absorb light themselves (unlike chromophores) but **deepen the colour** when mixed with coloured compounds — -OH, -NH₂, -Cl, -COOH.

Classification by Source

- **Natural dyes** — from plants (nature): **alizarin, indigo**.
- **Synthetic dyes** — man-made in the laboratory: malachite green, orange-I, orange-II, aniline yellow.

Constitution

- Nitro dyes — martius yellow; **azo dyes** — aniline yellow, methyl orange, orange-I, congo red; **triphenylmethane dyes** — malachite green, magenta; **indigoid dyes** — indigo, indigosol; **anthraquinone dyes** — alizarin; **phthalein dyes** — phenolphthalein.

Applications (by how they attach)

Class	Mechanism / note	Examples
Acid dyes	Sodium salts of azo dyes bearing sulphonie/carboxylic acid; dye wool, silk, polyurethane; not cotton; nylon's free amino groups give high affinity	Orange-I, orange-II, congo red, methyl orange, methyl red
Basic dyes	Salts of azo/triphenylmethane dyes with amino (auxochrome) groups; attach to anionic sites of fabric; for polyesters & reinforced nylons	Aniline yellow, butter yellow, malachite green, chrysoidine G
Direct dyes	Water-soluble; applied directly from aqueous solution; need H-bonding fabrics (cotton, wool, silk, rayon, nylon)	Congo red, martius yellow
Fibre-reactive dyes	Attach by irreversible reaction with -OH/-NH ₂ of fibre (cotton, wool, silk); very fast, long-lasting colour	2,4-dichloro-1,3,5-triazine derivatives
Insoluble azo dyes	Coupling of phenols/naphthols/aminophenols adsorbed on fabric (with polyurethanes, polyacrylonitrile, leather); ≈60 % of all dyes; toxic — never for food	Cosmetics, drugs, biological stains/indicators
Ingrain (developed) dyes	Insoluble in water; produced in situ on the fabric surface by coupling reactions	Para red
Vat dyes	Insoluble; first reduced to colourless soluble form (alkaline sodium hydrosulphite), applied, then re-oxidised to the coloured form (air/chromic acid); dye wool	Indigo, indigosol O
Mordant dyes	Fabric first treated with metal-ion mordant that binds dye & fibre; different metals → different colours; used for wool	Alizarin (rose-red / turkey red)

🔗 **De-dup note:** On source p. 150 the direct-dyes paragraph was accidentally printed twice — kept once. All classes A–H are preserved above.

12.3 Rocket Propellants

- **Rocket propellants** give the push that places rocket satellites into space. Each propellant combines two compounds: (a) an **explosive compound = fuel** and (b) an **oxidizer**.

Conditions for a Good Propellant

- (1) **Ashless** — burning leaves no ash; (2) burning produces a **large volume of gases per gram** of fuel; (3) combustion proceeds at a **fast rate**.

Classification (by physical state of fuel + oxidizer)

Type	Fuel	Oxidizer	Notes
Solid — composite	Polymeric binder: polyurethane or polybutadiene	Ammonium perchlorate—	
Solid — double base	Nitrocellulose + nitroglycerine (both in one mass)—	Once ignited they burn at a predetermined rate — no start/stop capability	
Liquid — monopropellant	Single liquid acting as fuel & oxidizer — nitromethane, methyl nitrate—	Higher thrust than solids; thrust controllable by switching flow on/off	
Liquid — bipropellant	Kerosene, alcohol, hydrazine, monomethyl hydrazine (MMH), liquid hydrogen	Liquid oxygen, nitrogen tetroxide, nitrous acid	Same advantages as monopropellants
Hybrid	Solid fuel — acrylic rubber	Liquid oxidizer — nitrous oxide—	

- **Specific impulse (I_s)** — defines the superiority/performance of a propellant: performance $\propto$ **flame temperature (T)** and inversely $\propto$ average molecular mass (M) of product gases — hotter flame & lighter exhaust gases $\rightarrow$ better propellant.

12.4 Chemicals in Food

- **Food additives** — chemicals that improve quality, appearance, taste, odour & nutritive value: preservatives, flavouring agents, artificial sweeteners, dyes, antioxidants, fortifiers, emulsifiers, antifoaming agents.
- **Antioxidants** — prevent oxidation of fats in packaged food (chips, biscuits, cereals, crackers); more reactive towards oxygen; slow free-radical ageing: **BHT** (butylated hydroxytoluene), **BHA** (butylated hydroxyanisole).
- **Artificial sweeteners** — for diabetics & calorie control (sucrose/fructose add calories & cause decay):

Sweetener	Key facts
Saccharin	Used as sodium/calcium salt; highly water-soluble; non-biodegradable; zero calorific value
Aspartame	Methyl ester of dipeptide (phenylalanine + aspartic acid); sugar substitute in soft drinks & cold foods
Alitame	Like aspartame but more stable
Sucralose	Trichloro derivative of sucrose; no calories, no tooth decay
Cyclamate	N-cyclohexyl sulfamate
L-Glucose	Sweet like D-sugars but no energy — our body lacks the enzymes to metabolise it

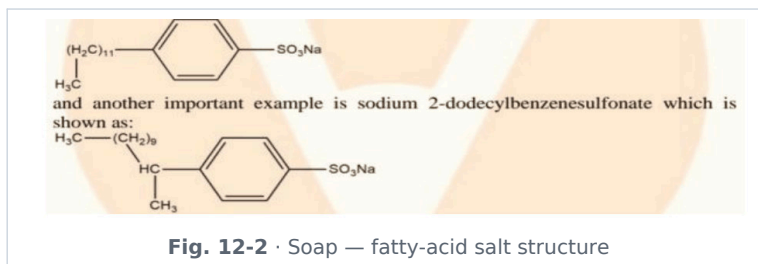
- **Preservatives** — protect food from bacteria, yeasts, moulds: **sodium benzoate**; sodium/potassium metabisulphite (jams, squashes, pickles — $\text{SO}_2 \rightarrow$ sulphurous acid inhibits growth); salts of propionic & sorbic acid.
- **Edible colours** — improve appearance, add no nutrition.

12.5 Soaps & Synthetic Detergents

Soaps

- Sodium/potassium salts of higher fatty acids — lauric ($\text{C}_{11}\text{H}_{23}\text{COOH}$), myristic ($\text{C}_{13}\text{H}_{27}\text{COOH}$), palmitic ($\text{C}_{15}\text{H}_{31}\text{COOH}$), stearic ($\text{C}_{17}\text{H}_{35}\text{COOH}$), oleic ($\text{C}_{17}\text{H}_{33}\text{COOH}$), linoleic ($\text{C}_{17}\text{H}_{31}\text{COOH}$), linolenic ($\text{C}_{17}\text{H}_{29}\text{COOH}$).

- **Biodegradable** — sewage microorganisms oxidise them fully to CO_2 → no water pollution. **Two disadvantages:**
 - Fail in **hard water** — $\text{Ca}^{2+}/\text{Mg}^{2+}$ form curdy white precipitate (scum).
 - Fail in **acidic solutions** — precipitate insoluble free fatty acids.



Synthetic Detergents

- Sodium salts of long-chain alkyl hydrogen sulphates / alkylbenzene sulphonates; work even in hard water (Ca/Mg salts soluble); not completely biodegradable → water pollution.

Fuels	Calorific Value (J/kg)
Cow dung cakes	6000 - 8000
Wood	17000 - 22000
Coal	25000 - 33000
Petrol	45000
Kerosene	45000
Diesel	45000
Methane	50000
CNG	52000

Fig. 12-3 · Classification of detergents

Type	Nature	Examples / facts
Anionic	Large anionic part	(a) Sodium alkyl sulphates — long-chain alcohols + conc. H_2SO_4 + NaOH ; purely biodegradable. (b) Alkylbenzene sulphonates — Friedel-Crafts alkylation + sulphonation + NaOH ; sodium dodecylbenzene sulphonate; effective in slightly acidic solutions (soaps fail there)
Cationic	Quaternary ammonium salts (Cl^- , Br^- , acetate); expensive; germicidal	Cetyltrimethylammonium chloride $[\text{CH}_3(\text{CH}_2)_{15}\text{N}^+(\text{CH}_3)_3]\text{Cl}^-$
Non-ionic	High-mass esters	Polyethylene glycol + stearic acid; or long-chain alcohols + excess ethylene oxide with base

- **Detergent pollution: branched** hydrocarbon chains — bacteria cannot attack the side chains → slow degradation & accumulation in lakes/ponds/river.
- **Biodegradable detergents** — straight chains (e.g. **sodium lauryl sulphate**) — microorganisms easily break them down.

🔗 **De-dup note:** The detergent/biodegradability block was repeated on source p. 153 — merged once above.

12.6 Fuels

- **Fuel** — any substance undergoing chemical/nuclear change to produce energy that can be converted to useful work.

- **Calorific value** — energy released per unit mass on complete combustion; **SI unit J/kg**.

Wood	
Coal	Coke
Biomass	Cow dung cake
Petroleum	Paraffin wax
Nuclear fuel	Uranium

Fig. 12-4 · Fuel classification

- **Renewable sources** — obtained back via biogeochemical cycles (running water → rain; solar).
- **Non-renewable** — fossil fuels take millions of years; use judiciously; prefer alternatives first.

Properties of an Ideal Fuel

- **Ignition temperature** — easily attainable (matchstick head ignites by friction heat).
- **High calorific value.**
- **Environment-friendly** — less harmful residue; complete combustion (CNG leaves almost none).
- **Moderate, constant rate of combustion** — rapid/explosive burning is undesirable.
- **Easy availability at low cost and easy storage & transport.**

Types of Fuels

- **By state:** solid / liquid / fuel gases (in their primary stage). **By occurrence:** natural, biofuels, fossil, artificial.
 - **Fossil fuels** — dead plants → **coal**; dead aquatic animals → **petroleum (crude oil)**; rich energy, non-renewable.
 - **Artificial fuels** — water gas & producer gas (laboratory/industrial conditions).

Fuel	Composition	Sources
Water Gas	Carbon monoxide + Hydrogen	Reaction of red-hot coke with steam
Producer Gas	Carbon monoxide + Nitrogen	Reaction of red-hot coke with insufficient air
Coal Gas	Hydrogen + Methane + ethylene + Acetylene + CO + N ₂	Destructive distillation of coal
Natural gas	Methane (83%) + Ethane	Petroleum deposits
Liquified petroleum gas (LPG)	Butane (C ₄ H ₁₀) 95%	Fractional distillation of Petroleum
Compressed natural gas (CNG)	Methane (CH ₄) 95%	Petroleum deposits
Biogas or Gobar Gas	Methane (CH ₄) + Carbon dioxide (CO ₂) + Hydrogen (H ₂) + Nitrogen (N ₂)	Organic wastes

Fig. 12-5 · Calorific values of common fuels

Solid	Liquid	Gaseous
Wood, coal, coke, cow-dung cake, paraffin wax, uranium (nuclear)	Petrol, diesel, kerosene, LPG, LNG	Natural gas (firedamp — mainly CH ₄ + higher hydrocarbons + traces of H ₂ S); biogas; blast-furnace gas; refinery gases; oil-gasification gas

- **Natural gas characteristics:** high calorific value; proper ignition temperature; easier to extract from petroleum deposits than coal; liquefiable → **LNG** transported by tankers.
- **Other gaseous fuel sources:** methane around coal beds; waste & biomass gases; industrial gases; petroleum-derived (LPG, refinery gases); fermentation (ethyl alcohol from sugarcane).

Octane Number & Knocking

- **Octane number** — fuel's ability to resist "**knocking**" (premature combustion in spark-ignition/petrol engines); scaled to **iso-octane = 100, n-heptane = 0**; measured as **RON** (Research Octane Number).
- **Definition:** % of iso-octane in an iso-octane/n-heptane mixture matching the fuel's anti-knock quality.
- **Knocking:** premature, uneven combustion of the fuel-air mix → pressure waves → efficiency loss & engine damage; more prevalent with low-octane fuel.
- **High vs low octane:** high-compression (luxury/sports) engines need 95–100; regular petrol 87–90; premium 91–100+; racing/aviation fuels often exceed 100.
- **India:** moving toward **95 RON** and **E20** (20 % ethanol) blends for efficiency & lower emissions; indigenous 97-octane reference fuel produced; E20 mandate also raises octane.
- **Global standards:** RON & MON; pumps in many countries display **AKI** (Anti-Knock Index = average of RON & MON).
- **Octane vs cetane:** octane → petrol engines (spark-ignition); **cetane number** → diesel engines (compression-ignition), measuring ignition delay.

Tetraethyl Lead (TEL)

- TEL was added to petrol as an **anti-knock agent** — suppresses premature ignition, raises octane, lets engines run at higher compression ratios & more power.
- **Phased out globally** — lead is toxic (neurotoxic, environmental hazard); remains only in some aviation gasoline (Avgas). Safer alternatives: **ethanol** and **MTBE** (methyl tertiary-butyl ether).

🔗 **De-dup note:** The octane/TEL discussion was repeated across source pp. 155–157 (some blocks printed twice); the facts are consolidated once above.

12.7 Quick Revision — Everyday Chemistry

One-line answers (from across the unit)

Antiseptic vs disinfectant — 0.2 % phenol antiseptic, 1 % disinfectant.

Aspirin — analgesic + antipyretic.

Bithional — antiseptic in soaps.

Saccharin — first artificial sweetener, zero calorie.

Sodium benzoate — most common food preservative.

BHT/BHA — antioxidants.

Cetyltrimethylammonium chloride — cationic detergent germicide.

Sodium lauryl sulphate — biodegradable (straight chain) detergent.

Iso-octane : n-heptane = 100 : 0 octane scale.

E20 — 20 % ethanol-blended petrol, mandated in India.